Information Value Is Not Policy Value: When LLM Warm-Starts Help Linear Contextual Bandits

author names withheld

Editor: Under Review for ALT 2027

Abstract

Routing the early rounds of a linear contextual bandit to a frozen LLM and then handing control to LinUCB can beat both, sometimes by a large factor, and the LLM's accuracy does not predict when. We explain why. The expert's routed rounds reach the learner in two ways: through *coverage*, the directions its actions explore, and through the *transfer*, what its observed rewards say beyond that coverage. Coverage alone cannot produce large gains when LinUCB explores as the theory requires: on any instance with a fixed finite arm set in which the best arm is always available, the saving it certifies over cold LinUCB is capped by the ratio of the prices the two pay for coverage, and the cap is attained. Under linear rewards with Gaussian noise, experts of equal coverage are interchangeable. The transfer is not capped. When LinUCB explores less than the theory requires, as it does in practice, we construct two experts with identical coverage, cost and number of right answers, the losing one possibly the more accurate, whose multipliers over cold LinUCB differ in most runs by a factor that grows with the horizon, and in expectation once the expert gives about $\log T$ right answers. On AmazonCat-13K, three LLM experts of similar, low accuracy multiply the router's accuracy over cold LinUCB by 1.71, 2.72 and 3.29, cutting its regret by 6.5 to 21%. Coverage alone saves about 2 to 3.5% of cold LinUCB's regret for each; the transfer, which steers the learner to the most frequent category, carries the rest and every significant difference between experts, and hiding the rewards of a handful of right answers on that category, while keeping their coverage, removes their gain. Information value is not policy value.

Keywords: linear contextual bandits, LinUCB, warm start, large language models, exploration, regret

1. Introduction

Large pretrained language models carry prior knowledge that can guide the early decisions of a sequential learner (Brown et al., 2020; Ouyang et al., 2022; Achiam et al., 2023; Alamdari et al., 2024; Yadkori et al., 2024), in recommendation (Li et al., 2010; Agarwal et al., 2016), healthcare (Tewari and Murphy, 2017; Svensson, 2023) and dialogue (Li et al., 2016). A striking empirical finding is that combining a frozen LLM with an adaptive linear bandit can strictly dominate both, sometimes by a large factor (Chen et al., 2024; Huang et al.): routing the early rounds to the LLM and then handing control to LinUCB beats the LLM alone and LinUCB started from scratch. The literature on warm starts and model selection (Bastani et al., 2021; Zhang et al., 2019; Agarwal et al., 2017) reports nothing comparable, and the LLM's accuracy is a poor guide: on AmazonCat-13K, three LLM experts that are right on 3.6–5.3% of contexts raise the accuracy of the same learner by factors from 1.71 to 3.29 (cutting its regret by 6.5 to 21%), and the least accurate is not the least useful. This paper explains where the gain comes from and why accuracy cannot predict it.

© 2027 .

Policy value and information value. The *policy value* of an expert is the reward its own actions earn; its *information value* is how much its actions teach a learner that keeps their data. The two can be wildly different. On the instance of Proposition 2, a constant policy that is wrong on every round produces data under which LinUCB's regret falls from linear to constant: no policy value, enormous information value. That data, however, is collected before the evaluated horizon and not paid for. When the expert's rounds count toward the regret, as in a handoff, the data it leaves behind is worth nothing on the same instance, for every expert: after the handoff LinUCB does exactly what cold LinUCB does (Proposition 3). The question is when an *online* handoff can multiply, and by how much.

The accounting. Write C for cold LinUCB, E for the frozen expert, and H_τ for the hybrid that follows E for τ rounds, feeding every observation to LinUCB, and then hands over. Then

$$R_T(\mathsf{C}) - R_T(\mathsf{H}_\tau) = A_\tau + I_\tau, \qquad R_T(\mathsf{E}) - R_T(\mathsf{H}_\tau) = B_\tau$$

(Proposition 1), where A_τ is the expert's advantage on the routed rounds, I_τ the regret LinUCB saves afterwards because of the data it inherited, and B_τ the learner's advantage over the expert on the tail. The inherited data enters LinUCB through two objects: the design matrix, which sets every confidence width, and the response sum, which sets the estimate. The first is *coverage*; beyond coverage, the second contributes a single expert-dependent term, the *transfer*.

Coverage has a price. With a valid confidence radius, LinUCB's regret is limited by the confidence widths of actions it would otherwise mistake for the best, and widths shrink only with coverage. Coverage is never free: the *price* of a direction is the least regret any action pays per unit of coverage of it, and cold LinUCB buys the same coverage on its own, at the price of the arms its optimism forces it to pull. Theorem 1 holds for LinUCB with a valid confidence radius, on any instance with a fixed finite arm set in which the best arm is always available: the multiplier of any policy over cold LinUCB is at most the ratio of cold LinUCB's price to the expert's price on the arms the warm start certifies, plus cold LinUCB's regret on the arms it does not certify. It thus caps the certified saving rather than the multiplier outright, and a large certified saving requires an action of small gap that covers a direction cold exploration can buy only dearly (Corollary 1). The cap is attained (Theorem 3), and a catalog of eleven instances (Appendix C) shows that removing any one of recurrence, cheapness, correct targeting, or the expert's suboptimality on the tail can remove the gain. Under valid exploration, then, coverage multiplies only through cheap informative actions, and it cannot explain experts of similar accuracy and similar coverage whose multipliers differ severalfold.

The transfer decides (main result). Under linear rewards with Gaussian noise, and for an expert whose choices do not depend on the noise, the response sum is determined in law by the Gram matrix (Proposition 4), so experts that cover the same directions are interchangeable (Corollary 2). This fails in practice for two reasons: rewards are not linear in the features, and an LLM reads text the features only summarize, so it can prefer an action on exactly the rounds where that action pays. Either way the estimate at handoff acquires a transfer term (Definition 1). Our main result (Theorem 2) shows that when LinUCB explores less than the theory requires, as it does in practice, the transfer can decide everything: two experts with identical coverage, cost and number of right answers on the routed rounds can have multipliers over cold LinUCB that differ, in most runs, by a factor growing with the horizon, according to where their right answers fall, and the losing expert

can be the more accurate one. A handful of right answers on an action that pays often settles the learner on it, with a failure probability that decays exponentially in their number; about $\log T$ of them make the separation hold in expectation. With a valid radius the effect disappears. The two theorems therefore describe two regimes: when exploration is valid, coverage governs and its value is capped by prices; when exploration is reduced, as in practice, the transfer can exceed that cap.

What the data shows. On AmazonCat-13K the three experts above cover the same directions to within 5%, and coverage alone saves about 2 to 3.5% of cold LinUCB's regret for each; every significant difference between them disappears when the transfer is removed. The transfer acts as Theorem 2 describes: one category is present on 30% of contexts, a learner that settles on it ends near 28% accuracy against cold LinUCB's 8%, and whether it settles is decided by a few of the expert's routed answers; adding 15 correct answers on that category to the weakest expert doubles its accuracy multiplier, while the same answers with their rewards hidden from the learner, which keeps their coverage, leave it unchanged. On OneShotWikiLinks, where no entity covers more than 0.36% of contexts, the Flan-T5 experts gain almost nothing (accuracy multiplier 1.01).

Contributions.

- **The transfer can decide everything** (Theorem 2): with reduced exploration, two experts equal in coverage, cost and number of routed right answers can differ by a factor of order the horizon, and the more accurate one can lose.
- **Coverage is capped by prices** (Theorems 1 and 3, Corollary 2): with a valid radius the certified saving of any policy at any cutoff is bounded by a price ratio, the bound is attained, and under linear rewards with Gaussian noise coverage is all there is; on the canonical instance an offline warm start multiplies by a factor of the horizon while an online handoff inherits nothing (Propositions 2 and 3).
- **Experiments that isolate the transfer** (Section 8): on two datasets and twelve configurations, with every tuning parameter chosen on seeds disjoint from the evaluation seeds, coverage gives a small gain that is the same for every expert, the transfer carries most of the gain and every significant difference between experts, and right answers on the most frequent category double the multiplier only when their rewards reach the learner.

Section 2 discusses related work, Section 3 fixes notation, Section 4 gives the accounting, Sections 5–7 the theory, and Section 8 the experiments. The catalog and all proofs are in the appendix.

2. Related Work

LLMs and bandits. Chen et al. (2024) combine an LLM policy with a contextual bandit through an online model-selection master and report reward multiplication; Huang et al. calibrate LLM-derived priors to set exploration bonuses; Bayley et al. (2026) warm-start a linear bandit from LLM-generated preferences through a single prior-error term; and Pershin et al. (2026) inject LLM pseudo-observations into LinUCB with a calibration-dependent weight. We separate the design matrix from the response sum, bound what the first can deliver by a price ratio, and give the expert-dependent remainder in closed form; our decomposition and bound hold for any routed set, so they apply to these rules as well. **Warm starts and model selection.** Warm-start analyses (Bastani et al., 2021; Zhang et al., 2019) bound the regret of a learner started from logged data through the quality of that data; they are offline in our sense, and our contribution relative to them is the online accounting,

in which acquiring the data is charged and can cost more than it is worth. In hybrid and offline-to-online reinforcement learning (Song et al., 2022; Wagenmaker and Pacchiano, 2023) offline data helps when it covers what the online learner needs, and with demonstrations its value depends on the demonstrator's quality (Hao et al., 2023); here the data is acquired online, and its value depends on which of the expert's answers are right rather than on its accuracy. Conservative bandits (Wu et al., 2016) keep a learner close to a baseline policy; here the baseline is a source of data. Corralling and model selection (Agarwal et al., 2017; Pacchiano et al., 2020; Marinov and Zimmert, 2021) compete with the best of several base learners; we ask when a combination beats both components, and our bound applies to a CORRAL master as to any other policy. **Geometry of exploration and misspecification.** The price of Section 5 is the single-direction shadow of the allocation programs behind instance-dependent lower bounds (Graves and Lai, 1997; Lattimore and Szepesvari, 2017); our use is comparative, with two policies paying different prices for the same coverage. Unlike a hint about $\theta^\star$ (Cutkosky et al., 2022), whose value grows with its accuracy, a warm start is data in the design matrix, and its value is not a function of accuracy. Work on misspecified linear bandits treats the part of the reward the linear model misses as a cost to be controlled (Ghosh et al., 2017; Lattimore et al., 2020; Foster et al., 2020); here that part, carried by the transfer, is the main source of the gain. Appendix A gives an extended discussion.

3. Setup

We work in a standard linear contextual bandit (Abbasi-Yadkori et al., 2011; Chu et al., 2011; Lattimore and Szepesvári, 2020). At each round $t = 1, \dots, T$, the learner observes a finite action set $\mathcal{A}_t$ where each $a \in \mathcal{A}_t$ has a feature vector $x_{t,a} \in \mathbb{R}^d$ with $\|x_{t,a}\|_2 \le L$, selects $A_t \in \mathcal{A}_t$, and observes $r_t = \langle \theta^\star, x_{t,A_t} \rangle + \eta_t$, where $\theta^\star$ is unknown with $\|\theta^\star\|_2 \le S$ and η_t is conditionally mean-zero and σ-sub-Gaussian. Write $\mu_t(a) = \langle \theta^\star, x_{t,a} \rangle$, $a_t^\star \in \arg\max_a \mu_t(a)$, and $\Delta_{t,a} = \mu_t(a_t^\star) - \mu_t(a)$ for the suboptimality gap of a at round t. For a policy π and an interval $I \subseteq [T]$ of rounds, $R_I(\pi) = \sum_{t \in I} \Delta_{t,\pi_t}$ is its pseudo-regret on I, and $R_T(\pi) = R_{[T]}(\pi)$. The language aspect of the applications we target enters only through the feature map: context and action embeddings from pretrained models are combined into $x_{t,a}$, and the frozen expert may be an LLM. Nothing in the theory uses language structure.

LinUCB and its warm start. LinUCB maintains $V_t = V_1 + \sum_{s<t} x_{s,A_s} x_{s,A_s}^\top$, $c_t = c_1 + \sum_{s<t} r_s x_{s,A_s}$, $\widehat{\theta}_t = V_t^{-1} c_t$, and plays $A_t \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a} \rangle + \beta_t \|x_{t,a}\|_{V_t^{-1}}$, where $\|x\|_{V^{-1}} = \sqrt{x^\top V^{-1} x}$ is the confidence width of x and β_t is the confidence radius. *Cold-started* LinUCB, written C, uses $V_1 = \lambda I_d$, $c_1 = 0$. A *warm start* (M, c) uses $V_1 = \lambda I_d + M$ and $c_1 = c$ with $M = \sum_s x_s x_s^\top \succeq 0$ and $c = \sum_s r_s x_s$ the Gram matrix and response sum of some logged data. We say the warm start is *offline* if that data is not counted in the evaluated horizon and *online* if it is.

Frozen expert and handoff hybrid. A frozen expert E proposes $a_t^{\mathsf{E}} \in \mathcal{A}_t$ at each round by a rule that is never updated (for an LLM, the action whose embedding is closest to the LLM's output; see Section 8.1). For $\tau \in \{0, \dots, T\}$ the *handoff hybrid* H_τ plays a_t^{E} for $t \le \tau$, updates (V_t, c_t) with every played action and observed reward including those rounds, and plays LinUCB for $t > \tau$. Thus $\mathsf{H}_0 = \mathsf{C}$ and $\mathsf{H}_T = \mathsf{E}$, and the LinUCB phase of H_τ is LinUCB with the online warm start $(M_\tau, c_\tau) = \big(\sum_{t \le \tau} x_{t,A_t} x_{t,A_t}^\top, \sum_{t \le \tau} r_t x_{t,A_t}\big)$. The deterministic router used in our experiments (Appendix J) chooses τ adaptively and may interleave the two; the theory below is for a fixed τ,

which is the right object for understanding why a multiplier appears; Appendix B.2 comments on adaptive τ.

Key terms. Six quantities recur. The *confidence width* of an action with features x under a design matrix V is $\|x\|_{V^{-1}}$; LinUCB plays a suboptimal action only while $2\beta_t$ times its width is at least its gap (Lemma 1). The *coverage* of a direction u by a sequence of played actions is $\sum_s \langle x_s, u\rangle^2$, the growth of the design matrix along u; widths along u fall only as its coverage grows (Lemma 3, Appendix B.1). The *price* of u is the least regret any available action pays per unit of coverage of u (Section 5). A design matrix *certifies* an action once 2β times its width is below its gap, so that the confidence bound alone rules the action out. The *acquisition regret* of a warm start is the regret of the routed rounds that produced it. Finally, the *multiplier* of a policy π over cold LinUCB is $R_T(\mathsf{C})/R_T(\pi)$; the experiments also report the ratio of accuracies used in the literature (Section 8).

4. Where the Gain Comes From

The catalog and the experiments are organized by one identity.

Proposition 1 (Decomposition identity) *For any instance, any expert* E*, any* τ*, and any realization of the noise,*

$$R_T(\mathsf{C}) - R_T(\mathsf{H}_\tau) = \underbrace{R_{[1,\tau]}(\mathsf{C}) - R_{[1,\tau]}(\mathsf{E})}_{A_\tau} + \underbrace{R_{[\tau+1,T]}(\mathsf{C}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)}_{I_\tau},$$
$$R_T(\mathsf{E}) - R_T(\mathsf{H}_\tau) = \underbrace{R_{[\tau+1,T]}(\mathsf{E}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)}_{B_\tau}.$$

The identity is elementary; its use is to name the three terms. A_τ is the expert's *direct* advantage: positive if the expert earns more than cold LinUCB would have on the routed rounds, negative if the expert's actions cost more regret than cold exploration. I_τ is the *information* value of the routed rounds: after round τ the hybrid runs LinUCB from the warm start (M_τ, c_τ) that the routed rounds left, while cold LinUCB runs the same algorithm from the state its own first τ rounds left, so I_τ compares the two starting states on the same future. B_τ is the adaptive learner's advantage on the tail and is large only when the expert remains substantially suboptimal late while the warm-started learner does not. We write $R_{[1,\tau]}(\mathsf{E}) = \sum_{t\le\tau} \Delta_{t,a_t^{\mathsf{E}}}$ for the *acquisition regret* of the warm start: the price, counted in the online horizon, of the data in M_τ.

Remark 1 The identities hold pathwise in the reward-table model, in which the k-th pull of an action always receives the same noise draw, so that two policies can be run on the same noise (Lattimore and Szepesvári, 2020, §4.6), and therefore also in expectation. They also hold when the routed rounds are not an initial segment: for any set $\mathcal{T} \subseteq [T]$ of routed rounds, fixed in advance or chosen adaptively (as by the router of Appendix J), replacing $[1,\tau]$ by $\mathcal{T}$ gives the same identities, because each side is a difference of sums over a partition of $[T]$. If the warm start (M, c) is built from data outside the horizon (an *offline* warm start H^{off}, LinUCB started from $(\lambda I + M, c)$), the routed rounds are not charged and the identities read $R_T(\mathsf{C}) - R_T(\mathsf{H}^{\text{off}}) =: I^{\text{off}}$ and $R_T(\mathsf{E}) - R_T(\mathsf{H}^{\text{off}}) =: B^{\text{off}}$. A multiplier over both base policies means that $A_\tau + I_\tau$ is almost all of $R_T(\mathsf{C})$ and B_τ almost all of $R_T(\mathsf{E})$; every diverging ratio in this paper is read along a sequence of instances.

What a warm start can change. Fix the confidence event $\mathcal{E}_\delta$ on which $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \le \beta_t$ for all rounds t at which LinUCB acts, where (V_t, c_t) are built from all data so far including any warm start (M, c) formed from the same kind of data; by the self-normalized bound of Abbasi-Yadkori et al. (2011) this event has probability at least $1-\delta$ with $\beta_t = \sigma\sqrt{2\log(\det(V_t)^{1/2}\det(\lambda I)^{-1/2}/\delta)} + \sqrt{\lambda}S$. Define, for a positive definite V,

$$\Phi_t(V) \;:=\; \max_{a\in\mathcal{A}_t} \min\Big\{\Delta_{t,a},\; 2\beta_t\|x_{t,a}\|_{V^{-1}}\Big\}. \tag{1}$$

Lemma 1 (Width-limited regret) *On $\mathcal{E}_\delta$, at any round t at which LinUCB acts with design matrix V_t, the instantaneous pseudo-regret satisfies $\Delta_{t,A_t} \le \Phi_t(V_t)$. Moreover Φ_t is monotone: $V \preceq V'$ implies $\Phi_t(V') \le \Phi_t(V)$ when the radius β_t is held fixed.*

A competitor with a large gap but a larger width is as dangerous as its gap, one with a tiny gap is cheap however wide it is, and the only way to lower the regret LinUCB can incur on a round is to shrink the widths of competitors whose widths exceed their gaps. Widths shrink through a warm start and through LinUCB's own observations, both by exact linear algebra (Lemmas 3 and 4, Appendix B). One consequence is that the determinant-ratio bound behind $\widetilde{O}(\sqrt{rT})$-type guarantees cannot certify a multiplier: a multiplier that a width-limited bound can certify must come from competitors whose widths the warm start pushes *below their gaps* (Proposition 5, Appendix B.2).

5. The Price Bound

Why a price. By Lemma 1, the routed rounds can lower LinUCB's later regret only by shrinking widths, and widths shrink only where the routed actions add coverage. Coverage is never free: covering a direction means playing actions that point along it, and those actions have gaps. Cold LinUCB buys the same coverage on its own—optimism makes it pull each arm whose width exceeds its gap until that width falls below the gap—so the expert and cold LinUCB buy the same thing, and the question is what each pays per unit. That exchange rate is the price. The theorem below says that no policy beats cold LinUCB by more than the ratio of the two prices on the arms it certifies, plus what cold LinUCB pays on the arms nobody certified.

Prices. For a direction $u \in \mathbb{R}^d$ and an action set $\mathcal{A}_t$, write

$$\kappa_t(u) \;:=\; \min\Big\{\frac{\Delta_{t,a}}{\langle x_{t,a}, u\rangle^2} :\; a\in\mathcal{A}_t,\; \langle x_{t,a}, u\rangle \neq 0\Big\}, \qquad \kappa(u) := \inf_t \kappa_t(u), \tag{2}$$

the least pseudo-regret any policy must pay at round t to add one unit to the coverage $\sum_s \langle x_{s,A_s}, u\rangle^2$ of u. Coverage is what reduces widths (Lemma 3), so $\kappa(u)$ is the exchange rate between regret and width reduction in direction u. The following lemma says that no policy, expert or otherwise, gets coverage for less.

Lemma 2 (No free coverage) *For any policy π, any interval I of rounds, and any direction u, $\sum_{t\in I}\langle x_{t,\pi_t}, u\rangle^2 \le R_I(\pi)/\kappa(u)$. Consequently, if H_τ hands off with design matrix $V_{\tau+1} = \lambda I + M_\tau$, then for every unit vector u, $\|u\|^2_{V_{\tau+1}^{-1}} \;\ge\; 1/\big(\lambda + R_{[1,\tau]}(\mathsf{E})/\kappa(u)\big)$.*

The lemma is one half of the theorem; the other is that cold LinUCB pays its own price for each arm and stops once the arm's width falls below its gap.

Setting for the theorem. In words: we fix a finite set of arms in which the best arm is always offered, compare cold LinUCB with an arbitrary policy π up to a cutoff t_0, and measure how much coverage π bought, at what price, and which arms that coverage lets the confidence bound rule out. Let $\mathcal{X} \subset \mathbb{R}^d$ be a finite set of arm vectors with $\|x\|_2 \le L$, let $\mathcal{A}_t \subseteq \mathcal{X}$ for every t, and suppose an arm $x^\star$ with $\mu(x^\star) = \max_{x\in\mathcal{X}} \mu(x)$ belongs to every $\mathcal{A}_t$. The gap of an arm is then the same at every round, $\Delta(x) := \mu(x^\star) - \mu(x)$; write $\mathcal{S} := \{x \in \mathcal{X} : \Delta(x) > 0\}$ and fix $\beta \ge \beta_t$ for every $t \le T$ (one exists because $\|x\|_2 \le L$ bounds $\det V_t$).

What cold LinUCB pays. For $x \in \mathcal{S}$ put $u_x := x/\|x\|_2$ and

$$\kappa_x := \frac{\Delta(x)}{\|x\|_2^2}, \qquad C_x^\star := \frac{4\beta^2\|x\|_2^2}{\Delta(x)^2},$$
$$n_x^\star := \Big(\frac{C_x^\star - \lambda}{\|x\|_2^2} + 1\Big)\mathbf{1}\{C_x^\star \ge \lambda\}, \qquad N_x := \#\{t \in [T] : x \in \mathcal{A}_t\}.$$

Here κ_x is the price of covering u_x by pulling x itself, $C_x^\star - \lambda$ is the least coverage of u_x at which the confidence bound can exclude x, and $n_x^\star$ bounds how often cold LinUCB pulls x before its own pulls exclude it.

What π pays. Write x_t^π for the arm π plays at round t and, for a set $U \subseteq \mathcal{S}$ of arms, $\mathrm{cov}_U(x) := \sum_{y\in U}\langle x, u_y\rangle^2$. The price π pays for coverage of U up to t_0, and the coverage of U it obtains for free by playing optimal arms, are

$$\kappa_U^\pi(t_0) := \min\Big\{\frac{\Delta(x_t^\pi)}{\mathrm{cov}_U(x_t^\pi)} : \ t \le t_0,\ \Delta(x_t^\pi) > 0,\ \mathrm{cov}_U(x_t^\pi) > 0\Big\},$$
$$C_U^{\mathrm{free}}(t_0) := \sum_{t\le t_0:\,\Delta(x_t^\pi)=0} \mathrm{cov}_U(x_t^\pi),$$

with the minimum over an empty set equal to $+\infty$. Finally $U_{\mathrm{cert}}(t_0)$ is the set of arms $x \in \mathcal{S}$ that π's first t_0 rounds certify: $2\beta\|x\|_{(V_{t_0+1}^\pi)^{-1}} < \Delta(x)$, with $V_{t_0+1}^\pi = \lambda I + \sum_{t\le t_0} x_t^\pi (x_t^\pi)^\top$. For the handoff hybrid with $t_0 = \tau$, $\kappa_U^\pi(\tau)$ is the expert's price and $R_{[1,t_0]}(\pi)$ its acquisition regret, but nothing below uses this.

Theorem 1 (Price bound for the online multiplier) *In the setting above, on the confidence event of cold LinUCB, let π be any policy, $t_0 \in \{0,\dots,T\}$ any cutoff and $U \subseteq U_{\mathrm{cert}}(t_0)$ any set of certified arms, and write $\kappa_U^{\mathsf{C}} := \max_{x\in U}\kappa_x$ ($:= 0$ if $U = \emptyset$) and $P_U := \kappa_U^{\mathsf{C}}/\kappa_U^\pi(t_0)$ for the price ratio. Then*

$$R_T(\mathsf{C}) \ \le\ P_U\, R_{[1,t_0]}(\pi) \ +\ \kappa_U^{\mathsf{C}}\, C_U^{\mathrm{free}}(t_0) \ +\ \sum_{x\in U}\Delta(x) \ +\ \sum_{x\in\mathcal{S}\setminus U}\Delta(x)\min\{n_x^\star, N_x\},$$

and consequently, whenever $R_T(\pi) > 0$,

$$\frac{R_T(\mathsf{C})}{R_T(\pi)} \ \le\ P_U \ +\ \frac{\kappa_U^{\mathsf{C}}\, C_U^{\mathrm{free}}(t_0) + \sum_{x\in U}\Delta(x) + \sum_{x\in\mathcal{S}\setminus U}\Delta(x)\min\{n_x^\star, N_x\}}{R_T(\pi)}. \tag{3}$$

For $\pi = \mathsf{H}_\tau$ and $t_0 = \tau$ this bounds the multiplier of the handoff hybrid by the price ratio between cold exploration and the expert on the arms the warm start certifies, plus a residual.

Proof sketch. The proof (Appendix D.9) has two steps. First, cold LinUCB pays its own price, once per arm: on its confidence event it plays a suboptimal arm x only while $2\beta\|x\|_{V_t^{-1}} \geq \Delta(x)$, and its own pulls of x shrink that width, so it pulls x at most $\min\{n_x^\star, N_x\}$ times (Lemma 6). Second, certification costs at least the price: by Cauchy–Schwarz, π certifies x only once its coverage of u_x exceeds $C_x^\star - \lambda$, and it buys that coverage at a price of at least $\kappa_U^\pi(t_0)$ per unit (Lemma 7). Splitting cold LinUCB's bill into certified and uncertified arms and paying for the certified ones at the rate κ_U^{C} gives the first display; since $R_T(\pi) \geq R_{[1,t_0]}(\pi)$, dividing by $R_T(\pi)$ gives the second.

Only the confidence event of cold LinUCB is used, and π is arbitrary, so the bound covers the fixed-τ hybrid, the adaptive router (take t_0 to be its last routed round), a CORRAL-style master, and cold LinUCB itself. It bounds the ratio of the two policies' regrets on the same instance, not one upper bound against another: two policies with the same asymptotic rate can still differ by a diverging factor, so a statement about the multiplier must bound one policy's regret in terms of the other's, as the theorem does. Appendix D.10 evaluates the bound on the instance of Theorem 3.

The two parts of the bound. The price ratio covers the arms the policy paid to certify. The residual is what nobody paid for, and it has three pieces: the coverage the policy obtains for free by playing optimal arms (zero when the optimal arm is orthogonal to the certified directions, as on every instance of the catalog), one pull of each certified arm, and cold LinUCB's regret on the arms whose widths the warm start does not push below their gaps. Only the last can be large, and it is counted arm by arm, so it can be loose when many arms share information. The theorem therefore controls the *certified saving*, the part of cold LinUCB's regret on arms the policy paid to certify, rather than the multiplier outright. On those arms the hybrid faces the same confidence bound as cold LinUCB, so no coverage argument can say which of the two does better there, and the term cannot be dropped (Corollary 1). When the arms cold LinUCB would otherwise pay for are certified, the residual is small and the price ratio is the bound, and Theorem 3 shows that it is then attained up to constants. Whether the residual can be large without a large price ratio is open. The bound holds whether the saving is collected during the routed rounds (A_τ) or after them (I_τ), and the confidence radius cancels in P_U because it scales $C_x^\star$, the coverage both policies must reach, identically.

Corollary 1 (Diverging online multipliers require vanishing prices) *Consider a sequence of instances, policies π and cutoffs t_0 along which $R_T(\mathsf{C})/R_T(\pi) \to \infty$ and, for some choice of $U \subseteq U_{\mathrm{cert}}(t_0)$ on each instance, the numerator of the residual in (3) is at most a fixed multiple of $R_T(\pi)$. Then $P_U \to \infty$: the price $\kappa_U^\pi(t_0)$ that π pays for the certified coverage vanishes relative to the price κ_U^{C} cold LinUCB pays for it, so π plays at least one action whose gap is negligible against the coverage it supplies. For the handoff hybrid this means the expert has access to such an action. On the canonical instance of Proposition 2 no online expert certifies any arm that is offered after the handoff, so all of cold LinUCB's regret on those arms is in the residual; that is the price-theoretic content of Proposition 3. Offline, where the acquisition is not charged, the same instance multiplies by a factor of the horizon.*

The proof is in Appendix D.11. Two relaxations of Theorem 1 and a discussion of when cheap informative actions exist are in Appendix B.

6. The Bound Is Attained, and Information Value Can Vanish

Theorem 1 is attained up to constants, and the information value it controls can vanish. On the canonical separating instance an offline warm start multiplies by a factor of the horizon while the data an online handoff inherits is worth nothing; on a modification of it the price ratio is attained.

The canonical instance: offline against online. The canonical instance, which we construct to model the phenomenon reported by Chen et al. (2024), uses horizon n, a single reward direction $g = e_1$, and n orthogonal "bad" arms $b_t = Le_{t+1}$ with $L > 1$.

Proposition 2 (Offline three-way separation) *Fix $n \geq 1$, $L > 1$, $\lambda > 0$. There exist $d \geq n+1$, $\theta^\star$, and two-action sets $\{\mathcal{A}_t\}_{t=1}^n$ such that, for any sub-Gaussian noise: (1) a constant policy π_{const} has pseudo-regret exactly n; (2) cold LinUCB also incurs regret n, selecting the same action as π_{const} each round; (3) for m large enough, LinUCB warm-started with the design matrix that π_{const} generates from mn* offline *rollouts incurs $O(1)$ regret with probability $\geq 1-\delta$.*

Cold LinUCB fails because at each round the fresh arm b_t has width $L/\sqrt{\lambda}$ whatever was played before, while g has width $1/\sqrt{\lambda}$ and estimated value 0, so optimism picks b_t every time. The offline design matrix $M = mL^2 \sum_{j\geq 2} e_j e_j^\top$ does not teach LinUCB that g is good; it teaches it that the *alternatives* are not uncertain, since $\|b_t\|^2_{(\lambda I+M)^{-1}} = L^2/(\lambda + mL^2)$ falls below the gap once m is moderate. Acquiring that data is not charged. Online it would be, and the instance is hostile to online acquisition: every round introduces a direction that no earlier observation touches.

Proposition 3 (No online gain on the canonical instance) *Consider the instance of Proposition 2 with horizon T and confidence radius β, and let $\zeta := \sigma\sqrt{2\log(2T/\delta)}/\sqrt{\lambda}$. If $\beta(L-1)/\sqrt{\lambda} > 1+\zeta$, then with probability at least $1-\delta$, for* every *expert policy E and every $\tau \in \{0,\dots,T\}$, the handoff hybrid H_τ plays b_t at every round $t > \tau$. Consequently $R_{[\tau+1,T]}(\mathsf{H}_\tau) = R_{[\tau+1,T]}(\mathsf{C}) = T - \tau$ and $I_\tau = 0$.*

No online router realizes the offline separation on this instance: a bad expert's data has $I_\tau = 0$, and a good expert has $B_\tau \leq 0$ because it is already optimal (Corollary 3, Appendix C). Online multiplication needs an expert whose routed actions are cheap *and* cover directions that recur.

The price ratio is attained. Theorem 1 is tight up to constants. We modify the canonical instance so that directions recur and, for a while, a cheap informative action is available: a *probe* of gap ε that covers a bad direction at rate γ^2 per round. The hybrid that routes the probes and then hands off beats cold LinUCB by a factor between $\gamma^2/(2220\,\varepsilon L^2)$ and $1 + \gamma^2/(\varepsilon L^2)$ (Theorem 3, Appendix C.2), the ratio of cold exploration's price $1/L^2$ to the probe's price ε/γ^2. A catalog of eleven instances built the same way (Table 2, Appendix C) shows that removing any one of recurrence, cheapness, correct targeting, or the expert's suboptimality on the tail can remove the gain: across these linear instances a gain over cold LinUCB appears only when the expert buys the directions cold LinUCB would otherwise pay for, more cheaply and early enough for them to recur.

7. Coverage and the Transfer

A warm start (M, c) enters LinUCB through two objects: the design matrix $V_1 = \lambda I + M$, which fixes every confidence width, and the response sum $c_1 = c$, which fixes the estimate $\widehat{\theta}_1 = V_1^{-1}c$.

Theorem 1 bounds the value of a warm start through the first. We now show that in the model of Section 3, with an expert that sees only what the learner sees, the first is the whole story, and that the second matters only when the logged rewards depart from that model, in which case it has an explicit form.

Proposition 4 (The response sum is determined by the design matrix) *Let the warm start be built from pairs $(x_s, r_s)_{s\le\tau}$ with $r_s = \langle\theta^\star, x_s\rangle + \eta_s$ and η_s i.i.d. $\mathcal{N}(0,\sigma^2)$, independent of which actions were logged and of which rounds were routed (as for a fixed handoff time and an expert whose choices depend only on the features). Then*

$$c = M\theta^\star + \xi, \qquad \xi := \sum_{s\le\tau} \eta_s x_s \sim \mathcal{N}(0, \sigma^2 M), \qquad \widehat{\theta}_1 - \theta^\star = -\lambda V_1^{-1}\theta^\star + V_1^{-1}\xi.$$

Hence the law of (V_1, c_1) given M does not depend on which policy generated the logged actions: two experts, or two routed sets, with the same Gram matrix leave warm starts with the same law, and the LinUCB phases of the corresponding hybrids have the same law; in particular they have the same $\mathbb{E}[I_\tau]$. (B_τ can still differ, since it also involves the expert's own actions after the handoff.)

For σ-sub-Gaussian noise the distributional identity is replaced by the observation that the self-normalized bound of Abbasi-Yadkori et al. (2011), the only property of ξ used in this paper, depends on the features only through M; every bound on the LinUCB phase in Sections 4–6 is therefore unchanged when one expert is exchanged for another of equal coverage, provided neither the experts' choices nor the routed rounds depend on the noise.

Corollary 2 (Coverage is complete under linear rewards) *Under the model of Section 3 with i.i.d. Gaussian noise, and with the logged actions and routed rounds independent of the noise, two experts whose routed rounds leave the same Gram matrix give hybrids whose LinUCB phases have the same law, and hence the same $\mathbb{E}[I_\tau]$. A systematic difference in I_τ between experts of equal coverage therefore comes from a departure from the linear model, from non-Gaussian noise, or from choices (of the expert or of the router) that depend on the noise.*

Proof Immediate from Proposition 4: equal Gram matrices give warm starts (V_1, c_1) of the same law, and the LinUCB phase is a deterministic function of (V_1, c_1) and of the future contexts and noise, which are independent of the warm start. Cold LinUCB does not depend on the expert, so $\mathbb{E}[I_\tau] = \mathbb{E}[R_{[\tau+1,T]}(\mathsf{C})] - \mathbb{E}[R_{[\tau+1,T]}(\mathsf{H}_\tau)]$ is the same for both. ■

Corollary 2 is a prediction that can be checked on data: experts whose routed rounds cover the same directions should leave warm starts of the same value. Section 8 checks it.

When the transfer appears. Suppose $r(x) = \langle\theta^\star, x\rangle + \varepsilon(x) + \eta$ for a fixed reference $\theta^\star$—for instance the best linear predictor—and a residual ε; the binary Precision@1 reward of Section 8 is of this form for every $\theta^\star$, and badly so, since no linear function of $\phi(x,a)$ reproduces an indicator. Regret is still measured against the true best action, with $\bar{r} := \langle\theta^\star, \cdot\rangle + \varepsilon(\cdot)$ and $\Delta_{t,a} = \bar{r}(x_{t,a_t^\star}) - \bar{r}(x_{t,a})$. A second departure concerns the expert rather than the reward. Proposition 4 assumes that the noise is independent of which actions were logged. An LLM reads the raw context, which the features only summarize, so it can prefer an action on exactly the rounds where that action pays; its logged rewards are then selected, and the response sum is no longer determined by the Gram matrix, even when $\varepsilon \equiv 0$.

Definition 1 (The transfer) Fix a reference $\theta^\star \in \mathbb{R}^d$ (under misspecification, the best linear predictor). The *transfer* of a warm start built from $(x_s, r_s)_{s\le\tau}$ is

$$m := V_1^{-1} \sum_{s\le\tau} \big(r_s - \langle \theta^\star, x_s\rangle\big)\, x_s, \qquad \text{so that} \qquad \widehat{\theta}_1 = \theta^\star - \lambda V_1^{-1}\theta^\star + m$$

exactly, since $c = M\theta^\star + V_1 m$ and $V_1^{-1}M = I - \lambda V_1^{-1}$. Given M, the transfer is the only part of the estimate at handoff that depends on which rounds were logged and what they returned.

Under the linear model with Gaussian noise, and with the logged actions and routed rounds independent of the noise, $m = V_1^{-1}\xi$ and its law is fixed by M (Proposition 4). It departs from that law through a residual $\varepsilon(x) = \mathbb{E}[r \mid x] - \langle \theta^\star, x\rangle$ (misspecification); through an expert whose choices depend on information about the reward that the features do not carry, as an LLM reading the raw text does, which makes $\mathbb{E}[r_s - \langle \theta^\star, x_s\rangle \mid x_s] \neq 0$ on the routed rounds even when $\varepsilon \equiv 0$; and through routed rounds chosen using past rewards. On AmazonCat the reward is a fixed function of the context, so these sources cannot be told apart there, and we measure their sum.

The transfer can help or hurt: Proposition 7 (Appendix B) bounds its effect on a round, and the bound improves when it moves the estimate toward a reference whose residual is smaller on the best action than on the actions LinUCB would otherwise play.

How far the transfer can go. Proposition 4 says that two experts of equal coverage are interchangeable when the logged noise is independent of the logged actions. The next result shows how far this fails once the expert sees more than the learner and LinUCB explores less than the theory requires, as in our experiments. It modifies the canonical instance of Proposition 2: the good action is right only on a fraction p of the rounds, which rounds is visible to the expert but not to the learner, a second action, right on a smaller fraction q, is added, and the fresh arm is right on a fraction f.

The instance. Fix $k \ge 1$, $0 \le q < p$ and $0 \le f < p$ with $p + q + f \le 1$, $L > 1$, $\lambda > 0$, a horizon $T > 4k$, and $d = T + 2$. At round t the learner is offered $D = e_1$, $R = e_2$ and a fresh arm $b_t = Le_{t+2}$. Each round has a type $y_t \in \{D, R, F, \varnothing\}$ that the learner does not see: on a round of type D or R the action of that name has reward 1, on a round of type F the fresh arm has reward 1, and every other action has reward 0. The first $\tau := 4k$ rounds form a window with k rounds of type D, k of type R and $2k$ of type $\varnothing$, in any order; later types are i.i.d. with $\mathbb{P}(D) = p$, $\mathbb{P}(R) = q$, $\mathbb{P}(F) = f$. After the window the learner's expected reward is linear in its features, $\mu(x) = \langle \theta^\star, x\rangle$ with $\theta^\star = pe_1 + qe_2 + (f/L)\sum_t e_{t+2}$, and regret is measured against it throughout: the gaps are 0 for D, $p - q$ for R and $p - f$ for every b_t. Two experts see the type. E_A plays D on rounds of types D, R and F and plays R on rounds of type $\varnothing$; E_B plays R on rounds of types D and R, the fresh arm on rounds of type F, and D on rounds of type $\varnothing$. After the window E_A is right with probability p and E_B with probability $q + f$, so E_B is the more accurate expert whenever $f > p - q$. LinUCB uses regularization λ and a constant radius $\beta > 0$; we write $\omega := \beta L/\sqrt{\lambda}$ for the index of an untried fresh arm, and break ties in favour of the fresh arm. The two experts are built so that everything coverage can see is the same: each plays D and R exactly $2k$ times in the window, pays the same regret, and is right exactly k times. Their wrong answers are padding that makes these totals equal, so the only difference left is which action their right answers fall on.

Theorem 2 (Equal coverage, unequal value) *On this instance the following hold.*

(i) (Same coverage, same cost, same right answers when routed.) $M_\tau^A = M_\tau^B = 2k(e_1e_1^\top + e_2e_2^\top)$, *both acquisition regrets equal* $2k(p-q)$*, and each expert is right on exactly* k *of its* τ *routed rounds. The two hybrids therefore hand off with the same design matrix, hence the same widths, certified arms and prices, and their estimates differ only through the transfer:* $\widehat{\theta}^A_{\tau+1} = \frac{k}{\lambda+2k}\, e_1$ *and* $\widehat{\theta}^B_{\tau+1} = \frac{k}{\lambda+2k}\, e_2$.

(ii) (Cold, and coverage alone.) *Cold LinUCB plays the fresh arm at every round, so* $R_T(\mathsf{C}) = (p-f)T$*. The same holds after round* τ *for either hybrid if the routed rounds contribute their covariance but not their rewards.*

(iii) (The expert whose right answers are rare.) H^B_τ *never plays* D *after round* τ*, so* $R_T(\mathsf{H}^B_\tau) \geq (T-2k)\min\{p-q,\, p-f\}$.

(iv) (The expert whose right answers are frequent.) *Suppose the exploration is small,* $\omega < p$*, and* $h := k(1-2\omega) - \omega\lambda > 0$*. With probability at least* $1 - e^{-8(p-\omega)h}$*,* H^A_τ *plays* D *at every round after* τ*, and then* $R_T(\mathsf{H}^A_\tau) = 2k(p-q)$*. If also* $\omega \leq \min\{p, \frac{1}{2}\}/2$*, this probability is at least* $1 - e^{-2p(k-2\omega\lambda)}$.

(v) (In expectation.) $\mathbb{E}R_T(\mathsf{H}^A_\tau) \leq 2k(p-q) + (p-f)T\, e^{-8(p-\omega)h}$*. In particular, if* $k \geq \big(\ln T/(8(p-\omega)) + \omega\lambda\big)/(1-2\omega)$*, then* $R_T(\mathsf{C})/\mathbb{E}R_T(\mathsf{H}^A_\tau) \geq (p-f)T/\big(2k(p-q)+p-f\big)$.

Consequently, with the probability in (iv),

$$\frac{R_T(\mathsf{C})}{R_T(\mathsf{H}^A_\tau)} = \frac{(p-f)T}{2k(p-q)}, \qquad \textit{while always} \qquad \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}^B_\tau)} \leq \frac{p-f}{\min\{p-q,\, p-f\}} \cdot \frac{T}{T-2k}.$$

Two experts that buy exactly the same coverage, at exactly the same price, with exactly as many right answers on the routed rounds, give warm starts whose multipliers over cold LinUCB differ, in most runs, by a factor of order T/k; when $f > p-q$ the expert that loses is the more accurate one. Nothing that depends on coverage alone—the design matrix, the widths, the prices—can tell the two hybrids apart, and accuracy ranks the experts the wrong way. What separates them is where their routed right answers fall: E_A's are on an action that is right on a fraction p of later rounds, E_B's on one that is right on a fraction q. The learner settles all or nothing: once H^A_τ leaves D it never returns, so for a fixed k it fails for good in a fixed fraction of runs, at most $e^{-8(p-\omega)h}$, and the average multiplier stays bounded. Part (v) shows that about $\log T$ right answers make that fraction small enough for the separation to hold in expectation as well, at order $T/\log T$.

That a learner which explores too little can lock onto whatever paid first is known: greedy and near-greedy linear bandits can succeed or fail depending on the data (Bastani et al., 2021), and the value of demonstrations to an online learner depends on the demonstrator's quality (Hao et al., 2023). What Theorem 2 adds is that, for an online warm start, lock-in makes the value of an expert's data depend on where its right answers fall rather than on its coverage, its cost or its accuracy: two experts equal in all three differ by a factor of order the horizon.

Remark 2 (Where the theorem sits) The instance violates the hypotheses of both earlier results, and each violation is needed. The experts see the type, so their logged rewards are not independent of their actions, which Proposition 4 assumes. And the radius is below any valid confidence radius, which Theorem 1 assumes. A valid radius has $\beta \geq \sqrt{\lambda}\,\|\theta^\star\|_2 \geq \sqrt{\lambda}\,p$, so $\omega = \beta L/\sqrt{\lambda} > p$ because $L > 1$, and the condition $\omega < p$ of (iv) cannot be met: as H^A_τ keeps playing D, its index for D tends to $p < \omega$, so after finitely many rounds it switches to fresh arms and, its statistics for D no longer

changing, never returns. Both hybrids then behave like cold LinUCB, and both multipliers tend to 1 as the horizon grows. With a valid radius coverage governs; with the reduced exploration used in practice (Section 8.1) the transfer can decide. The fresh arms serve only to make cold LinUCB fail and to make leaving D permanent; $d = T + 2$ is what they cost.

Remark 3 (The instance as a cartoon of AmazonCat) Each piece of the instance has a counterpart in Section 8.2. The action D plays the role of the category present on 29.8% of AmazonCat contexts, and R that of a rarer category. The fresh arms stand for the thousands of categories cold LinUCB keeps trying because their widths are large, which is why it settles on the dominant category in only 1 of 20 seeds. The number k of right answers on D corresponds to the number of rewarded routed answers on the dominant category: about 0.2 per run for Flan-T5-small and 3.7 and 4.0 for Flan-T5-base and Flan-T5-large. The all-or-nothing settling of part (iv) and its steep dependence on k appear in Tables 11 and 12. The instance is not fitted to the data; it isolates the mechanism the data display.

Because the transfer is not a function of coverage, an estimate of the multiplier built from coverage alone cannot tell the experts apart. A dry run on labeled held-out data ranks them correctly, because it replays the expert's right answers and so includes the transfer, but it underestimates the gain about threefold and needs labeled data and the expert's answers on it (Appendix K).

8. Experiments

Corollary 2 predicts that experts of equal coverage leave warm starts of equal value unless the transfer intervenes, and Theorem 2 predicts how it can intervene. We test both on real data, running the router, the LLM alone, and cold LinUCB on the same stream; the setup mirrors Chen et al. (2024).

Two scales. The literature reports *reward multiplication*, which with binary Precision@1 is a ratio of accuracies; we call it the *accuracy multiplier*. The theorems bound a ratio of regrets, and since every context has a correct action, $R_T(\pi) = T(1-\mathrm{acc}(\pi))$; we therefore also report the *regret ratio* $(1 - \max\{\mathrm{acc}_{\mathrm{LLM}}, \mathrm{acc}_{\mathrm{Cold}}\})/(1 - \mathrm{acc}_{\mathrm{Router}})$. At low accuracies the two differ sharply: on AmazonCat an accuracy multiplier of 3.29 from a base of 8.4% is a regret ratio of 1.27, or 21% less regret. A_τ, I_τ and B_τ are counted in correct answers and are the same on either scale.

8.1. Setup

Datasets. OneShotWikiLinks-311[1] ($T = 622{,}000$, $|\mathcal{A}| = 311$) pairs two model families (Flan-T5, Qwen2.5) with the same dataset, giving six configurations. **AmazonCat-13K** (Bhatia et al., 2016) ($T = 50{,}000$, $|\mathcal{A}| = 13{,}330$) pairs two embedding spaces (BERT-768d, BGE-1024d) with the same categories. Reward is binary Precision@1.

Policies. Each run tracks three policies on the same stream: the **Router** (Algorithm 1, Appendix J), which routes a round to the LLM when the LLM's width $\|\phi(x_t, a_t^{(1)})\|_2/\sqrt{c_{\mathrm{LLM}}}$ is below LinUCB's width $\|\phi(x_t, a_t^{(0)})\|_{V_t^{-1}}$ and updates (V_t, c_t) on every round with the played action and observed reward, so that its LinUCB component is the online-warm-started learner of Section 3; the **LLM** alone, which proposes the action whose embedding is closest to its textual output (Chen

1. https://www.kaggle.com/generall/oneshotwikilinks

et al., 2024); and **cold LinUCB**. We use $\lambda = 0.1$ and the exploration scale $\alpha_t = \max(0.1, 0.9985^t)$, which reaches its floor at $t \approx 1{,}500$. This is smaller than the confidence radius the theory requires, so the confidence event of Section 4 is not guaranteed in our runs: we use the theory to explain the experiments, not to certify them. [**TODO:** Phase B: add the tuned cold-LinUCB baselines (exploration schedule and floor, λ, category-term features) and the Router re-tuned against them.] A seed fixes the order of the stream and is shared by all policies, so every comparison is paired. The confidence parameter c_{LLM} of each configuration is chosen on seeds 11–20 by mean Router accuracy and evaluated on seeds 1–10, so tuning and evaluation never share a seed; this selects $c_{\mathrm{LLM}} = 100$ for AmazonCat/BERT, $c_{\mathrm{LLM}} = 0.1$ for AmazonCat/BGE and $c_{\mathrm{LLM}} = 10^4$ for WikiLinks/BERT (Appendix H). [**TODO:** WikiLinks sweep]

8.2. AmazonCat-13K

AmazonCat-13K has 13,330 categories and $T = 50{,}000$ rounds. We pair BERT-768d features with Flan-T5 experts and BGE-1024d features with Qwen2.5 experts (Table 1).

Table 1: AmazonCat-13K, $T = 50{,}000$, mean $\pm$ standard deviation over seeds 1–10. Acc. mult. is the seed-averaged Router accuracy over the seed-averaged $\max\{\text{LLM}, \text{cold}\}$; the regret ratio is $(1 - \max\{\text{LLM}, \text{cold}\})/(1 - \text{Router})$, the quantity Theorem 1 bounds. Cold LinUCB does not depend on the expert. BERT rows are at $c_{\mathrm{LLM}} = 100$ and BGE rows at $c_{\mathrm{LLM}} = 0.1$, each selected on the disjoint seeds 11–20 (Appendix H). On BGE the two larger experts gain significantly (paired $t = 3.92$ and 2.59 against cold) and the smallest does not ($t = 0.01$).

Features	LLM	LLM acc.	Cold LinUCB	Router	LLM%	Acc. mult.	Regret ratio
BERT-768d	Flan-T5-small	5.20%	$8.39 \pm 2.46\%$	$14.37 \pm 6.12\%$	2.8%	1.71×	1.07
	Flan-T5-base	3.61%	$8.39 \pm 2.46\%$	$22.80 \pm 9.00\%$	3.1%	2.72×	1.19
	Flan-T5-large	5.27%	$8.39 \pm 2.46\%$	$\mathbf{27.64 \pm 5.56\%}$	3.2%	**3.29×**	**1.27**
BGE-1024d	Qwen-0.5B	21.38%	$32.59 \pm 5.00\%$	$32.62 \pm 6.33\%$	0.04%	1.00×	1.00
	Qwen-3B	22.39%	$32.59 \pm 5.00\%$	$\mathbf{40.20 \pm 4.45\%}$	0.05%	**1.23×**	**1.13**
	Qwen-7B	27.17%	$32.59 \pm 5.00\%$	$38.71 \pm 6.38\%$	0.05%	1.19×	1.10

Equal coverage, unequal value. Held-out accuracy orders the three Flan-T5 experts differently from their multipliers (Table 1). The decomposition locates the difference: A_τ is 47, 27 and 53 correct answers and I_τ is 2,936, 7,178 and 9,571 (Table 5, Appendix H), and 96–100% of each Router's advantage accrues after its last routed round. The experts cover the directions cold LinUCB is trapped on to within 5%, at nearly the same prices, so their price ratios (1.73, 1.59, 1.58; an explanatory statistic measured after the run, Appendix H.2) are close to one another, ordered opposite to the multipliers, and above every observed regret ratio. In the linear model with Gaussian noise, Corollary 2 would make experts of equal coverage interchangeable; the real rewards are not linear, so we test directly whether the transfer carries the separation, and Figure 1 shows that it does. We replay each run with the routed rounds held fixed: all three versions of the learner route exactly the rounds the full Router chose and play the expert's answer there, so they have the same routed data and the same coverage, and differ only in what those rounds add to the response sum.

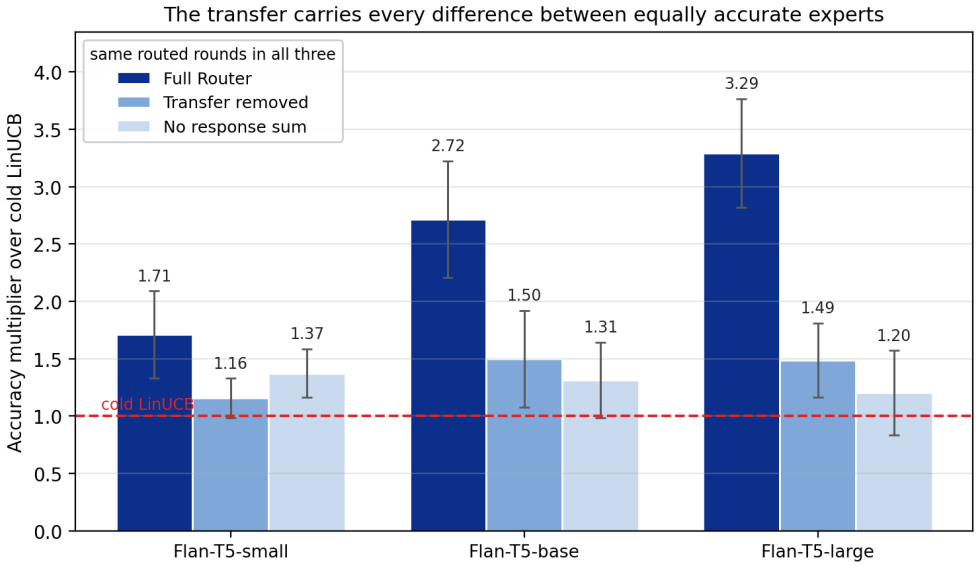


Figure 1: Coverage versus the transfer on AmazonCat/BERT-768d, ten seeds, standard errors over seeds. All three versions of the learner route exactly the same rounds, the ones the full Router chose, and play the expert's answer there; they differ only in what a routed round adds to the response sum. The three experts cover the same directions to within 5%. With the transfer present (dark bars) they span $1.71\times$ to $3.29\times$; with it removed they fall to $1.16\times$–$1.50\times$ and no pairwise difference survives. *Transfer removed* keeps the expert-independent part $M\theta_{\mathrm{or}}$ of the response sum and *no response sum* discards it entirely; the two do not differ significantly on any expert.

Feeding the learner $\langle \theta_{\mathrm{or}}, \phi \rangle$ instead of the observed reward on the routed rounds, which deletes m of Definition 1 with $\theta^\star = \theta_{\mathrm{or}}$, turns every significant pairwise difference into a null result (large over small: paired $t = 5.18$ becomes 1.33); dropping the response sum on the routed rounds gives the same collapse; and a linear reward makes the experts interchangeable, as Proposition 4 predicts. On the regret scale coverage alone saves about 2 to 3.5% of cold LinUCB's regret for every expert, while the full router saves 6.5%, 16% and 21% (Appendix H).

The dominant category. On AmazonCat one category is present on 29.8% of contexts, so a policy that knew this in advance and always predicted it would score 29.8%. An online learner does not know it, and finding it is not easy: context-free bandits that try categories without using the context reach 0.04% (UCB1) and 2.31% (optimistic greedy) over the same $50{,}000$ rounds and $13{,}330$ categories, seeds 1–10. The Routers that multiply are the ones that discover this category. LinUCB settles on one frequent category and predicts it on a large share of rounds, and its accuracy then tracks that category's frequency: runs that settle on the dominant category finish near 28%, while cold LinUCB settles on a category present on 11% of contexts, or on none, in 18 of 20 seeds. Whether a Router settles on the dominant category is decided by a handful of the expert's routed answers. Across 177 Router runs on BERT features (every c_{LLM} and CORRAL, excluding

configurations that route most rounds), a run whose expert gave no correct routed answer on this category reached it in 3 of 45 runs, and a run that received five or more reached it in all 54.

A controlled test. Adding 15 correct answers on this category to Flan-T5-small (+0.15 points of held-out accuracy) doubled its multiplier (1.33× to 2.62×, paired $p = 0.002$), while 15 correct answers on random categories gave no significant gain (1.65×, $p = 0.15$), and the same 15 answers with their rewards hidden from the learner, so that they add coverage along the category but no transfer, gave none either (1.48×, $p = 0.47$; against the same answers with their rewards, $p = 0.003$). Only about two of the 15 changed answers reach the learner in a typical run, because the Router consults the expert on few rounds, almost all of them early: two rewarded answers on the dominant category are enough to double the multiplier, and the same two answers with their rewards hidden do nothing.

On the evaluation stream. The same test on the 50,000-round stream and seeds of Table 1 gives the same result: 75 added answers on this category raise the multiplier from 1.71× to 2.56× ($p = 0.007$), and with their rewards hidden only to 1.84× ($p = 0.61$; Table 14). This is the pattern of Theorem 2. As in its part (ii), the routed covariance without the routed rewards buys nothing; every correct answer carries a residual near 1, but only a handful on a frequent action push the estimate in a direction that keeps paying, settling is all or nothing, and its probability rises steeply with the number of such answers. Appendix I gives the details.

BGE features and other routers. With BGE-1024d features cold LinUCB is much stronger and settles on the dominant category by itself in 7 of 20 seeds, so there is less to add; the two larger Qwen2.5 experts still gain significantly while routing 0.05% of rounds, again almost entirely through I_τ (Appendix H.4). The decomposition and the price bound hold for any routed set, so they apply to the randomized CORRAL master (Agarwal et al., 2017) used by Chen et al. (2024) as to the gate. Against CORRAL and offline warm starts on AmazonCat/BERT, no alternative is significantly better on any expert, and where one comes out ahead in the mean the margin costs 3–5× as many expert answers (Appendix H.7).

8.3. OneShotWikiLinks-311

On OneShotWikiLinks-311 (622,000 rounds, 311 entities) the most frequent entity covers 0.36% of contexts, so there is no category for a warm start to steer the learner toward. With Flan-T5 experts the Router gains 0.13–0.17 points over cold LinUCB (1.01×), although the routed data does reach useful directions (I_τ is 748, 832 and 886 correct answers and positive on every seed). With Qwen2.5 experts the expert is already better than the learner (29–39% against 19%), so $B_\tau < 0$ and the Router has little to add over the expert, as in row 3 of Table 2. [**TODO:** WikiLinks sweep] Full results are in Appendix H.

9. Conclusion

Reward multiplication in LLM–bandit hybrids has an exact accounting, and the data the expert leaves behind reaches the learner through coverage and through the transfer. The transfer is what decides the gain. When exploration is below the valid radius, as in practice, it can separate experts of identical coverage, cost and number of routed right answers by a factor of order the horizon, even when the losing expert is the more accurate. On AmazonCat-13K it carries every significant

difference between experts and most of the regret the router saves, through a handful of right answers on the most frequent category: with their rewards hidden from the learner, the same answers buy nothing. Coverage cannot play this role. With a valid radius the multiplier it can deliver is capped by a ratio of prices, the cap is attained on explicit instances, and the information value it controls vanishes on the canonical one; under linear rewards with Gaussian noise, for an expert whose choices ignore the noise, coverage is all there is; and on AmazonCat it saves about 2 to 3.5% of cold LinUCB's regret for every expert. A warm-start expert cannot be chosen by its accuracy or by its coverage: what decides its value is which of its answers are right. Information value is not policy value.

Limitations. The price bound assumes linear rewards and a fixed arm set with the best arm always available; real streams satisfy these only approximately, and the price ratio we report on them is a post-hoc explanatory statistic. Its residual, cold LinUCB's regret on arms the warm start does not certify, is not governed by a price, and we have no lower bound showing that it can be large without a large price ratio, so the theorem characterizes the certified saving rather than the multiplier outright. It also counts cold LinUCB's regret arm by arm, so with many arms that share directions, as on AmazonCat, the residual is large and the bound is loose: it explains why coverage cannot produce large gains under valid exploration rather than predicting the size of a gain. The constructions use a fixed radius, unoptimized constants and very large instances. Theorem 2 needs exploration below a valid confidence radius, as in our experiments, holds with high probability for a fixed number of right answers and in expectation only when that number grows like $\log T$, and compares two experts that both see the reward; how an informed expert compares with one that sees only the features is shown only numerically (Appendix F). We have no estimator that predicts the size of the multiplier before the expert is run. The large AmazonCat multipliers depend on a dominant category that the learner can be steered to, and we have not tested other datasets with a skewed label distribution. Only two real datasets are used, multi-expert behaviour is not stress-tested, reward models suited to binary feedback may change the split between coverage and the transfer, and Lemma 1 would need an analogue for Thompson sampling.

References

Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. Improved algorithms for linear stochastic bandits. *Advances in neural information processing systems*, 24, 2011.

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

Alekh Agarwal, Miroslav Dudík, Satyen Kale, John Langford, and Robert Schapire. Contextual bandit learning with predictable rewards. In *Artificial Intelligence and Statistics*, pages 19–26. PMLR, 2012.

Alekh Agarwal, Daniel Hsu, Satyen Kale, John Langford, Lihong Li, and Robert Schapire. Taming the monster: A fast and simple algorithm for contextual bandits. In *International conference on machine learning*, pages 1638–1646. PMLR, 2014.

Alekh Agarwal, Sarah Bird, Markus Cozowicz, Luong Hoang, John Langford, Stephen Lee, Jiaji Li, Dan Melamed, Gal Oshri, Oswaldo Ribas, et al. Making contextual decisions with low technical debt. *arXiv preprint arXiv:1606.03966*, 2016.

Alekh Agarwal, Haipeng Luo, Behnam Neyshabur, and Robert E. Schapire. Corralling a band of bandit algorithms. In *Proceedings of the 30th Conference on Learning Theory (COLT)*, volume 65 of *Proceedings of Machine Learning Research*, pages 1–27. PMLR, 2017. URL https://arxiv.org/abs/1612.06246v3.

Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? investigations with linear models. *arXiv preprint arXiv:2211.15661*, 2022.

Parand A Alamdari, Yanshuai Cao, and Kevin H Wilson. Jump starting bandits with llm-generated prior knowledge. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19821–19833, 2024.

Peter Auer, Nicolo Cesa-Bianchi, Yoav Freund, and Robert E Schapire. The nonstochastic multi-armed bandit problem. *SIAM journal on computing*, 32(1):48–77, 2002.

Hamsa Bastani, Mohsen Bayati, and Khashayar Khosravi. Mostly exploration-free algorithms for contextual bandits. *Management Science*, 67(3):1329–1349, 2021.

Adam Bayley, Xiaodan Zhu, Raquel Aoki, Yanshuai Cao, and Kevin H Wilson. Jump start or false start? a theoretical and empirical evaluation of llm-initialized bandits. *arXiv preprint arXiv:2604.02527*, 2026.

Kush Bhatia, Kunal Dahiya, Himanshu Jain, Purushottam Kar, Anshul Mittal, Yashoteja Prabhu, and Manik Varma. The extreme classification repository: Multi-label datasets and code, 2016.

Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.

Dingyang Chen, Qi Zhang, and Yinglun Zhu. Efficient sequential decision making with large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 9157–9170, 2024.

Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Misha Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. *Advances in neural information processing systems*, 34:15084–15097, 2021.

Wei Chu, Lihong Li, Lev Reyzin, and Robert Schapire. Contextual bandits with linear payoff functions. In *Proceedings of the fourteenth international conference on artificial intelligence and statistics*, pages 208–214. JMLR Workshop and Conference Proceedings, 2011.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, et al. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70):1–53, 2024.

Ashok Cutkosky, Chris Dann, Abhimanyu Das, and Qiuyi Zhang. Leveraging initial hints for free in stochastic linear bandits. In *International Conference on Algorithmic Learning Theory*, pages 282–318. PMLR, 2022.

Varsha Dani, Thomas P Hayes, and Sham M Kakade. Stochastic linear optimization under bandit feedback. In *21st Annual Conference on Learning Theory*, pages 355–366, 2008.

Chris Dann, Claudio Gentile, and Aldo Pacchiano. Data-driven online model selection with regret guarantees. In *International Conference on Artificial Intelligence and Statistics*, pages 1531–1539. PMLR, 2024.

Dylan Foster and Alexander Rakhlin. Beyond ucb: Optimal and efficient contextual bandits with regression oracles. In *International conference on machine learning*, pages 3199–3210. PMLR, 2020.

Dylan Foster, Alekh Agarwal, Miroslav Dudík, Haipeng Luo, and Robert Schapire. Practical contextual bandits with regression oracles. In *International Conference on Machine Learning*, pages 1539–1548. PMLR, 2018.

Dylan J Foster, Claudio Gentile, Mehryar Mohri, and Julian Zimmert. Adapting to misspecification in contextual bandits. *Advances in Neural Information Processing Systems*, 33:11478–11489, 2020.

Dylan J Foster, Sham M Kakade, Jian Qian, and Alexander Rakhlin. The statistical complexity of interactive decision making. *arXiv preprint arXiv:2112.13487*, 2021.

Shivam Garg, Dimitris Tsipras, Percy S Liang, and Gregory Valiant. What can transformers learn in-context? a case study of simple function classes. *Advances in neural information processing systems*, 35:30583–30598, 2022.

Claudio Gentile, Shuai Li, and Giovanni Zappella. Online clustering of bandits. In *International conference on machine learning*, pages 757–765. PMLR, 2014.

Avishek Ghosh, Sayak Ray Chowdhury, and Aditya Gopalan. Misspecified linear bandits. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, 2017.

Todd L Graves and Tze Leung Lai. Asymptotically efficient adaptive choice of control laws incontrolled markov chains. *SIAM journal on control and optimization*, 35(3):715–743, 1997.

Botao Hao, Rahul Jain, Tor Lattimore, Benjamin Van Roy, and Zheng Wen. Leveraging demonstrations to improve online learning: Quality matters. In *International Conference on Machine Learning*, pages 12527–12545. PMLR, 2023.

Chengkai Huang, Junda Wu, Hanqing Yang, Xutong Liu, Lei Shi, Tong Yu, Sungchul Kim, Carlee Joe-Wong, Julian McAuley, Quan Z Sheng, et al. Contextual bandits with llm-derived priors and adaptive calibration.

Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, et al. Qwen2. 5-coder technical report. *arXiv preprint arXiv:2409.12186*, 2024.

Akshay Krishnamurthy, Keegan Harris, Dylan J Foster, Cyril Zhang, and Aleksandrs Slivkins. Can large language models explore in-context? *Advances in Neural Information Processing Systems*, 37:120124–120158, 2024.

John Langford and Tong Zhang. The epoch-greedy algorithm for multi-armed bandits with side information. *Advances in neural information processing systems*, 20, 2007.

Tor Lattimore and Csaba Szepesvari. The end of optimism? an asymptotic analysis of finite-armed linear bandits. In *Artificial Intelligence and Statistics*, pages 728–737. PMLR, 2017.

Tor Lattimore and Csaba Szepesvári. *Bandit algorithms*. Cambridge University Press, 2020.

Tor Lattimore, Csaba Szepesvari, and Gellert Weisz. Learning with good feature representations in bandits and in rl with a generative model. In *International conference on machine learning*, pages 5662–5670. PMLR, 2020.

Alessandro Lazaric. Transfer in reinforcement learning: a framework and a survey. In *Reinforcement Learning: State-of-the-Art*, pages 143–173. Springer, 2012.

Jonathan Lee, Annie Xie, Aldo Pacchiano, Yash Chandak, Chelsea Finn, Ofir Nachum, and Emma Brunskill. Supervised pretraining can learn in-context reinforcement learning. *Advances in Neural Information Processing Systems*, 36:43057–43083, 2023.

Jiwei Li, Will Monroe, Alan Ritter, Dan Jurafsky, Michel Galley, and Jianfeng Gao. Deep reinforcement learning for dialogue generation. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, pages 1192–1202, 2016.

Lihong Li, Wei Chu, John Langford, and Robert E Schapire. A contextual-bandit approach to personalized news article recommendation. In *Proceedings of the 19th international conference on World wide web*, pages 661–670, 2010.

Teodor Vanislavov Marinov and Julian Zimmert. The pareto frontier of model selection for general contextual bandits. *Advances in Neural Information Processing Systems*, 34:17956–17967, 2021.

Giovanni Monea, Antoine Bosselut, Kianté Brantley, and Yoav Artzi. Llms are in-context bandit reinforcement learners. *arXiv preprint arXiv:2410.05362*, 2024.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35: 27730–27744, 2022.

Aldo Pacchiano, My Phan, Yasin Abbasi Yadkori, Anup Rao, Julian Zimmert, Tor Lattimore, and Csaba Szepesvari. Model selection in contextual stochastic bandit problems. *Advances in Neural Information Processing Systems*, 33:10328–10337, 2020.

Maksim Pershin, Ivan Golovanov, Pavel Baltabaev, and Natalia Trankova. Calibration-gated llm pseudo-observations for online contextual bandits. *arXiv preprint arXiv:2604.14961*, 2026.

Alexander Rakhlin and Karthik Sridharan. Online learning with predictable sequences. In *Conference on Learning Theory*, pages 993–1019. PMLR, 2013.

Sharath Chandra Raparthy, Eric Hambro, Robert Kirk, Mikael Henaff, and Roberta Raileanu. Generalization to new sequential decision making tasks with in-context learning. *arXiv preprint arXiv:2312.03801*, 2023.

Machel Reid, Yutaro Yamada, and Shixiang Shane Gu. Can wikipedia help offline reinforcement learning? *arXiv preprint arXiv:2201.12122*, 2022.

Carlos Riquelme, George Tucker, and Jasper Snoek. Deep bayesian bandits showdown: An empirical comparison of bayesian deep networks for thompson sampling. *arXiv preprint arXiv:1802.09127*, 2018.

Mark Rucker, Yinglun Zhu, and Paul Mineiro. Infinite action contextual bandits with reusable data exhaust. In *International Conference on Machine Learning*, pages 29259–29274. PMLR, 2023.

Paat Rusmevichientong and John N Tsitsiklis. Linearly parameterized bandits. *Mathematics of Operations Research*, 35(2):395–411, 2010.

Amr Sharaf and Hal Daumé III. Meta-learning for contextual bandit exploration. *arXiv preprint arXiv:1901.08159*, 2019.

David Simchi-Levi and Yunzong Xu. Bypassing the monster: A faster and simpler optimal algorithm for contextual bandits under realizability. *Mathematics of Operations Research*, 47(3): 1904–1931, 2022.

Marta Soare, Alessandro Lazaric, and Rémi Munos. Best-arm identification in linear bandits. *Advances in neural information processing systems*, 27, 2014.

Yuda Song, Yifei Zhou, Ayush Sekhari, J Andrew Bagnell, Akshay Krishnamurthy, and Wen Sun. Hybrid rl: Using both offline and online data can make rl efficient. *arXiv preprint arXiv:2210.06718*, 2022.

Niranjan Srinivas, Andreas Krause, Sham M Kakade, and Matthias W Seeger. Information-theoretic regret bounds for gaussian process optimization in the bandit setting. *IEEE transactions on information theory*, 58(5):3250–3265, 2012.

Yanchao Sun, Shuang Ma, Ratnesh Madaan, Rogerio Bonatti, Furong Huang, and Ashish Kapoor. Smart: Self-supervised multi-task pretraining with control transformers. *arXiv preprint arXiv:2301.09816*, 2023.

Hampus Gummesson Svensson. *Sequential Decision-Making for Drug Design: Towards Closed-Loop Drug Design*. Chalmers Tekniska Hogskola (Sweden), 2023.

Ambuj Tewari and Susan A Murphy. From ads to interventions: Contextual bandits in mobile health. In *Mobile health: sensors, analytic methods, and applications*, pages 495–517. Springer, 2017.

Michal Valko, Nathaniel Korda, Rémi Munos, Ilias Flaounas, and Nelo Cristianini. Finite-time analysis of kernelised contextual bandits. *arXiv preprint arXiv:1309.6869*, 2013.

Andrew Wagenmaker and Aldo Pacchiano. Leveraging offline data in online reinforcement learning. In *International Conference on Machine Learning*, pages 35300–35338. PMLR, 2023.

Yifan Wu, Roshan Shariff, Tor Lattimore, and Csaba Szepesvári. Conservative bandits. In *International conference on machine learning*, pages 1254–1262. PMLR, 2016.

Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-pack: Packed resources for general chinese embeddings. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, pages 641–649, 2024.

Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit bayesian inference. *arXiv preprint arXiv:2111.02080*, 2021.

Yasin A Yadkori, Ilja Kuzborskij, András György, and Csaba Szepesvari. To believe or not to believe your llm: Iterative prompting for estimating epistemic uncertainty. *Advances in Neural Information Processing Systems*, 37:58077–58117, 2024.

Chicheng Zhang, Alekh Agarwal, Hal Daumé III, John Langford, and Sahand N Negahban. Warm-starting contextual bandits: Robustly combining supervised and bandit feedback. *arXiv preprint arXiv:1901.00301*, 2019.

Mengxiao Zhang, Yuheng Zhang, Haipeng Luo, and Paul Mineiro. Efficient contextual bandits with uninformed feedback graphs. *arXiv preprint arXiv:2402.08127*, 2024.

Qinqing Zheng, Amy Zhang, and Aditya Grover. Online decision transformer. In *international conference on machine learning*, pages 27042–27059. PMLR, 2022.

Yinglun Zhu and Paul Mineiro. Contextual bandits with smooth regret: Efficient learning in continuous action spaces. In *International Conference on Machine Learning*, pages 27574–27590. PMLR, 2022.

Yinglun Zhu and Robert Nowak. Pareto optimal model selection in linear bandits. In *International Conference on Artificial Intelligence and Statistics*, pages 6793–6813. PMLR, 2022.

Yinglun Zhu, Julian Katz-Samuels, and Robert Nowak. Near instance optimal model selection for pure exploration linear bandits. In *International Conference on Artificial Intelligence and Statistics*, pages 6735–6769. PMLR, 2022.

Appendix A. Related Work

This appendix expands on Section 2. We cover (i) linear contextual bandits and confidence geometry, (ii) model selection and expert aggregation, (iii) warm starts and prior information, (iv) the geometry of information gain in linear bandits, and (v) large language models for sequential decision making.

Sequential decision making with contextual information has been studied extensively in the contextual bandit literature, where the learner must balance exploration and exploitation under partial feedback (Langford and Zhang, 2007; Agarwal et al., 2014; Foster et al., 2021; Lattimore and Szepesvári, 2020). In the linear setting, algorithms such as LinUCB and OFUL achieve regret bounds that depend explicitly on the geometry of the accumulated design matrix through self-normalized confidence ellipsoids (Chu et al., 2011; Abbasi-Yadkori et al., 2011). Subsequent work has developed statistically efficient contextual bandit algorithms under general function approximation assumptions (Agarwal et al., 2012; Foster et al., 2018; Foster and Rakhlin, 2020; Simchi-Levi

and Xu, 2022; Zhu and Mineiro, 2022; Rucker et al., 2023; Zhang et al., 2024). A central insight of this line of work is that the conditioning of the covariance matrix determines how quickly uncertainty shrinks and therefore how rapidly regret decreases. Our work builds on this perspective by showing that actions selected by a frozen predictor can reshape this geometry early in learning and accelerate subsequent adaptation.

Model selection and expert aggregation in bandits. A closely related direction studies how to combine multiple candidate policies or base algorithms online. Classical expert-learning methods provide regret guarantees relative to the best fixed expert in hindsight (Auer et al., 2002). In the bandit setting, CORRAL (Agarwal et al., 2017) uses Online Mirror Descent with a log-barrier regularizer to achieve regret competitive with the best base algorithm; subsequent refinements tighten the rates and extend the framework to contextual and stochastic settings (Pacchiano et al., 2020; Marinov and Zimmert, 2021; Zhu and Nowak, 2022; Zhu et al., 2022; Dann et al., 2024). These approaches are inherently *randomized* and target robustness: the master performs nearly as well as the best base algorithm regardless of which one that is. Our objective is different—we want a *deterministic* rule that exploits the complementarity between a good early frozen policy and a strong asymptotic adaptive learner to *strictly outperform* both. Correspondingly, our analysis does not bound a relative regret against the best base, but instead tracks how early actions selected by one policy reshape the design matrix and accelerate the statistical efficiency of another.

Warm starts and prior information in contextual bandits. Several works have investigated how prior information can be incorporated into contextual bandit algorithms through initialization, offline datasets, or pretrained predictors. Such approaches can substantially improve short-horizon performance by reducing uncertainty before online interaction begins (Bastani et al., 2021; Zhang et al., 2019). Related directions in transfer learning and structured bandits further show that leveraging information across tasks or user populations can accelerate adaptation and reduce exploration requirements (Lazaric, 2012; Gentile et al., 2014). Meta-learning approaches similarly learn priors that enable faster exploration in new environments (Sharaf and Daumé III, 2019). Empirical work on neural-linear bandit architectures also demonstrates that representation pretraining can significantly improve early decision quality (Riquelme et al., 2018). Our analysis formalizes these intuitions in a linear bandit framework: informative early actions contract the confidence ellipsoid on specific competitors (Lemmas 3 and 4), and the resulting saving is a difference of width-limited regrets rather than an improvement of the determinant-ratio rate, which Proposition 5 shows cannot by itself exceed a constant factor.

Geometry of information gain in linear bandits. A central theme in the analysis of linear bandit algorithms is that learning efficiency is governed by the geometry of the accumulated design matrix. Early work on stochastic linear optimization under bandit feedback showed that regret depends explicitly on the growth of the covariance matrix associated with the chosen actions and the resulting confidence ellipsoid (Dani et al., 2008). Subsequent analyses of linearly parameterized bandits further clarified how the eigenvalue structure of this matrix determines parameter identifiability and sample complexity (Rusmevichientong and Tsitsiklis, 2010). Later work established sharp self-normalized confidence bounds and finite-time regret guarantees expressed directly in terms of the design matrix and its inverse (Abbasi-Yadkori et al., 2011), while related results connected spectral growth of the covariance matrix to optimal arm identification in linear settings (Soare et al., 2014). From an information-theoretic perspective, regret can also be related to determinant growth and

effective dimension of the covariance operator (Srinivas et al., 2012; Valko et al., 2013). The price (2) is the single-direction shadow of the allocation programs behind asymptotic instance-dependent lower bounds, where regret is minimized subject to coverage constraints on every direction (Graves and Lai, 1997; Lattimore and Szepesvari, 2017); our use of it is non-asymptotic and comparative: two policies on the same instance pay different prices for the same coverage. Prior side information has also been analyzed as hints or predictable sequences in online learning (Rakhlin and Sridharan, 2013; Cutkosky et al., 2022). The comparison with Cutkosky et al. (2022) is the closest: there the learner receives a hint about $\theta^\star$ and the regret improves as a function of the hint's accuracy, with the standard rate recovered when the hint is poor. Our warm start is not a hint about $\theta^\star$ but data in the design matrix, and its value is not a function of accuracy: two experts of similar accuracy can leave design matrices of very different value (Section 8.2), and an expert of zero accuracy can leave one of enormous value (Proposition 2). What governs the value is the price ratio of Theorem 1. The warm-start analyses of Bastani et al. (2021); Zhang et al. (2019) bound the regret of a learner started from logged data through the quality of that data; they are offline in our sense (row 1 of Table 2), and our contribution relative to them is the online accounting, in which the acquisition of the data is charged and can exceed its value. Our work extends this geometric viewpoint by showing that a fixed policy can shape the spectrum of the design matrix during early rounds of interaction, thereby improving conditioning of the confidence ellipsoid and accelerating the learning rate of subsequent adaptive exploration.

Large language models for sequential decision making. Recent work has explored the use of large language models (LLMs) as policies in contextual bandit environments, demonstrating that frozen pretrained models can achieve strong short-horizon performance even without online updates (Chen et al., 2024). More broadly, LLMs exhibit strong in-context learning capabilities (Brown et al., 2020; Xie et al., 2021; Garg et al., 2022; Akyürek et al., 2022), and instruction tuning further improves their effectiveness in downstream decision tasks (Ouyang et al., 2022; Chung et al., 2024). Several recent approaches interpret sequential decision making itself as a sequence-modeling problem and adapt transformer architectures to interactive settings (Chen et al., 2021; Zheng et al., 2022; Reid et al., 2022; Sun et al., 2023; Raparthy et al., 2023; Lee et al., 2023). Closest to our work is the approach of Chen et al. (2024), who combine an LLM policy with an adaptive contextual bandit learner using an online model-selection master and empirically document the reward-multiplication phenomenon on large-scale recommendation datasets. Our contribution relative to that line is theoretical: we give a regret-based explanation for why such combinations can beat both components, and replace the randomized CORRAL-style master with a deterministic confidence-gated rule whose routing decision directly exposes the geometric warm-start mechanism. A concurrent line of work (Huang et al.) also uses LLM uncertainty to gate exploration, but does so by calibrating an embedding-based uncertainty estimate to adjust UCB-style exploration bonuses, rather than by comparing self-normalized confidence radii of a frozen predictor and an online learner; their analysis bounds the regret of a single calibrated agent through the accuracy of the LLM's reward predictions, whereas we give an exact decomposition of the hybrid's advantage, bound the multiplier over the cold learner on any finite-arm instance by a ratio of coverage prices, and show the bound is attained; in their setting the LLM's value enters through prediction error, in ours through the design matrix, and the two can disagree by a factor of the horizon (Proposition 2). Two concurrent works are closer still. Bayley et al. (2026) warm-start a linear bandit from LLM-generated preferences and bound the warm-started estimator's error through a single prior-error term measured in the geome-

try of the pretraining design matrix, which treats the design matrix and the response vector as one object; our analysis separates the two, shows that under linear rewards the second is determined by the first, and identifies the expert-dependent remainder in closed form, which is the part that differs across equally accurate experts on our data. Pershin et al. (2026) inject LLM pseudo-observations into LinUCB with a calibration-dependent weight on both the design matrix and the response vector, a routing rule close in spirit to ours, and analyze it through a bias–variance argument without a regret bound; the decomposition of Proposition 1 and the price bound of Theorem 1 apply to any such rule, since both hold for an arbitrary routed set. Additional evidence that LLMs can exhibit in-context reinforcement learning behavior (Monea et al., 2024; Krishnamurthy et al., 2024) and that LLM-derived priors can jump-start exploration (Alamdari et al., 2024; Yadkori et al., 2024) further motivates the study of principled mechanisms for combining frozen predictors with online learners.

Appendix B. Further Theory

This appendix collects results used in the main text but stated there only in passing: the two width identities, the regret of the handoff hybrid, the width-limited bound with the realized estimation error, what the transfer can buy, two relaxations of Theorem 1, and a discussion of when cheap informative actions exist.

B.1. How Widths Shrink

Lemma 1 is the familiar fact that an optimistic algorithm can only play an arm whose width exceeds half its gap, restated so that the gap and the width sit in one expression. It makes three facts visible at once. A competitor with a large gap but an even larger width is exactly as dangerous as its gap (Proposition 2: $\Delta = 1$, width $\beta L/\sqrt{\lambda} > 1/2$). A competitor with a tiny gap is cheap no matter how wide it is, so a heavy mass of near-ties does not by itself mean large regret. And the only way to reduce the regret LinUCB can incur on a given round is to reduce widths *on competitors whose widths exceed their gaps*; reducing the width of g in Proposition 2 would have achieved nothing.

There are exactly two ways widths get reduced: through data that arrives as a warm start, and through LinUCB's own observations. Both are exact linear algebra.

Lemma 3 (A warm start shrinks widths in the directions it covers) *For any $M \succeq 0$ and any x, $\|x\|_{(\lambda I+M)^{-1}} \le \|x\|_{(\lambda I)^{-1}} = \|x\|_2/\sqrt{\lambda}$. If P is an orthogonal projector, $M \succeq mP$, and $x \in \mathrm{Im}(P)$, then $\|x\|^2_{(\lambda I+M)^{-1}} \le \|x\|_2^2/(\lambda+m)$. If instead $x \perp \mathrm{Im}(M)$, then $\|x\|_{(\lambda I+M)^{-1}} = \|x\|_2/\sqrt{\lambda}$: a warm start does nothing for directions it does not cover.*

Lemma 4 (One observation shrinks the width of another direction by an exact factor) *For $V \succ 0$ and vectors x, z with $z \neq 0$,*

$$\|z\|^2_{(V+xx^\top)^{-1}} = \big(1-\rho_V(x,z)\big)\|z\|^2_{V^{-1}}, \qquad \rho_V(x,z) := \frac{(z^\top V^{-1}x)^2}{(1+\|x\|^2_{V^{-1}})\,\|z\|^2_{V^{-1}}} \in [0,1).$$

At a cold start, $\rho_{\lambda I}(x,z) = \cos^2(x,z)\cdot \|x\|_2^2/(\lambda+\|x\|_2^2)$.

Lemma 4 is where "feature coherence" enters, and it shows precisely how much of that notion survives. Observing x reduces the width of z by the squared cosine *in the V^{-1} geometry*, scaled

by how novel x itself is. The Euclidean cosine is the right quantity only at $V = \lambda I$; as V becomes anisotropic the relevant angle is measured in V^{-1}, and two vectors with positive Euclidean inner product can have ρ_V near zero (or vice versa). In Proposition 2 the Euclidean cosine between any two bad arms is exactly zero, so $\rho = 0$ and no observation ever helps a future round; that is what makes the instance hard, not the ambient dimension as such. A large ambient dimension is necessary for many mutually V^{-1}-orthogonal directions to exist, but it is the lack of carry-over between directions, not the dimension, that cold LinUCB pays for.

Dimension, coherence and gaps. Effective dimension, feature coherence and gap structure are all readings of Lemma 1 together with Lemmas 3 and 4: a uniform gap $\Delta_{\min}$ caps Φ_t at $\Delta_{\max}$ but saves nothing while widths exceed $\Delta_{\min}/2$; a small rank r of the span of played directions caps the elliptical potential $\sum_t \|x_{t,A_t}\|^2_{V_t^{-1}}$ at $2r\log(1 + L^2T/(r\lambda))$ and therefore the number of rounds at which any width can exceed a fixed level; and positive coherence lower-bounds $\rho_{\lambda I}$ at the start of learning only. None of the three is a condition under which a warm start cannot help by more than a constant factor, and we do not claim one here; what they control is how many rounds of cold exploration are needed before Φ_t falls below the level the warm start could have delivered immediately.

B.2. Regret of the Handoff Hybrid

Proposition 5 (Handoff regret in exact form) *Fix τ and an expert E. On the event $\mathcal{E}_\delta$ (probability at least $1-\delta$), the handoff hybrid satisfies*

$$R_T(\mathsf{H}_\tau) \;\le\; \underbrace{\sum_{t\le\tau} \Delta_{t,a_t^{\mathsf{E}}}}_{\textit{acquisition regret}} \;+\; \min\left\{ \sum_{t=\tau+1}^{T} \Phi_t(V_t),\;\; 2\beta_T \max\left\{1, \tfrac{L}{\sqrt{\lambda}}\right\} \sqrt{2(T-\tau)\,\log \frac{\det V_{T+1}}{\det V_{\tau+1}}} \right\}, \tag{4}$$

where $V_{\tau+1} = \lambda I + M_\tau$ and the widths inside Φ_t are those of the hybrid's own trajectory. The determinant ratio satisfies $\log\det V_{T+1}/\det V_{\tau+1} \le \sum_{j=1}^{d} \log\big(1 + L^2(T-\tau)/\sigma_j(V_{\tau+1})\big)$ with σ_j the eigenvalues of $V_{\tau+1}$, and cold LinUCB satisfies the same bound with $\tau = 0$ and $V_1 = \lambda I$.

We state the bound with the exact determinant ratio and make no asymptotic simplification of it. Two remarks explain why.

The determinant term cannot certify a multiplier. Suppose $M_\tau \succeq \nu\tau P$ on an r-dimensional subspace containing all actions, the situation analyzed in earlier work. Then the second term in (4) is at most $2\beta_T\sqrt{2r(T-\tau)\log(1 + L^2(T-\tau)/(\lambda+\nu\tau))}$, and its ratio to the cold bound $2\beta_T\sqrt{2rT\log(1+L^2T/\lambda)}$ is

$$\Psi(\tau,\nu,T) = \sqrt{\frac{T-\tau}{T}} \cdot \sqrt{\frac{\log\big(1 + L^2(T-\tau)/(\lambda+\nu\tau)\big)}{\log\big(1+L^2T/\lambda\big)}}.$$

For $\tau = o(T)$ both logarithms are $\Theta(\log T)$ and $\Psi \to 1$; for $\tau = T^\alpha$ with $\nu\tau \gg \lambda$ it tends to $\sqrt{1-\alpha}$, bounded away from zero. It is tempting to read Ψ as $\sqrt{\lambda/(\lambda+\nu\tau)}$ by comparing the arguments of the two logarithms; that reading is wrong. The correct conclusion is the opposite one: a warm start that improves only the $\sqrt{T}$-rate term buys at most a constant factor, and any multiplier this bound can certify must come from the first term in the minimum, that is, from competitors

whose widths the warm start pushes below their gaps so that Φ_t drops from the order of the gap to the order of the (now small) width or to zero. That is what happens in Proposition 2, where $\sum_t \Phi_t$ falls from T to $O(1)$ while the determinant bound barely moves.

What the bound says about multiplication. Combining (4) with Proposition 1, the hybrid beats cold LinUCB by more than a constant factor in this bound only if

$$\sum_{t>\tau}\big(\Delta_{t,A_t^{\mathsf{C}}} - \Phi_t(V_t^{\mathsf{H}})\big) \;\gg\; \sum_{t\le\tau}\Delta_{t,a_t^{\mathsf{E}}} - R_{[1,\tau]}(\mathsf{C}),$$

i.e., the width reduction that the routed data delivers on the competitors cold LinUCB would have played must exceed the net regret paid to acquire it; and it beats the expert only if the expert's tail regret exceeds the hybrid's. Since Φ_t only upper-bounds the hybrid's regret, these are conditions for the bound to certify a multiplier, not for a multiplier to occur. Neither condition is implied by low expert accuracy being "near-optimal early" or by an abstract coverage parameter. Both are properties of *which* actions the expert proposes, relative to the directions cold exploration leaves unresolved, and of what those actions cost.

Adaptive τ and interleaving. The router of Appendix J decides at each round whether to follow the expert or LinUCB by comparing confidence widths, so the expert rounds need not be contiguous. Proposition 5 applies verbatim with $\sum_{t\le\tau}$ replaced by the sum over expert-routed rounds and with the LinUCB-phase terms summed over LinUCB-routed rounds, since Lemma 1 is a round-level statement and the elliptical potential argument only uses that V_t is non-decreasing.

B.3. Width-Limited Regret with the Realized Estimation Error

Lemma 1 uses only the radius. With the realized estimation error in its place it reads as follows, which shows how the two objects of a warm start act on it.

Proposition 6 (Width-limited regret with estimation error) *At a round t at which LinUCB acts, let $\rho_t := \|\widehat{\theta}_t - \theta^\star\|_{V_t}$ be the realized estimation error, so $\rho_t \le \beta_t$ on $\mathcal{E}_\delta$. Then on $\mathcal{E}_\delta$*

$$\Delta_{t,A_t} \;\le\; \Phi_t(V_t,\rho_t) \;:=\; \max_{a\in\mathcal{A}_t}\min\big\{\Delta_{t,a},\; (\beta_t+\rho_t)\|x_{t,a}\|_{V_t^{-1}}\big\}, \tag{5}$$

and $\Phi_t(V,\rho)$ is non-increasing in V and non-decreasing in ρ. A warm start acts on the first argument through M and on the second through c. Since $0 \le \rho_t \le \beta_t$ we have $\Phi_t(V_t,\rho_t) \ge \frac{1}{2}\Phi_t(V_t)$: the response sum can tighten the width-limited bound by at most a factor of two, whereas the design matrix can reduce it by the price ratio, which is unbounded.

B.4. What the Transfer Can Buy

Under the model of Section 7, the transfer's effect on a single round is bounded as follows.

Proposition 7 (What the transfer can buy) *Under the misspecified model, at a round t at which LinUCB acts and for any reference $\widetilde{\theta} \in \mathbb{R}^d$, write $\widetilde{\rho}_t := \|\widehat{\theta}_t - \widetilde{\theta}\|_{V_t}$ and $\widetilde{\varepsilon}(x) := \bar{r}(x) - \langle\widetilde{\theta}, x\rangle$. Then*

$$\Delta_{t,A_t} \;\le\; (\beta_t+\widetilde{\rho}_t)\|x_{t,A_t}\|_{V_t^{-1}} \;+\; (\widetilde{\rho}_t-\beta_t)_+\|x_{t,a_t^\star}\|_{V_t^{-1}} \;+\; \big[\widetilde{\varepsilon}(x_{t,a_t^\star}) - \widetilde{\varepsilon}(x_{t,A_t})\big],$$

simultaneously for every reference. The design matrix fixes the widths; the response sum fixes $\widetilde{\rho}_t$ for every reference at once. The bound improves when the transfer moves $\widehat{\theta}$ toward a reference whose

residual is smaller on the optimal arm than on the arms LinUCB would otherwise play, and the bound is width-limited whenever some reference has $\widetilde{\rho}_t \le \beta_t$ and a nonpositive residual difference on the played arm.

Proposition 7 is a statement about what the transfer *can* deliver, not a prediction of what a given expert will deliver: m depends on the residuals at the actions the expert chooses, which are not known before it is run. Theorem 2 shows how large the difference between two experts can be on a constructed instance, and Section 8.2 measures m on real data, finds that it accounts for essentially the whole estimate at handoff, and shows that removing it collapses the differences between experts of similar accuracy.

B.5. Two Relaxations of the Price Bound

Remark 4 (Two relaxations) (i) The factor 2 in the definition of U_{cert} is the crude one. On the confidence event with the true self-normalized radius $\beta_t^{\mathrm{true}} \le \beta$, the proof of Lemma 1 gives $\Delta_{t,A_t} \le (\beta_t + \beta_t^{\mathrm{true}})\|x_{t,A_t}\|_{V_t^{-1}}$ for the played arm, so the theorem holds with $C_x^\star$ replaced by $(\beta + \beta^{\mathrm{true}})^2\|x\|_2^2/\Delta(x)^2$ throughout and U_{cert} enlarged to the arms with $(\beta + \beta^{\mathrm{true}})\|x\| < \Delta(x)$; P_U is unchanged. (ii) If the optimal arm is not offered at every round, so that the gap of x varies with t, write $\Delta^-(x)$ and $\Delta^+(x)$ for its smallest and largest values over rounds at which x is offered. Lemma 6 holds with $C_x^\star$ computed from $\Delta^-(x)$ and each pull charged $\Delta^+(x)$, and Lemma 7 with $\Delta^-(x)$; the price ratio then carries a factor $\Delta^+(x)/\Delta^-(x)$, which is 1 under the standing assumption.

B.6. When Do Cheap Probes Exist?

The positive rows need an action of small gap whose feature vector carries a non-negligible component along a direction that cold LinUCB can otherwise cover only by pulling an arm of large gap. Two observations make this less special than it looks. First, the probe need not be a single designated arm: any near-optimal action whose embedding overlaps the bad direction serves, and in the embedding spaces used in practice (Section 8) near-optimal actions overlap many others. Second, the expert does not need to know that the probe is informative; it needs only to prefer it, which any policy with prior knowledge of the reward does when the probe is nearly as good as the best arm. What the expert supplies, and cold optimism cannot, is the *order* in which coverage is bought: optimism ranks a fresh bad arm above a cheap probe because the bad arm's width is larger, and pays the bad arm's price first (Lemma 12 in the appendix); a policy that already believes the bad arm is bad takes the probe at once. The price ratio $\gamma^2/(\varepsilon L^2)$ of rows 5 and 5′ is the exchange rate between those two orders, and Theorem 1 says it is the only exchange rate available: an expert whose actions are not cheaper per unit of certified coverage than cold exploration's cannot multiply, however accurate it is.

Appendix C. The Price Ratio Attained, and the Catalog

This appendix states the rows of Table 2 that the main text only summarizes. The canonical instance (rows 1–2) is defined in Section 6 and the template in Appendix C.2 below; proofs are in Appendix E.

The catalog has eleven rows rather than the $2^3 = 8$ of a binary table because several terms admit qualitatively distinct failure modes—for instance, I_τ can vanish because the expert buys the same directions cold exploration buys at the same price (row 4), because it buys the wrong directions (row 7), or because it buys the right directions too dearly (row 8)—because the offline/online distinction (rows 1–2) and the two ways of killing the tail term (rows 3 and 6) each warrant separate entries, and because the positive case appears twice, with the saving collected in the routed window (row 5) and on the tail (row $5'$).

C.1. Standing Conventions

Noise is i.i.d. σ-sub-Gaussian and independent of the actions; we use the reward-table model in which the m-th pull of a given action vector receives the m-th entry of an i.i.d. sequence attached to that vector (Lattimore and Szepesvári, 2020, §4.6), which has the same law as the sequential model and lets us compare two policies on the same noise. We fix

$$L \geq 8, \quad \gamma \in (0,1], \quad 0 < \varepsilon < \gamma/L, \quad \lambda \geq \max\{\sigma^2 d\,\ell_0,\ 8\sigma^2\ell_1,\ 60\}, \quad \beta \equiv 2\sqrt{\lambda}, \tag{6}$$

with $\ell_0 := \log(1 + TL^2) + 2\log(1/\delta)$ and $\ell_1 := \log(2dT/\delta)$. The confidence radius is constant. The lower bound on λ makes $2\sqrt{\lambda}$ at least the radius of Abbasi-Yadkori et al. (2011) at every round, so the confidence event of Section 4 holds for each policy with probability $\geq 1 - \delta$, and it makes the noise on any $\lambda/30$ pulls of one arm smaller than a quarter of that arm's confidence bonus. The constant $m_0 := \lfloor \lambda/30 \rfloor \geq 2$ is the number of pulls of a fresh bad arm that optimism forces before the good arm or the probe can be preferred to it. Conditions on T appear in each statement; they only say that the horizon is long enough for the claimed rounds to take place.

C.2. The Template, and the Price Ratio Attained

All instances below are built from the canonical one by three modifications: directions may recur, a cheap informative action may be available, and that action may disappear. Fix $K \geq 1$ and set $d = K + 1$ (or $d = 2K + 1$ when nuisance directions are present). The parameter is $\theta^\star = e_1$. Directions cycle: $j(t) := 2 + ((t-1) \bmod K)$. Three actions may be offered at round t:

$$g := e_1 \ (\mu = 1), \qquad b_t := Le_{j(t)} \ (\mu = 0), \qquad p_t := (1-\varepsilon)e_1 + \gamma\, e_{j(t)} \ (\mu = 1 - \varepsilon),$$

the *good* arm, the *bad* arm, and the *probe*. The instance $\mathcal{I}(K, L, \gamma, \varepsilon, \tau_0)$ has $\mathcal{A}_t = \{g, b_t, p_t\}$ for $t \leq \tau_0$ and $\mathcal{A}_t = \{g, b_t\}$ for $t > \tau_0$. The price of e_j, $j \geq 2$, is $\min\{1/L^2, \varepsilon/\gamma^2\}$ while probes are offered and $1/L^2$ afterwards. The expert is the frozen linear policy E_w that plays $\arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_{\mathsf{E}}, x_{t,a} \rangle$ with $\widehat{\theta}_{\mathsf{E}} = e_1 + w \sum_{j \geq 2} e_j$; ties do not occur for the values of w we use. Throughout, $\tau_0 = Kn$ for an integer n, so each direction is probed exactly n times by a policy that plays every probe.

We work under the standing conventions (6) of Appendix C.1: $L \geq 8$, $\gamma \in (0,1]$, $0 < \varepsilon < \gamma/L$, $\lambda \geq \max\{\sigma^2 d\,\ell_0, 8\sigma^2\ell_1, 60\}$ with ℓ_0, ℓ_1 logarithmic in T/δ, and the constant radius $\beta = 2\sqrt{\lambda}$, in the reward-table noise model of Remark 1; we write $m_0 := \lfloor \lambda/30 \rfloor$ for the number of pulls of a fresh bad arm that optimism forces.

On $\mathcal{I}(K, L, \gamma, \varepsilon, \tau_0)$ the expert E_w with $w \in (1/L, (1-\varepsilon)/(L-\gamma))$ plays every probe while probes are offered and every bad arm afterwards (Lemma 14). Its routed actions cost ε each, cover $e_{j(t)}$ at rate γ^2 per round, and recur; after τ_0 it is the worst possible policy. The hybrid that hands off exactly when the probes disappear therefore isolates the price ratio.

Theorem 3 (Online multiplication, and the price ratio as the multiplier) *Let* $\mathcal{I} = \mathcal{I}(K, L, \gamma, \varepsilon, Kn)$ *under* (6)*, with* $\mathsf{E} = \mathsf{E}_w$ *for some* $w \in (1/L, (1-\varepsilon)/(L-\gamma))$*, and* $\tau = \tau_0 = Kn$*. With probability at least* $1 - 3\delta$ *the following hold simultaneously.*

(i) (Expert.) $R_T(\mathsf{E}) = \varepsilon Kn + (T - Kn)$.

(ii) (Cold, lower bound.) *If* $n \geq m_0$ *then* $R_{[1,Kn]}(\mathsf{C}) \geq Km_0$*; in particular* $A_{\tau_0} \geq Km_0 - \varepsilon Kn$.

(iii) (Cold, upper bound.) *For every* $n \geq 1$, $R_T(\mathsf{C}) \leq K(8\beta^2 + 1) + \varepsilon Kn$.

(iv) (Hybrid.) *If* $n \geq 36\lambda L^2/\gamma^2$ *and* $K \geq 36L^2$*, then* H_{τ_0} *plays* g *at every round* $t > \tau_0$*, so* $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$*,* $I_{\tau_0} = R_{[Kn+1,T]}(\mathsf{C}) \geq 0$*, and* $B_{\tau_0} = T - Kn$.

Consequently, for $n \geq 36\lambda L^2/\gamma^2$ *and* $K \geq 36L^2$*,*

$$\frac{m_0}{\varepsilon n} \leq \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \leq 1 + \frac{8\beta^2 + 1}{\varepsilon n}, \qquad \frac{R_T(\mathsf{E})}{R_T(\mathsf{H}_{\tau_0})} = 1 + \frac{T - Kn}{\varepsilon Kn}, \tag{7}$$

and at the smallest admissible $n = \lceil 36\lambda L^2/\gamma^2 \rceil$ *the first ratio is* $\Theta(\gamma^2/(\varepsilon L^2))$*, with explicit constants* $\gamma^2/(2220\,\varepsilon L^2) \leq R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \leq 1 + \gamma^2/(\varepsilon L^2)$ *(the constants of this section are not optimized; see Appendix E, "Remarks on the constants").*

The bracket (7) is the price ratio: cold exploration buys coverage L^2 of a bad direction per unit of regret (price $1/L^2$), the probe buys γ^2 per ε (price ε/γ^2), and the multiplier over cold LinUCB is their ratio $\gamma^2/(\varepsilon L^2)$ up to constants. The conditions on n and K in (iv) are not about prices alone. Enough coverage of e_j is necessary for the handoff matrix to certify b_j but not sufficient: every probe is orthogonal to $v = \gamma\, e_1 - (1-\varepsilon)\sum_{j\geq 2} e_j$, so the width of e_j stays at least $(1-\varepsilon)/\sqrt{\lambda(\gamma^2 + K(1-\varepsilon)^2)}$ however many probes are played, and certification needs K of order L^2; the price ratio does not see this geometric condition. The instances are large by design: orthogonal bad directions stand in for an action space too large for cold exploration to exhaust (AmazonCat has 13,330 categories), and at the smallest admissible sizes the window alone exceeds 3×10^8 rounds. The multiplier over the expert, $(T - \tau_0)/(\varepsilon\tau_0)$, diverges as the tail lengthens. Here the saving is collected during the routed rounds, as A_{τ_0}; keeping cold LinUCB busy elsewhere during the window moves it after them, into I_{τ_0}, with the same price ratio (Proposition 9, row $5'$, below).

The catalog. Theorem 3 is one row of a catalog of eleven instances, each obtained from the canonical one by changing a single ingredient (Table 2). Exactly two online rows multiply over both base policies, rows 5 and $5'$, and they realize the same price ratio, one during the routed rounds and one after them.

Across these linear instances, apart from the direct advantage of an expert that is simply better than cold LinUCB while it plays (row 3), a gain over cold LinUCB appears only when the expert buys the directions cold LinUCB would otherwise pay for, more cheaply and early enough for them to recur, and a gain over the expert only when, in addition, the expert is worse than the warm-started learner on the tail.

C.3. The Other Rows

Row 3: an oracle expert on the canonical instance.

Corollary 3 (Oracle expert on the canonical instance) *On the instance of Proposition 3, under its hypothesis, let* E *play* g *at every round. With probability at least* $1 - \delta$*, for every* τ*:* $R_T(\mathsf{E}) = 0$*,*

#	Instance	A_τ	I_τ	B_τ	Ratio to C	Ingredient isolated
1	canonical, offline (Prop. 2)	not charged	$I^{\mathrm{off}} \asymp T$	$B^{\mathrm{off}} \asymp T$	$\asymp T$	data is free
2	canonical, online (Prop. 3)	$\le \tau$	0	≤ 0	$\le T/(T-\tau)$	directions never recur
3	canonical, oracle (Cor. 3)	τ	0	$-(T-\tau)$	additive	expert too good
4	recurring, bad-arm expert (Prop. 8)	0	0	≥ 0	≤ 961	same price as cold
5	recurring, cheap probes (Thm. 3)	$\ge Km_0 - \varepsilon\tau_0$	≥ 0	$T-\tau_0$	$\asymp \gamma^2/(\varepsilon L^2)$	price ratio, in the window
5′	as 5, cold kept busy in the window (Prop. 9)	$-\varepsilon\tau_0$	$\ge Km_0$	$T-\tau_0$	$\asymp \gamma^2/(\varepsilon L^2)$	price ratio, on the tail
6	as 5, expert optimal on tail (Prop. 10)	as 5	as 5	0	as 5	expert optimal late
7	nuisance probes (Prop. 11)		$< R_{[\tau_0+1,T]}(\mathsf{C})$		≤ 65	wrong directions
8	as 5 with $\varepsilon \ge \gamma^2/L^2$ (Cor. 4)				≤ 2	probe too dear
9	cold already cheap (Prop. 12)				$o(1)$ avg. reward	nothing to buy
10	bad arms with gap η (Prop. 13)				$\le 1 + \gamma^2/(\varepsilon L^2)$	gap rescales prices

Table 2: The catalog: what the three terms of Proposition 1 can be (statements in Appendix C, proofs in Appendix E). Row 1 is the offline identity, in which A is not charged; rows 2–10 are online handoffs. Rows 3 and 6 kill B_τ from the two sides. Rows 4, 7 and 8 kill the multiplier over cold LinUCB: the expert buys what cold buys at the same price, buys directions nobody needs, or buys the right directions too dearly. Row 9 is the case in which cold LinUCB's bill is already negligible, and row 10 shows that the gap rescales both prices. Rows 5 and 5′ realize the same price ratio, collected as A_τ and as I_τ. Blank cells are not claimed; constants are not optimized.

$R_T(\mathsf{H}_\tau) = T - \tau$, $A_\tau = \tau$, $I_\tau = 0$, $B_\tau = -(T-\tau)$. *The hybrid improves on cold LinUCB by exactly the rounds it routes and on the expert by nothing.*

Row 4: an expert that mimics cold exploration.

Proposition 8 (Expert mimics cold exploration: same price, no gain) *Let $\mathcal{I}(K, L, \cdot, \cdot, 0)$ be the instance with no probes ($\mathcal{A}_t = \{g, b_t\}$ for all t, $d = K+1$), and let E be any policy that plays b_t at every round (for instance E_w with $w > 1/L$). With probability at least $1 - 2\delta$: (i) for every $\tau \le Km_0$, H_τ and C play the same action at every round, so $A_\tau = I_\tau = 0$ and $R_T(\mathsf{H}_\tau) = R_T(\mathsf{C})$; (ii) for every $\tau > Km_0$, $R_T(\mathsf{H}_\tau) \ge \tau > Km_0$ while $R_T(\mathsf{C}) \le K(4\beta^2+1)$, so $R_T(\mathsf{C})/R_T(\mathsf{H}_\tau) \le (16\lambda+1)/m_0 \le 961$.*

Row 5: discussion of Theorem 3. The bracket (7) is the price ratio. Cold exploration buys coverage L^2 of a bad direction per unit of regret, so its price is $1/L^2$; the probe buys γ^2 per ε, price ε/γ^2; the ratio of the two prices is $\gamma^2/(\varepsilon L^2)$, and the multiplier over cold LinUCB is that ratio up to constants, and is $O(1)$ when the ratio is. The confidence radius does not appear: both policies must push the same widths below the same level, and β scales the required coverage identically for both. The multiplier over the expert is $(T-\tau_0)/(\varepsilon\tau_0)$ and diverges as the tail lengthens, because the expert keeps paying for bad arms that the warm-started learner has learned to avoid. The lower bound in (ii) is attained inside the routed window, so on this instance $A_{\tau_0} > 0$ whenever $\varepsilon n < m_0$: the expert's probes are cheaper than the exploration cold LinUCB is forced into at the same rounds, and the data they leave is what prevents that exploration from resuming after τ_0. On this instance the saving is carried by A_{τ_0}, not by I_{τ_0}, and for a structural reason: the certificate in (iv) needs coverage of order $\beta^2 L^2$ of each bad direction, which the probe supplies at γ^2 per round while cold LinUCB's forced pulls supply it at L^2 per round. The window long enough for the expert to certify ($n \ge 36\lambda L^2/\gamma^2$ rounds per direction) is therefore longer by a factor of order L^2/γ^2 than the exploration cold LinUCB is forced into ($m_0 \le$ pulls $\le 8\beta^2 + 1$ per direction), and cold

LinUCB completes that exploration inside the window whenever the bad arm is offered there (exact-arithmetic simulations of the template, at the reduced sizes of Appendix G, give $R_{[Kn+1,T]}(\mathsf{C}) = 0$). Theorem 1 does not distinguish the two: the certified saving is bounded by the same price ratio wherever it is collected. Row $5'$ shows that the same ratio can be collected after the routed rounds when the routed window keeps cold LinUCB away from the directions it will be trapped on later.

Remark 5 (What certification costs on the template) On the template, the certificate used in Theorem 3(iv), namely that the confidence bound alone excludes the bad arm in direction e_j, $2\beta L\|e_j\|_{V_{\tau+1}^{-1}} < 1$, requires by Lemma 2 that $R_{[1,\tau]}(\mathsf{E}) > \kappa(e_j)\,(4\beta^2 L^2 - \lambda)$ for that direction, hence a total acquisition regret exceeding $K \min\{1/L^2, \varepsilon/\gamma^2\}(4\beta^2 L^2 - \lambda)$ for all K directions. This is Lemma 7 written out for the template.

Row $5'$: the same price ratio, collected on the tail. For the saving to appear as I_τ, the routed window must contain rounds at which cold LinUCB is *not* offered the bad arms it will later be trapped on, and is busy with something else; otherwise, as just explained, it completes its forced exploration inside the window. The following instance keeps cold LinUCB busy with *distractors*: arms of gap zero whose feature vectors carry a large component in a fresh direction, so that optimism prefers them to everything else while the expert probes. Fix $K, K', n \in \mathbb{N}$, put $d = 1 + K + K'$, $\theta^\star = e_1$, $L' := 12$, and $\tau_0 := Kn$. Directions $J := \{2, \dots, K+1\}$ are the bad directions and $J' := \{K+2, \dots, K+K'+1\}$ the distractor directions, cycled by $j(t) := 2 + ((t-1) \bmod K)$ and $j'(t) := K + 2 + ((t-1) \bmod K')$. The arms are

$$g = e_1\,(\mu = 1), \quad p_j = (1-\varepsilon)e_1 + \gamma\, e_j\,(\mu = 1-\varepsilon), \quad b_j = Le_j\,(\mu = 0), \quad b'_{j'} = e_1 + L' e_{j'}\,(\mu = 1),$$

and the instance $\mathcal{J}(K, K', L, \gamma, \varepsilon, n)$ offers $\mathcal{A}_t = \{g, p_{j(t)}, b'_{j'(t)}\}$ for $t \le \tau_0$ and $\mathcal{A}_t = \{g, b_{j(t)}\}$ for $t > \tau_0$. The expert plays the probe while probes are offered and the bad arm afterwards; the linear policy $\widehat{\theta}_{\mathsf{E}} = e_1 + w \sum_{j \in J} e_j - \sum_{j' \in J'} e_{j'}$ with $w \in (1/L, (1-\varepsilon)/(L-\gamma))$ does this, by the computation of Lemma 14 together with $\langle \widehat{\theta}_{\mathsf{E}}, b'_{j'} \rangle = 1 - L' < 1$. The prices of e_j, $j \in J$, are ε/γ^2 in the window and $1/L^2$ on the tail, as in Theorem 3; the distractor directions have price 0 and no bad arm uses them.

Proposition 9 (The price ratio collected as I_τ) *Let $\mathcal{J} = \mathcal{J}(K, K', L, \gamma, \varepsilon, n)$ under (6) strengthened to $L \ge 13$ and $\lambda \ge 16\sigma^2 d\,\ell_0$, with $n \ge 36\lambda L^2/\gamma^2$, $K \ge 36L^2$, $K' \ge L'^2 Kn/\lambda$, and $T - \tau_0 \ge Km_0$. With probability at least $1 - 3\delta$ the following hold simultaneously.*

(i) (Expert.) $R_T(\mathsf{E}) = \varepsilon Kn + (T - Kn)$.

(ii) (Cold.) *Cold LinUCB plays the distractor at every round $t \le \tau_0$ and the bad arm at each of the first m_0 rounds at which every tail direction is offered, so $R_{[1,\tau_0]}(\mathsf{C}) = 0$ and $Km_0 \le R_{[\tau_0+1,T]}(\mathsf{C}) \le K(4\beta^2 + 1)$.*

(iii) (Hybrid.) H_{τ_0} *plays g at every round $t > \tau_0$, so $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$.*

Consequently $A_{\tau_0} = -\varepsilon Kn < 0$, $Km_0 \le I_{\tau_0} \le K(16\lambda + 1)$, $B_{\tau_0} = T - Kn$, and

$$\frac{m_0}{\varepsilon n} \le \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \le \frac{16\lambda + 1}{\varepsilon n},$$

which at the smallest admissible n lies between $\gamma^2/(2220\,\varepsilon L^2)$ and $\gamma^2/(2\varepsilon L^2)$: the bracket of Theorem 3 up to its constant, now carried entirely by I_{τ_0}.

The proof (Appendix E.7) has one new ingredient: on the confidence event with the true radius, which the strengthened condition on λ makes at most $\frac{5}{8}\beta$, every index satisfies $U_t(x) \geq \mu(x) + \frac{3}{8}\beta\|x\|_{V_t^{-1}}$, and the distractor's width is at least $L'/\sqrt{2\lambda}$ as long as it has been pulled at most λ/L'^2 times, which the condition on K' guarantees. Its index then exceeds 7 while the indices of g and of the probe are below 4.3 and 5.6. Cold LinUCB never probes, reaches the tail with every bad direction untouched, and is trapped there exactly as in Theorem 3(ii); the hybrid's window is the same as in Theorem 3(iv). In the language of Theorem 1, rows 5 and 5′ have the same certified set $U = \{e_j : j \in J\}$, the same prices, and the same $P_U = \gamma^2/(\varepsilon L^2)$; they differ only in which rounds cold LinUCB's counterfactual exploration falls on. On the real streams of Section 8.2, too, the saving is collected after routing stops, as in row 5′, although there its source is mostly the transfer of Section 7.

Row 6: the same expert, optimal on the tail.

Proposition 10 (Expert optimal on the tail: large I_τ, no gain over the expert) *Let $\mathcal{I} = \mathcal{I}(K, L, \gamma, \varepsilon, Kn)$ with $n \geq 36\lambda L^2/\gamma^2$ and $K \geq 36L^2$, and let $\mathsf{E} = \mathsf{E}_w$ with $w \in (\varepsilon/\gamma, 1/L)$, which plays every probe and then g at every round. With probability at least $1 - 3\delta$, $R_T(\mathsf{E}) = R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$ and $B_{\tau_0} = 0$, while A_{τ_0} and I_{τ_0} are exactly as in Theorem 3.*

Row 7: probes that cover the wrong directions.

Proposition 11 (Probes cover the wrong directions) *Let $d = 2K + 1$ and replace the probe by $p'_t := (1-\varepsilon)e_1 + \gamma\, e_{K+j(t)}$, whose second coordinate is a nuisance direction that no bad arm uses, with $\mathcal{A}_t = \{g, b_t, p'_t\}$ for $t \leq \tau_0 = Kn$ and $\{g, b_t\}$ afterwards. Let E play p'_t for $t \leq \tau_0$ and b_t afterwards (for instance the linear policy with $\widehat{\theta}_{\mathsf{E}} = e_1 + w\sum_{j\geq 2} e_j$, $w \in (1/L, (1-\varepsilon)/(L-\gamma))$). Assume $n \geq 36\lambda L^2/\gamma^2$, $K \geq 36L^2$, and $T - \tau_0 \geq K\lfloor\lambda/2\rfloor$. With probability at least $1 - 3\delta$,*

$$R_T(\mathsf{H}_{\tau_0}) \;\geq\; \varepsilon Kn + K\lfloor\lambda/2\rfloor, \qquad R_T(\mathsf{C}) \;\leq\; K(4\beta^2+1) + \varepsilon Kn, \qquad \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \leq 65.$$

The principal angle between the expert's Gram matrix and the span of the bad directions is 90° and $I_{\tau_0} \leq R_{[\tau_0+1,T]}(\mathsf{C}) - K\lfloor\lambda/2\rfloor$.

Row 8: probes that are too dear.

Corollary 4 (Row 8: probes no cheaper than cold exploration) *In the setting of Theorem 3, if $\varepsilon \geq \gamma^2/L^2$ (the probe's price ε/γ^2 is at least cold exploration's price $1/L^2$) then $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \leq 2$ for every $n \geq 36\lambda L^2/\gamma^2$.*

Row 9: cold LinUCB already cheap.

Proposition 12 (Cold LinUCB already cheap: no multiplier in average reward) *On the no-probe instance $\mathcal{I}(K, L, \cdot, \cdot, 0)$, with probability at least $1-\delta$, $R_T(\mathsf{C}) \leq K(4\beta^2+1) = K(16\lambda+1)$. Hence for* every *policy π, $(R_T(\mathsf{C}) - R_T(\pi))/T \leq K(16\lambda+1)/T$, which with $\lambda = \sigma^2 d\ell_0$ is $O(\sigma^2 K^2 \ell_0/T)$.*

Row 10: the gap.

Proposition 13 (The gap enters only through the price) *Fix $\eta \in (0,1]$ and replace g by $g_\eta := \eta e_1$ ($\mu = \eta$) and p_t by $p_{t,\eta} := (1-\varepsilon)\eta e_1 + \gamma\, e_{j(t)}$ ($\mu = (1-\varepsilon)\eta$), so that the gaps of b_t and p_t are η and $\varepsilon\eta$ and the prices of e_j are η/L^2 and $\varepsilon\eta/\gamma^2$. Let $\mathsf{E} = \mathsf{E}_w$ with $w \in (\eta/L, (1-\varepsilon)\eta/(L-\gamma))$. Under (6), with $n \geq 36\lambda L^2/(\eta^2\gamma^2)$, $K \geq 36L^2/\eta^2$, and $\tau = \tau_0 = Kn$, with probability at least $1-3\delta$,*

$$\frac{\eta K m_0}{\varepsilon\eta K n} \;\leq\; \frac{R_T(\mathsf{C})}{R_T(\mathsf{H}_{\tau_0})} \;\leq\; 1 + \frac{8\beta^2/\eta^2 + 1}{\varepsilon n} \;\leq\; 1 + \frac{\gamma^2}{\varepsilon L^2},$$

and at the smallest admissible n the left-hand side is at least $\eta^2\gamma^2/(2220\,\varepsilon L^2)$. The upper bound is the price ratio $(\eta/L^2)/(\varepsilon\eta/\gamma^2) = \gamma^2/(\varepsilon L^2)$, in which η cancels; the lower bound loses a factor η^2 because our lower bound on cold LinUCB counts only the m_0 pulls per direction that optimism forces regardless of the gap.

Where the experiments sit. The experiments of Section 8 relate to the rows as follows. AmazonCat with Flan-T5 has the timing of row $5'$: multipliers $1.71\times$, $2.72\times$ and $3.29\times$, routed rounds concentrated in the first few thousand, and 96–100% of the saving accruing after the last routed round (Figure 2). Its source, however, is mostly the transfer of Section 7, which lies outside the linear model of the catalog: the three experts buy the same coverage at the same price and still differ (Section 8.2). AmazonCat with BGE features shows the same timing with a smaller gain (up to $1.23\times$ on held-out seeds), since cold LinUCB already reaches about a third of the correct answers. WikiLinks with Flan-T5 does not multiply ($1.01\times$): the routed data does reach useful directions ($I_\tau > 0$ on every seed), but no entity is frequent enough for the expert to steer the learner toward (Section 8.3). [**TODO:** WikiLinks sweep: place the Qwen configurations once the sweep is complete.]

Appendix D. Proofs

Section D.1 proves the decomposition identity. Section D.2 proves Proposition 2. Section D.3 proves Proposition 3. Sections D.4–D.6 prove Lemma 1 and the two width identities of Appendix B.1, Section D.7 proves Proposition 5, and the remaining sections prove the results of Sections 5 and 7, including Theorem 1.

D.1. Proof of Proposition 1

Proof Split $[T]$ into $[1,\tau]$ and $[\tau+1,T]$. On $[1,\tau]$ the hybrid plays a_t^{E}, so $R_{[1,\tau]}(\mathsf{H}_\tau) = R_{[1,\tau]}(\mathsf{E})$. Hence

$$R_T(\mathsf{C}) - R_T(\mathsf{H}_\tau) = \big(R_{[1,\tau]}(\mathsf{C}) - R_{[1,\tau]}(\mathsf{E})\big) + \big(R_{[\tau+1,T]}(\mathsf{C}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)\big) = A_\tau + I_\tau,$$

and

$$R_T(\mathsf{E}) - R_T(\mathsf{H}_\tau) = \big(R_{[1,\tau]}(\mathsf{E}) - R_{[1,\tau]}(\mathsf{E})\big) + \big(R_{[\tau+1,T]}(\mathsf{E}) - R_{[\tau+1,T]}(\mathsf{H}_\tau)\big) = B_\tau.$$

Both are identities between sums of gaps, so they hold for every noise realization and therefore in expectation. ■

D.2. Proof of Proposition 2

We first record the standard subgaussian concentration lemma used in the high-probability part of the argument, and then construct the linear bandit instance and verify the three claims of Proposition 2.

Lemma 5 (Subgaussian concentration; Lattimore and Szepesvári, 2020, Corollary 5.5) *Assume that $X_1 - \mu, \dots, X_n - \mu$ are independent, σ-subgaussian random variables. Then for any $\varepsilon \geq 0$,*

$$\mathbb{P}(\hat{\mu} \geq \mu + \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right), \qquad \mathbb{P}(\hat{\mu} \leq \mu - \varepsilon) \leq \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right),$$

where

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} X_t.$$

Problem setup. Let the dimension satisfy $d \geq n + 1$, where n is the horizon. The unknown parameter is

$$\theta^\star = e_1 \in \mathbb{R}^d,$$

where $\{e_i\}_{i=1}^d$ denotes the standard basis. Rewards follow the stochastic linear model

$$X_t = \langle \theta^\star, A_t \rangle + \eta_t,$$

where η_t are independent mean-zero σ-subgaussian random variables.

At each round $t = 1, \dots, n$, the learner is offered two actions,

$$g := e_1, \qquad b_t := Le_{t+1},$$

where $L > 1$ is a fixed constant. Their expected rewards are

$$\langle \theta^\star, g \rangle = 1, \qquad \langle \theta^\star, b_t \rangle = 0.$$

Both LinUCB variants below use the rule

$$\mathrm{UCB}_t(a) = a^\top \hat{\theta}_{t-1} + \beta \|a\|_{V_{t-1}^{-1}},$$

with

$$V_{t-1} = V_0 + \sum_{s<t} A_s A_s^\top, \qquad \hat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s<t} A_s X_s.$$

Algorithm 1: a constant policy. Consider a policy π_{const} that selects an action by maximizing $\langle \hat{\theta}_{\mathrm{const}}, a \rangle$ for some fixed parameter estimate $\hat{\theta}_{\mathrm{const}} \in \mathbb{R}^d$ that is never updated. For concreteness, take

$$\hat{\theta}_{\mathrm{const}} = \frac{1}{L}\sum_{j=2}^{n+1} e_j,$$

so that

$$\langle \hat{\theta}_{\mathrm{const}}, g \rangle = 0, \qquad \langle \hat{\theta}_{\mathrm{const}}, b_t \rangle = 1 \quad \text{for every } t \leq n.$$

Then π_{const} selects b_t at every round t. Since each pull of b_t yields expected reward 0 while g has expected reward 1, the cumulative pseudo-regret of π_{const} on the horizon n is exactly n. This proves part (1) of Proposition 2.

The failure mode is structural rather than statistical: π_{const} never queries direction e_1, so no amount of data can refine $\hat{\theta}_{\text{const}}$ in the reward-relevant direction. The policy has no learning mechanism. We will see, however, that the trajectory it produces is precisely what suppresses the misplaced optimism of LinUCB in the bad directions.

Algorithm 2: LinUCB with isotropic initialization. The first LinUCB variant uses

$$V_0 = \lambda I_d, \qquad \hat{\theta}_0 = 0.$$

At $t = 1$,

$$\|g\|^2_{V_0^{-1}} = \frac{1}{\lambda}, \qquad \|b_1\|^2_{V_0^{-1}} = \frac{L^2}{\lambda}.$$

Since $\hat{\theta}_0 = 0$, we have

$$\mathrm{UCB}_1(b_1) > \mathrm{UCB}_1(g),$$

so the algorithm selects b_1.

Now suppose inductively that the algorithm has selected $b_1, \dots, b_{t-1}$ in the first $t-1$ rounds. Then

$$V_{t-1} = \lambda I_d + \sum_{s=1}^{t-1} b_s b_s^\top = \lambda I_d + L^2 \sum_{j=2}^{t} e_j e_j^\top.$$

Hence V_{t-1} is diagonal in the standard basis. In particular, for the fresh direction e_{t+1},

$$e_{t+1}^\top V_{t-1}^{-1} e_{t+1} = \frac{1}{\lambda}, \qquad \|b_t\|^2_{V_{t-1}^{-1}} = \frac{L^2}{\lambda}, \qquad \|g\|^2_{V_{t-1}^{-1}} = \frac{1}{\lambda}.$$

The key point is that although the observed rewards on $b_1, \dots, b_{t-1}$ contain noise, the estimate

$$\hat{\theta}_{t-1} = V_{t-1}^{-1} \sum_{s=1}^{t-1} b_s X_s$$

lies entirely in $\mathrm{span}\{e_2, \dots, e_t\}$. Therefore,

$$g^\top \hat{\theta}_{t-1} = 0, \qquad b_t^\top \hat{\theta}_{t-1} = 0,$$

because both $g = e_1$ and the fresh arm $b_t = L e_{t+1}$ are orthogonal to $\mathrm{span}\{e_2, \dots, e_t\}$.

Thus, even in the stochastic-reward setting, the comparison between g and b_t is determined purely by the confidence terms:

$$\mathrm{UCB}_t(g) = \frac{\beta}{\sqrt{\lambda}}, \qquad \mathrm{UCB}_t(b_t) = \frac{\beta L}{\sqrt{\lambda}}.$$

Since $L > 1$, we have $\mathrm{UCB}_t(b_t) > \mathrm{UCB}_t(g)$ for all $t \le n$. Hence the isotropically initialized algorithm selects b_t at every round and, like π_{const}, incurs cumulative pseudo-regret n. This proves part (2) of Proposition 2.

Algorithm 3: LinUCB with an anisotropic warm start. We now define an initialization that is realized by exactly the trajectory π_{const} would produce. Fix an integer $m \geq 1$, and let

$$V_0' = \lambda I_d + mL^2 \sum_{j=2}^{n+1} e_j e_j^\top, \qquad \hat{\theta}_0' = 0.$$

This corresponds to having observed each bad direction Le_j, $j = 2, \ldots, n+1$, exactly m times before the online interaction begins — precisely the kind of data π_{const} generates if rolled out for mn rounds on a sequence of action sets containing each bad direction. Thus V_0' is a genuine design matrix obtainable from a collection of outer products produced by a constant logging policy.

At $t = 1$,

$$\|g\|^2_{(V_0')^{-1}} = \frac{1}{\lambda}, \qquad \|b_1\|^2_{(V_0')^{-1}} = L^2 \cdot \frac{1}{\lambda + mL^2} = \frac{L^2}{\lambda + mL^2}.$$

Hence, if

$$m > \frac{\lambda(L^2 - 1)}{L^2},$$

then $\frac{1}{\lambda} > \frac{L^2}{\lambda + mL^2}$, and therefore

$$\mathrm{UCB}_1'(g) > \mathrm{UCB}_1'(b_1).$$

So the warm-started algorithm selects g at round 1.

Suppose now that the algorithm continues to select g for the first s rounds. Then

$$V_s' = \lambda I_d + mL^2 \sum_{j=2}^{n+1} e_j e_j^\top + s\, e_1 e_1^\top.$$

Since each reward observed on g has mean 1,

$$\sum_{i=1}^{s} X_i = s + \sum_{i=1}^{s} \eta_i,$$

and therefore

$$g^\top \hat{\theta}_s' = e_1^\top (V_s')^{-1} \sum_{i=1}^{s} g X_i = \frac{1}{\lambda + s} \sum_{i=1}^{s} X_i.$$

Applying Lemma 5 with $\mu = 1$, we obtain that with probability at least $1 - \delta$, simultaneously for all $1 \leq s \leq n$,

$$\frac{1}{s} \sum_{i=1}^{s} X_i \geq 1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}.$$

On this event,

$$g^\top \hat{\theta}_s' \geq \frac{s}{\lambda + s} \left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right).$$

For every fresh bad arm $b_{s+1} = Le_{s+2}$,

$$b_{s+1}^\top \hat{\theta}_s' = 0,$$

since the online trajectory has only sampled the direction e_1, and the warm-start contributes only to the design matrix, not to the response vector. Moreover,

$$\|b_{s+1}\|^2_{(V'_s)^{-1}} = \frac{L^2}{\lambda + mL^2}, \qquad \|g\|^2_{(V'_s)^{-1}} = \frac{1}{\lambda + s}.$$

On the same high-probability event,

$$\mathrm{UCB}'_{s+1}(g) \geq \frac{s}{\lambda + s}\left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right) + \frac{\beta}{\sqrt{\lambda + s}},$$

while

$$\mathrm{UCB}'_{s+1}(b_{s+1}) = \frac{\beta L}{\sqrt{\lambda + mL^2}}.$$

Consequently, if m is chosen large enough so that for every $1 \leq s \leq n$,

$$\frac{s}{\lambda + s}\left(1 - \sqrt{\frac{2\sigma^2 \log(2n/\delta)}{s}}\right) + \frac{\beta}{\sqrt{\lambda + s}} > \frac{\beta L}{\sqrt{\lambda + mL^2}}, \tag{8}$$

then, with probability at least $1 - \delta$, the warm-started algorithm selects g at every round, and the cumulative pseudo-regret over the n online rounds is 0 on this event. This proves part (3) of Proposition 2; condition (8) is the explicit "m sufficiently large" requirement referenced in the proposition statement. The condition is satisfiable by a finite m whenever the left-hand side is positive for every s, which holds for any $\beta \geq \sigma\sqrt{2\log(2n/\delta)}$ (in particular for the radius of Abbasi-Yadkori et al., 2011), since then $\beta/\sqrt{\lambda + s} \geq \sigma\sqrt{2\log(2n/\delta)}\,\sqrt{s}/(\lambda + s)$. ■

Remark 6 The warm-started algorithm in the proof uses $\widehat{\theta}'_0 = 0$, i.e., it carries the offline Gram matrix but not the offline response sum. Because every offline observation is of an arm with expected reward 0, the offline response sum has mean zero in each bad direction and carrying it would only add a mean-zero perturbation to $\widehat{\theta}$ in directions orthogonal to g; the comparison between g and b_{s+1} is unchanged up to the noise term already absorbed in the choice of m, so we omit it for simplicity.

D.3. Proof of Proposition 3

Proof Work on the instance of Section D.2 with horizon T: $\theta^\star = e_1$, $\mathcal{A}_t = \{g, b_t\}$ with $g = e_1$ and $b_t = Le_{t+1}$. Fix an arbitrary expert E and a handoff time τ. Whatever the expert plays during rounds $1, \ldots, \tau$, and whatever LinUCB plays afterwards, the set of arms played before round t is contained in $\{g\} \cup \{b_s : s < t\}$, all of which are orthogonal to e_{t+1}. Hence for every t, the design matrix $V_t = \lambda I + \sum_{s<t} x_{s,A_s} x_{s,A_s}^\top$ is block diagonal with the block on e_{t+1} equal to λ, and $\widehat{\theta}_t = V_t^{-1} c_t$ lies in $\mathrm{span}\{e_1, \ldots, e_t\}$. Therefore

$$\|b_t\|^2_{V_t^{-1}} = \frac{L^2}{\lambda}, \qquad \langle \widehat{\theta}_t, b_t \rangle = 0, \qquad \text{for every } t \text{ and every trajectory.}$$

So $\mathrm{UCB}_t(b_t) = \beta L/\sqrt{\lambda}$ exactly, for all t.

Now bound $\mathrm{UCB}_t(g)$. Let s_t be the number of pulls of g before round t and $X_1, \dots, X_{s_t}$ the rewards observed on those pulls; since V_t is diagonal with entry $\lambda + s_t$ on e_1 and the rewards on bad arms enter c_t only in directions orthogonal to e_1,

$$\langle \widehat{\theta}_t, g \rangle = \frac{1}{\lambda + s_t} \sum_{i=1}^{s_t} X_i, \qquad \|g\|^2_{V_t^{-1}} = \frac{1}{\lambda + s_t} \le \frac{1}{\lambda}.$$

By Lemma 5 and a union bound over $s \in \{1, \dots, T\}$, with probability at least $1 - \delta$ we have $\sum_{i \le s} X_i \le s + \sqrt{2\sigma^2 s \log(2T/\delta)}$ for all $s \le T$ simultaneously. On that event,

$$\langle \widehat{\theta}_t, g \rangle \le \frac{s_t}{\lambda + s_t} + \frac{\sqrt{2\sigma^2 s_t \log(2T/\delta)}}{\lambda + s_t} \le 1 + \frac{\sigma\sqrt{2\log(2T/\delta)}}{\sqrt{\lambda + s_t}} \le 1 + \zeta,$$

using $\sqrt{s_t}/(\lambda + s_t) \le 1/\sqrt{\lambda + s_t} \le 1/\sqrt{\lambda}$. Hence $\mathrm{UCB}_t(g) \le 1 + \zeta + \beta/\sqrt{\lambda}$. Under the hypothesis $\beta(L-1)/\sqrt{\lambda} > 1 + \zeta$ this is strictly less than $\beta L/\sqrt{\lambda} = \mathrm{UCB}_t(b_t)$, so LinUCB, whenever it acts, plays b_t. The argument did not use anything about rounds $1, \dots, \tau$ beyond the fact that they cannot produce coverage of e_{t+1} for $t > \tau$, so it holds for every expert and every τ. Since $\Delta_{t,b_t} = 1$, $R_{[\tau+1,T]}(\mathsf{H}_\tau) = T - \tau$. Cold LinUCB plays b_t on every round by part (2) of Proposition 2 (no condition on β is needed there because $\widehat{\theta}_t \perp g$ on its own trajectory), so $R_{[\tau+1,T]}(\mathsf{C}) = T - \tau$ and $I_\tau = 0$. ∎

D.4. Proof of Lemma 1

Proof Fix a round t at which LinUCB acts, and write $x^\star = x_{t,a_t^\star}$, $x = x_{t,A_t}$, $w(u) = \|u\|_{V_t^{-1}}$. On $\mathcal{E}_\delta$, Cauchy–Schwarz gives $|\langle \widehat{\theta}_t - \theta^\star, u \rangle| \le \beta_t w(u)$ for every u. By the selection rule, $\langle \widehat{\theta}_t, x \rangle + \beta_t w(x) \ge \langle \widehat{\theta}_t, x^\star \rangle + \beta_t w(x^\star) \ge \langle \theta^\star, x^\star \rangle$, and $\langle \widehat{\theta}_t, x \rangle \le \langle \theta^\star, x \rangle + \beta_t w(x)$. Combining, $\Delta_{t,A_t} = \langle \theta^\star, x^\star - x \rangle \le 2\beta_t w(x)$. Thus $\Delta_{t,A_t} = \min\{\Delta_{t,A_t}, 2\beta_t w(x_{t,A_t})\} \le \max_a \min\{\Delta_{t,a}, 2\beta_t w(x_{t,a})\} = \Phi_t(V_t)$. Monotonicity: $V \preceq V'$ implies $V'^{-1} \preceq V^{-1}$, so every width is non-increasing and so is each term in the maximum. ∎

D.5. Proof of Lemma 3

Proof $\lambda I + M \succeq \lambda I$ gives $(\lambda I + M)^{-1} \preceq (\lambda I)^{-1}$, which is the first claim. For the second, $\lambda I + M \succeq \lambda I + mP$, so $x^\top (\lambda I + M)^{-1} x \le x^\top (\lambda I + mP)^{-1} x$, and for $x \in \mathrm{Im}(P)$ the right-hand side is $\|x\|_2^2/(\lambda + m)$ because $(\lambda I + mP)^{-1}$ acts as $(\lambda + m)^{-1}$ on $\mathrm{Im}(P)$. For the third, if $x \perp \mathrm{Im}(M)$ then x is an eigenvector of $\lambda I + M$ with eigenvalue λ, so $x^\top (\lambda I + M)^{-1} x = \|x\|_2^2/\lambda$. ∎

D.6. Proof of Lemma 4

Proof By the Sherman–Morrison formula, $(V + xx^\top)^{-1} = V^{-1} - \dfrac{V^{-1} x x^\top V^{-1}}{1 + x^\top V^{-1} x}$. Multiplying on both sides by z,

$$z^\top (V + xx^\top)^{-1} z = z^\top V^{-1} z - \frac{(z^\top V^{-1} x)^2}{1 + \|x\|^2_{V^{-1}}} = \|z\|^2_{V^{-1}} \Big(1 - \frac{(z^\top V^{-1} x)^2}{(1 + \|x\|^2_{V^{-1}})\|z\|^2_{V^{-1}}}\Big).$$

The bracket is at least $1 - \dfrac{\|x\|_{V^{-1}}^2 \|z\|_{V^{-1}}^2}{(1+\|x\|_{V^{-1}}^2)\|z\|_{V^{-1}}^2} > 0$ by Cauchy–Schwarz in the V^{-1} inner product, so $\rho_V \in [0,1)$. At $V = \lambda I$, $z^\top V^{-1} x = \langle z, x\rangle/\lambda$, $\|x\|_{V^{-1}}^2 = \|x\|_2^2/\lambda$, $\|z\|_{V^{-1}}^2 = \|z\|_2^2/\lambda$, and substituting gives $\rho_{\lambda I}(x,z) = \dfrac{\langle x, z\rangle^2}{\|x\|_2^2\|z\|_2^2} \cdot \dfrac{\|x\|_2^2}{\lambda + \|x\|_2^2}$. ■

D.7. Proof of Proposition 5

Proof The acquisition term is the definition of $R_{[1,\tau]}(\mathsf{H}_\tau) = R_{[1,\tau]}(\mathsf{E})$. For the LinUCB phase we give two bounds and take the minimum.

First bound. By Lemma 1 applied at each $t > \tau$, $\sum_{t>\tau} \Delta_{t,A_t} \le \sum_{t>\tau} \Phi_t(V_t)$.

Second bound. From the proof of Lemma 1, $\Delta_{t,A_t} \le 2\beta_t \|x_{t,A_t}\|_{V_t^{-1}} \le 2\beta_T \|x_{t,A_t}\|_{V_t^{-1}}$ for $t > \tau$ (the radius β_t is non-decreasing in t). Cauchy–Schwarz over $t = \tau+1, \dots, T$ gives

$$\sum_{t>\tau} \Delta_{t,A_t} \le 2\beta_T \sqrt{(T-\tau) \sum_{t>\tau} \|x_{t,A_t}\|_{V_t^{-1}}^2}.$$

By the elliptical potential lemma (Abbasi-Yadkori et al., 2011, Lemma 11),

$$\sum_{t>\tau} \min\{1, \|x_{t,A_t}\|_{V_t^{-1}}^2\} \le 2 \log \frac{\det V_{T+1}}{\det V_{\tau+1}};$$

since $\|x_{t,A_t}\|_{V_t^{-1}}^2 \le L^2/\lambda$, the truncation at 1 costs at most a factor $\max\{1, L^2/\lambda\}$ on the sum of squares, which is the factor $\max\{1, L/\sqrt{\lambda}\}$ in (4) after the square root; with $L^2 \le \lambda$ nothing is lost. This gives the second term of (4).

Determinant ratio. Let $V_{\tau+1} = \sum_j \sigma_j u_j u_j^\top$ be an eigendecomposition and $G = \sum_{t>\tau} x_{t,A_t} x_{t,A_t}^\top$. Then $\det(V_{\tau+1}+G)/\det V_{\tau+1} = \det(I + V_{\tau+1}^{-1/2} G V_{\tau+1}^{-1/2}) \le \prod_j (1 + u_j^\top G u_j/\sigma_j)$ by Hadamard's inequality for positive definite matrices, and $u_j^\top G u_j = \sum_{t>\tau} \langle u_j, x_{t,A_t}\rangle^2 \le L^2(T-\tau)$. Taking logarithms gives the stated bound. If all played actions lie in an r-dimensional subspace $\mathrm{Im}(P)$ then G is supported there and only the r eigenvalues of $V_{\tau+1}$ restricted to $\mathrm{Im}(P)$ appear, which is how the factor r in the remark after the theorem arises; if in addition $V_{\tau+1} \succeq (\lambda + \nu\tau)P$ on that subspace, each $\sigma_j \ge \lambda + \nu\tau$.

The cold bound is the case $\tau = 0$, $V_1 = \lambda I$, $\sigma_j = \lambda$. The ratio Ψ in the text is the quotient of the two isotropic bounds and is stated only to show that it does not tend to zero. ■

Remark 7 (Validity of the confidence event with an online warm start) The hybrid's LinUCB phase uses $(V_{\tau+1}, c_{\tau+1})$ built from the expert-routed rounds together with the observed rewards on those rounds. The self-normalized bound of Abbasi-Yadkori et al. (2011) applies to any sequence of (feature, reward) pairs in which the feature at round s is measurable with respect to the history before s, whoever chose it, so the event $\mathcal{E}_\delta$ covers the hybrid's trajectory. If instead the warm start carried M_τ but dropped c_τ ("passive" updates), the estimate $\widehat{\theta}_t$ would be biased toward zero in the directions of M_τ and the event would not be guaranteed; the Router in our experiments updates both, and Algorithm 1 reflects this.

D.8. Proofs for Section 7

Proof of Proposition 6 Write $x^\star = x_{t,a_t^\star}$, $x = x_{t,A_t}$, $w(u) = \|u\|_{V_t^{-1}}$. Cauchy–Schwarz in the V_t inner product gives $|\langle \widehat{\theta}_t - \theta^\star, u\rangle| \le \rho_t w(u)$ for every u; this is the inequality in the proof of Lemma 1 with the realized error in place of its bound. The selection rule gives

$$\langle \widehat{\theta}_t, x\rangle + \beta_t w(x) \;\ge\; \langle \widehat{\theta}_t, x^\star\rangle + \beta_t w(x^\star) \;\ge\; \langle \theta^\star, x^\star\rangle + (\beta_t - \rho_t) w(x^\star) \;\ge\; \langle \theta^\star, x^\star\rangle,$$

using $\rho_t \le \beta_t$, and $\langle \widehat{\theta}_t, x\rangle \le \langle \theta^\star, x\rangle + \rho_t w(x)$. Subtracting, $\Delta_{t,A_t} \le (\beta_t + \rho_t) w(x)$, and taking the maximum over a of the minimum with the gap gives (5). Monotonicity in V is Lemma 1; in ρ it is immediate. Finally $\min\{\Delta, (\beta+\rho)w\} \ge \min\{\Delta, \beta w\} \ge \frac{1}{2}\min\{\Delta, 2\beta w\}$. ∎

Proof of Proposition 4 $c = \sum_s r_s x_s = \sum_s (\langle \theta^\star, x_s\rangle + \eta_s) x_s = \big(\sum_s x_s x_s^\top\big)\theta^\star + \sum_s \eta_s x_s = M\theta^\star + \xi$. Conditionally on the features, ξ is Gaussian with mean zero and covariance $\sigma^2 \sum_s x_s x_s^\top = \sigma^2 M$. Then $\widehat{\theta}_1 - \theta^\star = V_1^{-1}(M\theta^\star + \xi) - \theta^\star = V_1^{-1}(M - V_1)\theta^\star + V_1^{-1}\xi = -\lambda V_1^{-1}\theta^\star + V_1^{-1}\xi$. Both $V_1 = \lambda I + M$ and the law of ξ are functions of M alone, so the conditional law of (V_1, c_1) given M is the same whatever policy produced the actions. The LinUCB phase is a deterministic function of (V_1, c_1) and of the future contexts and noise, which are independent of the warm start; hence the LinUCB phases agree in law, and so does the expected tail regret of the hybrid. B_τ is not covered, since $R_{[\tau+1,T]}(\mathsf{E})$ depends on the expert's own actions. ∎

Proof of Proposition 7 Fix $\widetilde{\theta}$ and write $w(u) = \|u\|_{V_t^{-1}}$, $x^\star = x_{t,a_t^\star}$, $x = x_{t,A_t}$. Cauchy–Schwarz gives $|\langle \widehat{\theta}_t - \widetilde{\theta}, u\rangle| \le \widetilde{\rho}_t w(u)$. By the definition of $\widetilde{\varepsilon}$,

$$\bar{r}(x^\star) = \langle \widetilde{\theta}, x^\star\rangle + \widetilde{\varepsilon}(x^\star) \le \langle \widehat{\theta}_t, x^\star\rangle + \widetilde{\rho}_t w(x^\star) + \widetilde{\varepsilon}(x^\star).$$

The selection rule gives $\langle \widehat{\theta}_t, x^\star\rangle \le \langle \widehat{\theta}_t, x\rangle + \beta_t w(x) - \beta_t w(x^\star)$, and $\langle \widehat{\theta}_t, x\rangle \le \langle \widetilde{\theta}, x\rangle + \widetilde{\rho}_t w(x) = \bar{r}(x) - \widetilde{\varepsilon}(x) + \widetilde{\rho}_t w(x)$. Chaining the three displays,

$$\bar{r}(x^\star) - \bar{r}(x) \le (\beta_t + \widetilde{\rho}_t) w(x) + (\widetilde{\rho}_t - \beta_t) w(x^\star) + \widetilde{\varepsilon}(x^\star) - \widetilde{\varepsilon}(x),$$

and $(\widetilde{\rho}_t - \beta_t) w(x^\star) \le (\widetilde{\rho}_t - \beta_t)_+ w(x^\star)$. ∎

D.9. Proof of Theorem 1

We work in the setting of Section 5, on the confidence event $\mathcal{E}_\delta$ of cold LinUCB, and prove the two steps of the proof sketch as lemmas.

Lemma 6 (Cold LinUCB pays its own price, once per arm) $R_T(\mathsf{C}) \le \sum_{x \in \mathcal{S}} \Delta(x) \min\{n_x^\star, N_x\}$.

Proof Fix $x \in \mathcal{S}$ and suppose cold LinUCB plays x at round t after i earlier pulls of x. On $\mathcal{E}_\delta$ the index of x is at most $\mu(x) + 2\beta_t \|x\|_{V_t^{-1}}$ and the index of $x^\star \in \mathcal{A}_t$ is at least $\mu(x^\star)$, so playing x requires $\Delta(x) \le 2\beta \|x\|_{V_t^{-1}}$ (this is Lemma 1). The i earlier pulls are among the summands of V_t, so $V_t \succeq \lambda I + i\, x x^\top$; inversion reverses the Loewner order, and the Sherman–Morrison identity gives $\|x\|^2_{V_t^{-1}} \le \|x\|_2^2/(\lambda + i\|x\|_2^2)$. Hence $\Delta(x)^2(\lambda + i\|x\|_2^2) \le 4\beta^2 \|x\|_2^2$, that is, $\lambda + i\|x\|_2^2 \le C_x^\star$. If $C_x^\star < \lambda$ this fails already at $i = 0$, and x is never played. Otherwise every pull has

$i \le (C_x^\star - \lambda)/\|x\|_2^2$, so x is pulled at most $n_x^\star$ times, and at most N_x times because it is offered on N_x rounds. Each pull costs $\Delta(x)$ because $x^\star \in \mathcal{A}_t$, and optimal pulls cost nothing. Summing over $x \in \mathcal{S}$ gives the lemma. ∎

Lemma 7 (Certification costs at least the price) *For every policy π, every cutoff $t_0 \in \{0, \dots, T\}$ and every set of arms $U \subseteq U_{\mathrm{cert}}(t_0)$,*

$$\sum_{x \in U} (C_x^\star - \lambda)_+ \;\le\; \sum_{x \in U} \sum_{t \le t_0} \langle x_t^\pi, u_x \rangle^2 \;\le\; \frac{R_{[1,t_0]}(\pi)}{\kappa_U^\pi(t_0)} + C_U^{\mathrm{free}}(t_0),$$

with $1/\kappa_U^\pi(t_0) := 0$ when $\kappa_U^\pi(t_0) = +\infty$.

Proof Fix $x \in U$, write $V := V_{t_0+1}^\pi$, and let $C(u_x) := \sum_{t \le t_0} \langle x_t^\pi, u_x \rangle^2$ be the coverage of u_x after round t_0. Then $x^\top V x = \|x\|_2^2(\lambda + C(u_x))$, and Cauchy–Schwarz in the V inner product, $\|x\|_2^4 = \langle x, x \rangle^2 \le \|x\|_{V^{-1}}^2 \, x^\top V x$, gives $\|x\|_{V^{-1}}^2 \ge \|x\|_2^2/(\lambda + C(u_x))$. Certification of x, $4\beta^2 \|x\|_{V^{-1}}^2 < \Delta(x)^2$, therefore forces $\lambda + C(u_x) > C_x^\star$; since also $C(u_x) \ge 0$, $C(u_x) \ge (C_x^\star - \lambda)_+$, which summed over U is the first inequality. For the second, exchange the order of summation: $\sum_{x \in U} C(u_x) = \sum_{t \le t_0} \mathrm{cov}_U(x_t^\pi)$. The rounds with $\Delta(x_t^\pi) = 0$ contribute $C_U^{\mathrm{free}}(t_0)$ by definition. On a round with $\Delta(x_t^\pi) > 0$, the definition of $\kappa_U^\pi(t_0)$ as a minimum gives $\mathrm{cov}_U(x_t^\pi) \le \Delta(x_t^\pi)/\kappa_U^\pi(t_0)$, trivially when $\mathrm{cov}_U(x_t^\pi) = 0$ (including the case $\kappa_U^\pi(t_0) = +\infty$). Summing, these rounds contribute at most $R_{[1,t_0]}(\pi)/\kappa_U^\pi(t_0)$. ∎

Proof of Theorem 1 Split the sum of Lemma 6 into $x \in U$ and $x \in \mathcal{S} \setminus U$. For $x \in U$, $\min\{n_x^\star, N_x\} \le n_x^\star \le (C_x^\star - \lambda)_+/\|x\|_2^2 + 1$, so $\Delta(x) \min\{n_x^\star, N_x\} \le \kappa_x (C_x^\star - \lambda)_+ + \Delta(x) \le \kappa_U^{\mathsf{C}} (C_x^\star - \lambda)_+ + \Delta(x)$. Summing over U and applying Lemma 7 bounds these terms by $\kappa_U^{\mathsf{C}} \big(R_{[1,t_0]}(\pi)/\kappa_U^\pi(t_0) + C_U^{\mathrm{free}}(t_0) \big) + \sum_{x \in U} \Delta(x)$, and $\kappa_U^{\mathsf{C}}/\kappa_U^\pi(t_0) = P_U$; the terms with $x \in \mathcal{S} \setminus U$ are carried unchanged. Finally $R_{[1,t_0]}(\pi) \le R_T(\pi)$ because pseudo-regret is nonnegative on every round, so dividing by $R_T(\pi)$ gives (3). For $\pi = \mathsf{H}_\tau$ and $t_0 = \tau$, $R_{[1,\tau]}(\mathsf{H}_\tau) = R_{[1,\tau]}(\mathsf{E})$ because the hybrid plays the expert's actions on $[1, \tau]$. ∎

D.10. The Price Bound on the Template

Here we check the bound of Theorem 1 on the template instance.

On the instance of Theorem 3 take for U the K bad arms $b_j = L e_j$, all of which are certified at τ_0 by (iv). Then $\kappa_{b_j} = 1/L^2$, so $\kappa_U^{\mathsf{C}} = 1/L^2$; the expert plays only probes in the window, $\mathrm{cov}_U(p_j) = \gamma^2$ and $\Delta(p_j) = \varepsilon$, so $\kappa_U^\pi(\tau_0) = \varepsilon/\gamma^2$ and $P_U = \gamma^2/(\varepsilon L^2)$; $C_U^{\mathrm{free}} = 0$ because $g \perp U$; $\sum_U \Delta = K$; and the uncertified arms are the probes, for which $\Delta(p_j) \min\{n_{p_j}^\star, N_{p_j}\} \le \varepsilon n$. The theorem gives $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \le \gamma^2/(\varepsilon L^2) + (K + \varepsilon K n)/(\varepsilon K n) = \gamma^2/(\varepsilon L^2) + 1 + 1/(\varepsilon n)$, consistent with the upper bound $1 + (8\beta^2 + 1)/(\varepsilon n) \le 1 + \gamma^2/(\varepsilon L^2)$ proved directly. The same computation applies verbatim to Proposition 9, where the distractors have gap 0 and contribute nothing.

D.11. Proof of Corollary 1

Proof From (3), $P_U \geq R_T(\mathsf{C})/R_T(\pi) - C$ for the fixed multiple C, so $P_U \to \infty$; since $\kappa_U^{\mathsf{C}} \leq \max_{x \in \mathcal{S}} \kappa_x$ is a property of the instance, this is $\kappa_U^{\pi}(t_0) \to 0$ relative to κ_U^{C}. On the canonical instance the arms offered after round τ are $b_t = Le_{t+1}$, $t > \tau$, and no arm played at or before round τ has a component along e_{t+1}, so $\|b_t\|_{V_{\tau+1}^{-1}} = L/\sqrt{\lambda}$ and $2\beta L/\sqrt{\lambda} > 1 = \Delta(b_t)$ under the hypothesis of Proposition 3; hence no arm offered after round τ is certified. ■

Appendix E. Proofs for the Catalog of Section 6

Throughout this appendix the conventions (6) are in force, $\alpha := 1-\varepsilon$, $m_0 := \lfloor \lambda/30 \rfloor$, and "LinUCB acts at round t" means that the action at t is chosen by the LinUCB rule (every round for C; rounds $t > \tau$ for H_τ). Ties in the LinUCB rule are broken arbitrarily; every comparison we use is strict. For a policy π and a round t we write $s(t)$, $m_j(t)$, $n_j(t)$ for the numbers of pulls of g, b in direction e_j, and p in direction e_j before round t (all of them counted from the start, including expert-routed rounds), and we drop the argument t when no confusion arises. In the variant of Proposition 13 the roles of g and p are played by g_η and $p_{t,\eta}$ and $\alpha := (1-\varepsilon)\eta$; we prove everything for general $\eta \in (0,1]$ and set $\eta = 1$ outside that proposition.

E.1. Events and their probabilities

Lemma 8 (Good events) *Fix a policy π on any instance of the template. Let $\mathcal{E}_{\mathrm{conf}}(\pi)$ be the event that $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \leq \beta$ for every $t \in [T]$, where (V_t, c_t) are formed from all of π's data before t. Let $\mathcal{E}_{\mathrm{noise}}(\pi)$ be the event that for every direction $j \in \{2, \dots, d\}$ and every $m \in [T]$, the sum $S_j^{(m)}$ of the noises on the first m pulls of b in direction e_j by π satisfies $|S_j^{(m)}| \leq \sigma\sqrt{2m\ell_1}$. Then $\mathbb{P}(\mathcal{E}_{\mathrm{conf}}(\pi)) \geq 1-\delta$ and $\mathbb{P}(\mathcal{E}_{\mathrm{noise}}(\pi)) \geq 1-\delta$. Moreover, under (6), $\beta \geq 4\sigma\sqrt{2\ell_1}$.*

Proof All arm vectors of the template have Euclidean norm at most L: $\|g_\eta\| = \eta \leq 1$, $\|b_t\| = L$, $\|p_{t,\eta}\|^2 = \alpha^2 + \gamma^2 \leq 2 \leq L^2$. Also $\|\theta^\star\| = 1$. By Theorem 2 of Abbasi-Yadkori et al. (2011), with probability at least $1-\delta$, simultaneously for all t, $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \leq \sigma\sqrt{2\log\big(\det(V_t)^{1/2}\det(\lambda I)^{-1/2}/\delta\big)} + \sqrt{\lambda}$. Since $V_t \preceq (\lambda + tL^2)I$, $\det(V_t)^{1/2}\det(\lambda I)^{-1/2} \leq (1+tL^2/\lambda)^{d/2}$, so the radius is at most $\sigma\sqrt{d\log(1+TL^2/\lambda) + 2\log(1/\delta)} + \sqrt{\lambda} \leq \sigma\sqrt{d\,\ell_0} + \sqrt{\lambda} \leq 2\sqrt{\lambda} = \beta$, using $\lambda \geq 1$ and $\lambda \geq \sigma^2 d\ell_0$. This proves the first claim. The theorem of Abbasi-Yadkori et al. (2011) only requires the noise to be conditionally sub-Gaussian given the past and the current action, which holds in the reward-table model, and it applies to the data of any policy, not only to LinUCB's own.

For the second claim, in the reward-table model the noises on successive pulls of b in direction e_j form an i.i.d. σ-sub-Gaussian sequence $(\eta_{j,i})_{i \geq 1}$ regardless of the policy, and $S_j^{(m)} = \sum_{i \leq m} \eta_{j,i}$. By Hoeffding's inequality for sub-Gaussian sums, $\mathbb{P}(|S_j^{(m)}| > \sigma\sqrt{2m\ell_1}) \leq 2e^{-\ell_1} = \delta/(dT)$. A union bound over $d-1 < d$ directions and T values of m gives probability at most δ for the complement. Finally $\beta = 2\sqrt{\lambda} \geq 2\sqrt{8\sigma^2\ell_1} = 4\sigma\sqrt{2\ell_1}$. ■

Lemma 9 (Confidence consequences) *On $\mathcal{E}_{\mathrm{conf}}(\pi)$, for every t and every $x \in \mathbb{R}^d$, $|\langle \widehat{\theta}_t - \theta^\star, x\rangle| \leq \beta\|x\|_{V_t^{-1}}$. Hence the index $U_t(x) := \langle \widehat{\theta}_t, x\rangle + \beta\|x\|_{V_t^{-1}}$ satisfies $\mu(x) \leq U_t(x) \leq \mu(x) + 2\beta\|x\|_{V_t^{-1}}$, where $\mu(x) = \langle \theta^\star, x\rangle$.*

Proof Cauchy–Schwarz in the V_t inner product: $|\langle \widehat{\theta}_t - \theta^\star, x\rangle| \le \|\widehat{\theta}_t - \theta^\star\|_{V_t}\|x\|_{V_t^{-1}} \le \beta\|x\|_{V_t^{-1}}$. ■

E.2. The state of LinUCB on the template

Every action vector of the template is g_η, some b in direction e_j, or some p_η in direction e_j, and the probe is the only one that couples two coordinates. The design matrix therefore has an arrowhead structure whose inverse is explicit.

Lemma 10 (Arrowhead inverse) *Let $V = \lambda I + Se_1e_1^\top + \sum_{j=2}^{d}\big(M_je_je_j^\top + n_jq_jq_j^\top\big)$ with $S, M_j, n_j \ge 0$ and $q_j = \alpha e_1 + \gamma e_j$. Put*

$$D_j := \lambda + M_j + n_j\gamma^2, \qquad u_j := \alpha\gamma n_j,$$

$$Q := \lambda + S + \alpha^2\sum_{j\ge2} n_j - \sum_{j\ge2}\frac{u_j^2}{D_j} = \lambda + S + \alpha^2\sum_{j\ge2}\frac{n_j(\lambda+M_j)}{D_j}.$$

Then $Q \ge \lambda$ and

$$(V^{-1})_{11} = \frac{1}{Q}, \qquad (V^{-1})_{1j} = -\frac{u_j}{D_jQ},$$

$$(V^{-1})_{jj} = \frac{1}{D_j} + \frac{u_j^2}{D_j^2Q}, \qquad (V^{-1})_{jj'} = \frac{u_ju_{j'}}{D_jD_{j'}Q}\ \ (j \ne j'),$$

for $j, j' \ge 2$. In particular, if $n_j = 0$ then $(V^{-1})_{jj} = 1/D_j$ and $(V^{-1})_{1j} = (V^{-1})_{jj'} = 0$ for all $j' \ne j$; and if $n_j \ge 1$ and $M_j \ge 1$ then $(V^{-1})_{jj} \le 1/D_j + 1/M_j$.

Proof Write $V = \begin{pmatrix} a & u^\top \\ u & D \end{pmatrix}$ with $a = \lambda + S + \alpha^2\sum_j n_j$, $u = (u_2,\dots,u_d)^\top$, and $D = \mathrm{diag}(D_2,\dots,D_d)$; this is the matrix in the statement because $q_jq_j^\top = \alpha^2e_1e_1^\top + \alpha\gamma(e_1e_j^\top + e_je_1^\top) + \gamma^2e_je_j^\top$. The Schur complement of D is $a - u^\top D^{-1}u = Q$, and the block inverse formula gives $(V^{-1})_{11} = Q^{-1}$, $(V^{-1})_{1,\cdot} = -Q^{-1}u^\top D^{-1}$, and $(V^{-1})_{\cdot,\cdot} = D^{-1} + D^{-1}uu^\top D^{-1}Q^{-1}$ on the lower block, which are the displayed entries. The second expression for Q follows from $\alpha^2n_j - u_j^2/D_j = \alpha^2n_j(D_j - \gamma^2n_j)/D_j = \alpha^2n_j(\lambda+M_j)/D_j$, and it shows $Q \ge \lambda$. If $n_j = 0$ then $u_j = 0$ and the special case follows. For the last claim, $Q \ge \alpha^2n_j(\lambda+M_j)/D_j \ge \alpha^2n_jM_j/D_j$, so $u_j^2/(D_j^2Q) \le \alpha^2\gamma^2n_j^2D_j/(D_j^2\alpha^2n_jM_j) = \gamma^2n_j/(D_jM_j) \le 1/M_j$, using $\gamma^2n_j \le D_j$. ■

Lemma 11 (State of a policy on the template) *Let π be any policy on an instance of the template and t a round. Then V_t is the matrix of Lemma 10 with $S = \eta^2 s$, $M_j = m_jL^2$, n_j the probe counts, and $q_j = p_\eta$'s direction vector. For a direction $j \ge 2$ with $n_j = 0$:*

(a) $\|e_j\|^2_{V_t^{-1}} = 1/(\lambda + m_jL^2)$ and $\widehat{\theta}_{t,j} = LS_j^{(m_j)}/(\lambda + m_jL^2)$, where $S_j^{(m_j)}$ is the sum of the noises on the m_j pulls of b in direction e_j;

(b) $\|g_\eta\|^2_{V_t^{-1}} = \eta^2/Q \le 1/\lambda$ and $\|p_\eta\|^2_{V_t^{-1}} = \alpha^2/Q + \gamma^2/(\lambda + m_jL^2) \le 2/\lambda$ for the probe in direction e_j.

For any direction with $m_j \geq 1$, $\|e_j\|^2_{V_t^{-1}} \leq 2/(m_j L^2)$.

Proof The form of V_t is immediate from the definitions. For (a), Lemma 10 with $n_j = 0$ gives $(V_t^{-1})_{jj} = 1/D_j = 1/(\lambda + m_j L^2)$ and that the j-th row of V_t^{-1} vanishes off the diagonal. The j-th coordinate of the response vector c_t is $L\sum r$ over the pulls of b in direction e_j plus $\gamma\sum r$ over the pulls of p in that direction; the second sum is empty, and since $\mu(b) = 0$ the first is $LS_j^{(m_j)}$. Hence $\widehat{\theta}_{t,j} = (V_t^{-1})_{jj} L S_j^{(m_j)}$. For (b), $\|g_\eta\|^2_{V_t^{-1}} = \eta^2 (V_t^{-1})_{11} = \eta^2/Q$ and $\|p_\eta\|^2_{V_t^{-1}} = \alpha^2 (V_t^{-1})_{11} + 2\alpha\gamma (V_t^{-1})_{1j} + \gamma^2 (V_t^{-1})_{jj} = \alpha^2/Q + \gamma^2/D_j$, using $(V_t^{-1})_{1j} = 0$; both bounds use $Q \geq \lambda$, $D_j \geq \lambda$, $\alpha \leq 1$, $\gamma \leq 1$, $\eta \leq 1$. The last claim is the last claim of Lemma 10 with $M_j = m_j L^2 \geq L^2 \geq 1$ together with $1/D_j \leq 1/(m_j L^2)$, and it holds trivially when $n_j = 0$. ■

E.3. Three reusable facts

Lemma 12 (Optimism trap) *Let π be a policy on an instance of the template and suppose LinUCB acts for π at round t. On $\mathcal{E}_{\mathrm{conf}}(\pi) \cap \mathcal{E}_{\mathrm{noise}}(\pi)$, if the direction $j = j(t)$ satisfies $n_j(t) = 0$ and $m_j(t) < m_0$, then*

$$U_t(b_t) \;\geq\; \tfrac{3}{4}\,\frac{\beta L}{\sqrt{\lambda + m_j(t) L^2}} \;>\; \max\{U_t(g_\eta), U_t(p_{t,\eta})\},$$

where the probe term is omitted if the probe is not offered at t. In particular LinUCB plays b_t.

Proof Write $m = m_j(t)$ and $D = \lambda + mL^2$. By Lemma 11(a), $U_t(b_t) = L\widehat{\theta}_{t,j} + \beta L \|e_j\|_{V_t^{-1}} = L^2 S_j^{(m)}/D + \beta L/\sqrt{D}$. On $\mathcal{E}_{\mathrm{noise}}$, $|L^2 S_j^{(m)}/D| \leq L^2 \sigma \sqrt{2m\ell_1}/D \leq L\sigma\sqrt{2\ell_1}/\sqrt{D}$, since $L\sqrt{m} \leq \sqrt{D}$; by Lemma 8 this is at most $(\beta/4) L/\sqrt{D}$. This gives the first inequality. For the second, by Lemma 9 and Lemma 11(b), $U_t(g_\eta) \leq \eta + 2\beta\eta/\sqrt{Q} \leq 1 + 2\beta/\sqrt{\lambda}$ and $U_t(p_{t,\eta}) \leq (1-\varepsilon)\eta + 2\beta\sqrt{2/\lambda} \leq 1 + 2\sqrt{2}\,\beta/\sqrt{\lambda}$. Since $\beta = 2\sqrt{\lambda}$, $1 \leq \beta/(2\sqrt{\lambda})$, so both indices are at most $(\tfrac{1}{2} + 2\sqrt{2})\beta/\sqrt{\lambda}$. It remains to check $\tfrac{3}{4} L/\sqrt{D} > (\tfrac{1}{2} + 2\sqrt{2})/\sqrt{\lambda}$, i.e. $D < \lambda L^2 \,(3/4)^2/(\tfrac{1}{2} + 2\sqrt{2})^2$. Numerically $(3/4)^2/(\tfrac{1}{2} + 2\sqrt{2})^2 = 0.5625/11.0784 > 0.0507$. Since $m \leq m_0 - 1 < \lambda/30$, $D < \lambda + \lambda L^2/30 = \lambda L^2 (1/L^2 + 1/30) \leq \lambda L^2 (1/64 + 1/30) < 0.0491\,\lambda L^2$, using $L \geq 8$. The inequality holds with room to spare. ■

Lemma 13 (Bad-pull cap) *Let π be a policy on an instance of the template. On $\mathcal{E}_{\mathrm{conf}}(\pi)$, at any round t at which LinUCB acts for π and plays b_t with direction j, $m_j(t) \leq 8\beta^2/\eta^2$; if moreover $n_j(t) = 0$, then $m_j(t) \leq 4\beta^2/\eta^2$. Consequently the number of rounds at which LinUCB acting for π plays b in direction e_j is at most $8\beta^2/\eta^2 + 1$, and at most $4\beta^2/\eta^2 + 1$ if no probe in direction e_j is ever played by π.*

Proof By Lemma 1, a bad arm with gap η is played only if $\eta \leq 2\beta L \|e_j\|_{V_t^{-1}}$, i.e. $\|e_j\|^2_{V_t^{-1}} \geq \eta^2/(4\beta^2 L^2)$. If $m_j(t) \geq 1$, Lemma 11 gives $\|e_j\|^2_{V_t^{-1}} \leq 2/(m_j(t) L^2)$, so $m_j(t) \leq 8\beta^2/\eta^2$; with $n_j(t) = 0$ the sharper bound $\|e_j\|^2_{V_t^{-1}} = 1/(\lambda + m_j(t) L^2) \leq 1/(m_j(t) L^2)$ gives $m_j(t) \leq 4\beta^2/\eta^2$. Since $m_j(\cdot)$ increases by one at each such round, the number of rounds at which b in direction e_j is played with $m_j(t) \leq M$ is at most $M + 1$. ■

Lemma 14 (Expert routing) *Let $\varepsilon L < \gamma$ and let E_w be the frozen linear policy with $\widehat{\theta}_{\mathsf{E}} = e_1 + w\sum_{j\ge 2} e_j$ on the instance with $g_\eta, b_t, p_{t,\eta}$.*

(a) If $w \in (\eta/L, (1-\varepsilon)\eta/(L-\gamma))$, a nonempty interval, then E_w plays $p_{t,\eta}$ whenever the probe is offered and b_t otherwise.

(b) If $w \in (\varepsilon\eta/\gamma, \eta/L)$, a nonempty interval, then E_w plays $p_{t,\eta}$ whenever the probe is offered and g_η otherwise.

All three comparisons are strict.

Proof The scores are $\langle\widehat{\theta}_{\mathsf{E}}, g_\eta\rangle = \eta$, $\langle\widehat{\theta}_{\mathsf{E}}, b_t\rangle = wL$, and $\langle\widehat{\theta}_{\mathsf{E}}, p_{t,\eta}\rangle = (1-\varepsilon)\eta + w\gamma$. The probe beats g_η iff $w\gamma > \varepsilon\eta$ and beats b_t iff $w < (1-\varepsilon)\eta/(L-\gamma)$; b_t beats g_η iff $w > \eta/L$. The interval in (a) is nonempty iff $(1-\varepsilon)\eta L > \eta(L-\gamma)$, i.e. $\gamma > \varepsilon L$, and every w in it satisfies $w > \eta/L > \varepsilon\eta/\gamma$ because $\gamma > \varepsilon L$. The interval in (b) is nonempty iff $\varepsilon\eta/\gamma < \eta/L$, again $\gamma > \varepsilon L$; for w in it, $w < \eta/L < (1-\varepsilon)\eta/(L-\gamma)$. ■

Lemma 15 (Widths at handoff after n probes in each direction) *Let $V = \lambda I + n\sum_{j=2}^{K+1} q_j q_j^\top$ with $q_j = \alpha e_1 + \gamma e_j$, $\alpha > 0$, $n \ge 1$. Then for every $j \in \{2, \dots, K+1\}$,*

$$\|e_j\|_{V^{-1}}^2 \;\le\; \frac{1}{n\gamma^2} + \frac{1}{K\lambda}, \qquad \textit{and} \qquad Q = \lambda + \frac{\alpha^2 K n\lambda}{\lambda + n\gamma^2} \;\ge\; \lambda\Big(1 + \frac{\alpha^2 K}{2\gamma^2}\Big) \;\textit{if}\; n\gamma^2 \ge \lambda.$$

Proof Apply Lemma 10 with $S = 0$, $M_j = 0$, $n_j = n$: $D_j = D := \lambda + n\gamma^2$, $u_j = \alpha\gamma n$, and $Q = \lambda + \alpha^2 K n\lambda/D$. Then $(V^{-1})_{jj} = 1/D + (\alpha\gamma n)^2/(D^2 Q)$. The first term is at most $1/(n\gamma^2)$. For the second, $Q \ge \alpha^2 K n\lambda/D$, so $(\alpha\gamma n)^2/(D^2 Q) \le (\alpha\gamma n)^2 D/(D^2\alpha^2 K n\lambda) = \gamma^2 n/(DK\lambda) \le 1/(K\lambda)$, using $\gamma^2 n \le D$. The bound on Q follows from $D \le 2n\gamma^2$ when $n\gamma^2 \ge \lambda$. ■

E.4. Proof of Lemma 2

Fix u. At a round $t \in I$ with $\langle x_{t,\pi_t}, u\rangle = 0$ the contribution to the coverage is zero. At a round with $\langle x_{t,\pi_t}, u\rangle \ne 0$, the definition (2) gives $\Delta_{t,\pi_t}/\langle x_{t,\pi_t}, u\rangle^2 \ge \kappa_t(u) \ge \kappa(u)$, i.e. $\langle x_{t,\pi_t}, u\rangle^2 \le \Delta_{t,\pi_t}/\kappa(u)$ (if $\kappa(u) = 0$ the bound is vacuous). Summing over I gives the first claim. For the second, let u be a unit vector and $V = V_{\tau+1} = \lambda I + M_\tau$. By Cauchy–Schwarz in the V inner product, $1 = (u^\top u)^2 = (u^\top V^{1/2} V^{-1/2} u)^2 \le (u^\top V u)(u^\top V^{-1} u)$, so $\|u\|_{V^{-1}}^2 \ge 1/(\lambda + u^\top M_\tau u)$, and $u^\top M_\tau u = \sum_{t\le\tau}\langle x_{t,A_t}, u\rangle^2 \le R_{[1,\tau]}(\mathsf{E})/\kappa(u)$ by the first claim applied to the expert on $[1,\tau]$. ■

E.5. Proof of Theorem 3

We work on $\mathcal{E}_{\mathrm{conf}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{noise}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{conf}}(\mathsf{H}_{\tau_0})$, which has probability at least $1 - 3\delta$ by Lemma 8. Set $\eta = 1$.

(i) The expert. By Lemma 14(a), E_w plays p_t at every $t \le \tau_0$ (gap ε) and b_t at every $t > \tau_0$ (gap 1), so $R_T(\mathsf{E}) = \varepsilon Kn + (T - Kn)$.

(ii) Cold LinUCB, lower bound. We claim that at every round $t \in [T]$ with $m_{j(t)}(t) < m_0$, cold LinUCB plays b_t. We argue by induction on t. Suppose the claim holds for all rounds before t and let $j = j(t)$ with $m_j(t) < m_0$. If cold LinUCB had played the probe in direction e_j at some earlier round t' (so $j(t') = j$), then $m_j(t') \le m_j(t) < m_0$ and the claim at t' says it played $b_{t'}$, a contradiction; hence $n_j(t) = 0$. Lemma 12 now gives that LinUCB plays b_t. (The base case $t = 1$ is the same argument with no earlier rounds.) Direction e_j is offered at exactly n rounds in $[1, Kn]$, and at each of the first $\min\{n, m_0\}$ of them $m_j < m_0$, so cold LinUCB plays the bad arm, paying 1 each time. With $n \ge m_0$ this gives $R_{[1,Kn]}(\mathsf{C}) \ge Km_0$, and $A_{\tau_0} = R_{[1,Kn]}(\mathsf{C}) - R_{[1,Kn]}(\mathsf{E}) \ge Km_0 - \varepsilon Kn$ by (i).

(iii) Cold LinUCB, upper bound. Cold LinUCB acts at every round. By Lemma 13 it plays b in each of the K directions at most $8\beta^2 + 1$ times over the horizon, each costing 1. Probes exist only at rounds $t \le Kn$ and cost ε each, so probe pulls cost at most εKn. Pulls of g cost nothing. Hence $R_T(\mathsf{C}) \le K(8\beta^2 + 1) + \varepsilon Kn$.

(iv) The hybrid. During $t \le \tau_0$ the hybrid plays the expert's actions, which by (i) are the probes, each direction exactly n times. Hence $V_{\tau_0+1} = \lambda I + n\sum_{j=2}^{K+1} p^{(j)}p^{(j)\top}$ with $p^{(j)} = \alpha e_1 + \gamma e_j$, and Lemma 15 gives, for every bad direction, $\|e_j\|^2_{V_{\tau_0+1}^{-1}} \le 1/(n\gamma^2) + 1/(K\lambda)$. With $n\gamma^2 \ge 36\lambda L^2 = 9\beta^2 L^2$ and $K\lambda \ge 36\lambda L^2 = 9\beta^2 L^2$ this is at most $2/(9\beta^2 L^2)$, so $2\beta L\|e_j\|_{V_{\tau_0+1}^{-1}} \le 2\sqrt{2}/3 < 1$. Now fix $t > \tau_0$. The hybrid's design matrix satisfies $V_t \succeq V_{\tau_0+1}$, so by Lemma 3 $\|e_{j(t)}\|_{V_t^{-1}} \le \|e_{j(t)}\|_{V_{\tau_0+1}^{-1}}$, and by Lemma 9, $U_t(b_t) \le 0 + 2\beta L\|e_{j(t)}\|_{V_t^{-1}} < 1 \le U_t(g)$. Since $\mathcal{A}_t = \{g, b_t\}$, LinUCB plays g, with gap 0. Therefore $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) = 0$, $R_T(\mathsf{H}_{\tau_0}) = R_{[1,\tau_0]}(\mathsf{E}) = \varepsilon Kn$, $I_{\tau_0} = R_{[Kn+1,T]}(\mathsf{C}) - 0 \ge 0$, and $B_{\tau_0} = (T - Kn) - 0$.

The ratios. From (ii), (iii), (iv): $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \ge Km_0/(\varepsilon Kn) = m_0/(\varepsilon n)$ and $\le (K(8\beta^2 + 1) + \varepsilon Kn)/(\varepsilon Kn) = 1 + (8\beta^2+1)/(\varepsilon n)$; the expert ratio is (i) over (iv). At $n = \lceil 36\lambda L^2/\gamma^2 \rceil$, since $36\lambda L^2/\gamma^2 \ge 36 \cdot 60 \cdot 64 > 1$ we have $n \le 37\lambda L^2/\gamma^2$, and $m_0 \ge \lambda/30 - 1 \ge \lambda/60$ because $\lambda \ge 60$; hence $m_0/(\varepsilon n) \ge (\lambda/60)\gamma^2/(37\varepsilon\lambda L^2) = \gamma^2/(2220\,\varepsilon L^2)$. For the upper bound, $8\beta^2 + 1 = 32\lambda + 1$ and $n \ge 36\lambda L^2/\gamma^2$ give $(8\beta^2 + 1)/(\varepsilon n) \le (32\lambda + 1)\gamma^2/(36\varepsilon\lambda L^2) \le \gamma^2/(\varepsilon L^2)$ since $\lambda \ge 1$. ■

E.6. Proof of Corollary 4

Let $\varepsilon \ge \gamma^2/L^2$ and $n \ge 36\lambda L^2/\gamma^2$, so that $\varepsilon n \ge (\gamma^2/L^2)(36\lambda L^2/\gamma^2) = 36\lambda$. With $\beta = 2\sqrt{\lambda}$ from (6) we have $8\beta^2 + 1 = 32\lambda + 1$, hence

$$\frac{8\beta^2 + 1}{\varepsilon n} \le \frac{32\lambda + 1}{36\lambda} = \frac{8}{9} + \frac{1}{36\lambda} \le \frac{8}{9} + \frac{1}{2160} < 1,$$

using $\lambda \ge 60$. The upper bound of (7) then gives $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \le 1 + (8\beta^2 + 1)/(\varepsilon n) < 2$. ■

E.7. Proof of Proposition 9

Events. All arm vectors of $\mathcal{J}$ have norm at most L: $\|b'_{j'}\|_2^2 = 1 + L'^2 = 145 \le L^2$ since $L \ge 13$, and the others are as in Lemma 8. Let $\mathcal{E}'_{\text{conf}}(\mathsf{C})$ be the event that $\|\widehat{\theta}_t - \theta^\star\|_{V_t} \le \frac{5}{4}\sqrt{\lambda}$ for every t on cold LinUCB's trajectory. By Theorem 2 of Abbasi-Yadkori et al. (2011) and the computation in the proof of Lemma 8, the self-normalized radius is at most $\sigma\sqrt{d\ell_0} + \sqrt{\lambda} \le \frac{1}{4}\sqrt{\lambda} + \sqrt{\lambda}$ under $\lambda \ge 16\sigma^2 d\ell_0$, so $\mathbb{P}(\mathcal{E}'_{\text{conf}}(\mathsf{C})) \ge 1 - \delta$, and $\mathcal{E}'_{\text{conf}}(\mathsf{C}) \subseteq \mathcal{E}_{\text{conf}}(\mathsf{C})$ since $\frac{5}{4}\sqrt{\lambda} \le \beta = 2\sqrt{\lambda}$. We

work on $\mathcal{E}'_{\mathrm{conf}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{noise}}(\mathsf{C}) \cap \mathcal{E}_{\mathrm{conf}}(\mathsf{H}_{\tau_0})$, of probability at least $1-3\delta$ (Lemma 8; the noise event concerns the pulls of b in the directions e_j, $j \in J$, and its proof is unchanged). On $\mathcal{E}'_{\mathrm{conf}}(\mathsf{C})$, Cauchy–Schwarz gives $|\langle \widehat{\theta}_t - \theta^\star, x\rangle| \le \frac{5}{4}\sqrt{\lambda}\|x\|_{V_t^{-1}} = \frac{5}{8}\beta\|x\|_{V_t^{-1}}$ for every x, hence for the index $U_t(x) = \langle \widehat{\theta}_t, x\rangle + \beta\|x\|_{V_t^{-1}}$,

$$\mu(x) + \tfrac{3}{8}\beta\|x\|_{V_t^{-1}} \;\le\; U_t(x) \;\le\; \mu(x) + \tfrac{13}{8}\beta\|x\|_{V_t^{-1}}. \tag{9}$$

(i) The expert. The expert plays $p_{j(t)}$ (gap ε) at every $t \le \tau_0$ and $b_{j(t)}$ (gap 1) at every $t > \tau_0$, so $R_T(\mathsf{E}) = \varepsilon K n + (T - Kn)$.

(ii) Cold LinUCB in the window. We show by induction on $t \le \tau_0$ that cold LinUCB plays $b'_{j'(t)}$ at round t. Suppose this holds before t; then no probe and no bad arm has been played, so for every $j \in J$ no played vector has a component along e_j, $V_t e_j = \lambda e_j$, and $V_t^{-1} e_j = e_j/\lambda$. Three width bounds follow. First, $\|g\|^2_{V_t^{-1}} = (V_t^{-1})_{11} \le 1/\lambda$ because $V_t \succeq \lambda I$. Second, with $j = j(t)$, $\|p_j\|^2_{V_t^{-1}} = (1-\varepsilon)^2 (V_t^{-1})_{11} + 2(1-\varepsilon)\gamma\, e_1^\top V_t^{-1} e_j + \gamma^2 (V_t^{-1})_{jj} = (1-\varepsilon)^2 (V_t^{-1})_{11} + \gamma^2/\lambda \le (1+\gamma^2)/\lambda \le 2/\lambda$, using $e_1^\top V_t^{-1} e_j = e_1^\top e_j/\lambda = 0$. Third, with $j' = j'(t)$ and m the number of earlier pulls of $b'_{j'}$: the only played vector with a component along $e_{j'}$ is $b'_{j'}$ itself, so $e_{j'}^\top V_t e_{j'} = \lambda + mL'^2$, and Cauchy–Schwarz in the V_t inner product gives $\|b'_{j'}\|^2_{V_t^{-1}} \ge \langle b'_{j'}, e_{j'}\rangle^2 / (e_{j'}^\top V_t e_{j'}) = L'^2/(\lambda + mL'^2)$. Direction j' is offered at $\lceil \tau_0/K' \rceil$ rounds in the window, so $m \le \tau_0/K' \le \lambda/L'^2$ by the condition on K', and $\|b'_{j'}\|^2_{V_t^{-1}} \ge L'^2/(2\lambda)$. Now apply (9) with $\beta = 2\sqrt{\lambda}$:

$$U_t(b'_{j'}) \;\ge\; 1 + \tfrac{3}{8}\cdot 2\sqrt{\lambda}\cdot \frac{L'}{\sqrt{2\lambda}} = 1 + \frac{3L'}{4\sqrt{2}} > 7.3,$$

$$U_t(g) \;\le\; 1 + \tfrac{13}{8}\cdot 2\sqrt{\lambda}\cdot \frac{1}{\sqrt{\lambda}} = 4.25, \qquad U_t(p_j) \;\le\; 1 - \varepsilon + \tfrac{13}{8}\cdot 2\sqrt{\lambda}\cdot \frac{\sqrt{2}}{\sqrt{\lambda}} < 5.6,$$

using $L' = 12$. So cold LinUCB plays $b'_{j'(t)}$, whose gap is 0. The base case $t = 1$ is the same computation with $m = 0$. Hence $R_{[1,\tau_0]}(\mathsf{C}) = 0$, and at round $\tau_0 + 1$ every direction e_j, $j \in J$, is untouched: $m_j = n_j = 0$.

(ii) Cold LinUCB on the tail. For $t > \tau_0$ the action set is $\{g, b_{j(t)}\}$ and the directions e_j, $j \in J$, are touched only by the bad arms, so for each of them e_j is an eigenvector of V_t with eigenvalue $\lambda + m_j(t)L^2$, the j-th coordinate of the response vector is $LS_j^{(m_j(t))}$, and the conclusions of Lemma 11(a) hold verbatim; Lemma 11(b)'s bound $U_t(g) \le 1 + 2\beta/\sqrt{\lambda}$ holds on $\mathcal{E}_{\mathrm{conf}}(\mathsf{C})$ by Lemma 9 and $\|g\|_{V_t^{-1}} \le 1/\sqrt{\lambda}$. These are the only properties of the template that the proofs of Lemmas 12 and 13 use for a direction with $n_j = 0$, so both lemmas apply to cold LinUCB on $\mathcal{J}$ at every tail round (with no probe term). The induction of Theorem 3(ii) now shows that cold LinUCB plays b_t at every tail round with $m_{j(t)}(t) < m_0$; since $T - \tau_0 \ge Km_0$ and the tail directions cycle, each direction is offered at least m_0 times on the tail, giving $R_{[\tau_0+1,T]}(\mathsf{C}) \ge Km_0$. Lemma 13 with $n_j = 0$ bounds the pulls of b in each direction by $4\beta^2 + 1$, and g costs nothing, so $R_{[\tau_0+1,T]}(\mathsf{C}) \le K(4\beta^2 + 1) = K(16\lambda + 1)$.

(iii) The hybrid. During $t \le \tau_0$ the hybrid plays the expert's actions, which by (i) are the probes, each direction $j \in J$ exactly n times; no distractor and no bad arm is played. Hence $V_{\tau_0+1} = \lambda I + n\sum_{j\in J} p_j p_j^\top$, which is the matrix of Theorem 3(iv), and the argument there (Lemma 15,

Lemma 3, Lemma 9 on $\mathcal{E}_{\text{conf}}(\mathsf{H}_{\tau_0})$) shows that $U_t(b_t) < 1 \le U_t(g)$ at every $t > \tau_0$, so the hybrid plays g and $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) = 0$. Therefore $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$.

The terms and the ratios. $A_{\tau_0} = R_{[1,\tau_0]}(\mathsf{C}) - R_{[1,\tau_0]}(\mathsf{E}) = 0 - \varepsilon Kn$; $I_{\tau_0} = R_{[\tau_0+1,T]}(\mathsf{C}) - 0 \in [Km_0, K(16\lambda+1)]$; $B_{\tau_0} = (T - Kn) - 0$. Since $R_T(\mathsf{C}) = R_{[\tau_0+1,T]}(\mathsf{C})$ and $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$, the ratio lies between $m_0/(\varepsilon n)$ and $(16\lambda+1)/(\varepsilon n)$. At $n = \lceil 36\lambda L^2/\gamma^2 \rceil$ the lower bound is at least $\gamma^2/(2220\,\varepsilon L^2)$ exactly as in the proof of Theorem 3, and $(16\lambda+1)/(\varepsilon n) \le (16\lambda+1)\gamma^2/(36\varepsilon\lambda L^2) \le \gamma^2/(2\varepsilon L^2)$ since $\lambda \ge 1$. ■

E.8. Proof of Proposition 8

Work on $\mathcal{E}_{\text{conf}}(\mathsf{C}) \cap \mathcal{E}_{\text{noise}}(\mathsf{C})$ (probability $\ge 1 - 2\delta$). There are no probes, so $n_j \equiv 0$ for every policy.

(i) Let $t \le Km_0$ and $j = j(t)$. Direction e_j was offered at $\lceil t/K \rceil - 1 \le m_0 - 1$ earlier rounds, so $m_j(t) \le m_0 - 1 < m_0$ for cold LinUCB, and Lemma 12 (with the probe term absent) says cold LinUCB plays b_t. Hence cold LinUCB plays b_t at every $t \le Km_0$, and so does E. Fix $\tau \le Km_0$. The two policies C and H_τ play the same action vectors in the same order during $[1,\tau]$; in the reward-table model they therefore observe the same rewards, so their states $(V_{\tau+1}, c_{\tau+1})$ coincide. From round $\tau+1$ on both run LinUCB from that state on the same action sets and the same reward table, so they play the same actions at every round. Thus $R_I(\mathsf{C}) = R_I(\mathsf{H}_\tau)$ for every interval I, $A_\tau = R_{[1,\tau]}(\mathsf{C}) - R_{[1,\tau]}(\mathsf{E}) = \tau - \tau = 0$, and $I_\tau = 0$.

(ii) For any τ, $R_T(\mathsf{H}_\tau) \ge R_{[1,\tau]}(\mathsf{E}) = \tau$. By Lemma 13 with $n_j \equiv 0$, cold LinUCB plays each bad direction at most $4\beta^2 + 1$ times, so $R_T(\mathsf{C}) \le K(4\beta^2+1) = K(16\lambda+1)$. For $\tau > Km_0$, $R_T(\mathsf{C})/R_T(\mathsf{H}_\tau) \le K(16\lambda+1)/(Km_0) \le (16\lambda+1)/(\lambda/60) = 960 + 60/\lambda \le 961$. ■

E.9. Proof of Proposition 10

Work on the event of Theorem 3. By Lemma 14(b), E_w plays every probe during $[1,\tau_0]$ and g afterwards, so $R_T(\mathsf{E}) = \varepsilon Kn$. The hybrid's routed rounds are the same probes as in Theorem 3(iv), so its state at handoff and its subsequent play are identical to those established there: $R_T(\mathsf{H}_{\tau_0}) = \varepsilon Kn$ and $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) = 0$. Hence $B_{\tau_0} = 0 - 0 = 0$. Cold LinUCB does not depend on the expert, so A_{τ_0} and I_{τ_0} are as in Theorem 3. ■

E.10. Proof of Proposition 11

Work on $\mathcal{E}_{\text{conf}}(\mathsf{C}) \cap \mathcal{E}_{\text{conf}}(\mathsf{H}_{\tau_0}) \cap \mathcal{E}_{\text{noise}}(\mathsf{H}_{\tau_0})$ (probability $\ge 1 - 3\delta$). Here $d = 2K+1$; coordinates $2, \dots, K+1$ are the bad directions and $K+2, \dots, 2K+1$ the nuisance directions. The probe p'_t couples e_1 with a nuisance coordinate only, so Lemma 10 applies with the nuisance coordinates playing the role of the probed j's and the bad coordinates having $n_j = 0$ for every policy at every round.

Cold LinUCB. Since $n_j \equiv 0$ on bad coordinates, Lemma 13 gives at most $4\beta^2 + 1$ bad pulls in each of the K directions, costing 1 each; probe pulls occur only at $t \le Kn$ and cost ε each; so $R_T(\mathsf{C}) \le K(4\beta^2+1) + \varepsilon Kn$.

The hybrid. During $[1,\tau_0]$ it plays the Kn probes, paying εKn. At handoff, $V_{\tau_0+1} = \lambda I + n\sum_i q_i q_i^\top$ over the K nuisance directions with $q_i = \alpha e_1 + \gamma e_{K+1+i}$, and the bad coordinates are untouched: by Lemma 10 (with $M_j = n_j = 0$ on them), $\|e_j\|^2_{V_{\tau_0+1}^{-1}} = 1/\lambda$ for every bad direction and the estimate has zero j-th coordinate. By Lemma 15 applied to the nuisance block,

$Q \geq \lambda(1+\alpha^2 K/(2\gamma^2))$ once $n\gamma^2 \geq \lambda$, which holds since $n \geq 36\lambda L^2/\gamma^2$. As $\alpha = 1-\varepsilon \geq 1-1/L \geq 7/8$ and $\gamma \leq 1$, $Q \geq \lambda(1+49K/128) \geq 16\lambda = 4\beta^2$ because $K \geq 36L^2 \geq 2304$. Hence at every $t > \tau_0$, since Q only grows, $U_t(g) \leq 1 + 2\beta/\sqrt{Q} \leq 2$ by Lemma 9. On the other hand, for $t > \tau_0$ and $j = j(t)$ with $m_j(t) = m$, Lemma 11(a) and $\mathcal{E}_{\text{noise}}(\mathsf{H}_{\tau_0})$ give, exactly as in the proof of Lemma 12, $U_t(b_t) \geq \frac{3}{4}\beta L/\sqrt{\lambda + mL^2}$, which exceeds 2 as soon as $\lambda + mL^2 < \frac{9}{64}\beta^2 L^2 = \frac{9}{16}\lambda L^2$, i.e. $m < \frac{9}{16}\lambda - \lambda/L^2$. With $L \geq 8$, $\frac{9}{16}\lambda - \lambda/L^2 \geq \frac{35}{64}\lambda > \lambda/2$, so this holds whenever $m \leq \lfloor \lambda/2 \rfloor - 1$. Thus at every round $t > \tau_0$ at which $m_{j(t)}(t) < \lfloor \lambda/2 \rfloor$ the hybrid plays the bad arm. Each direction is offered at least $\lfloor (T-\tau_0)/K \rfloor \geq \lfloor \lambda/2 \rfloor$ times after τ_0, so each is pulled at least $\lfloor \lambda/2 \rfloor$ times, and $R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) \geq K\lfloor \lambda/2 \rfloor$. Hence $R_T(\mathsf{H}_{\tau_0}) \geq \varepsilon K n + K\lfloor \lambda/2 \rfloor$.

The ratio and I_{τ_0}. Since $\lambda \geq 60$, $\lfloor \lambda/2 \rfloor \geq \lambda/4$, and for $x \geq 0$, $(a+x)/(b+x) \leq \max\{a/b, 1\}$; with $a = K(16\lambda+1)$, $b = K\lambda/4$, $x = \varepsilon K n$ this gives $R_T(\mathsf{C})/R_T(\mathsf{H}_{\tau_0}) \leq 64 + 4/\lambda \leq 65$. Finally $I_{\tau_0} = R_{[\tau_0+1,T]}(\mathsf{C}) - R_{[\tau_0+1,T]}(\mathsf{H}_{\tau_0}) \leq R_{[\tau_0+1,T]}(\mathsf{C}) - K\lfloor \lambda/2 \rfloor$. The expert's Gram matrix $n\sum_i q_i q_i^\top$ has range in $\operatorname{span}\{e_1, e_{K+2}, \ldots, e_{2K+1}\}$, which is orthogonal to the span of the bad directions, so the principal angle is $90°$. ∎

E.11. Proof of Proposition 12

On $\mathcal{E}_{\text{conf}}(\mathsf{C})$, Lemma 13 with $n_j \equiv 0$ gives $R_T(\mathsf{C}) \leq K(4\beta^2+1) = K(16\lambda+1)$; since $R_T(\pi) \geq 0$ for every π, $(R_T(\mathsf{C}) - R_T(\pi))/T \leq K(16\lambda+1)/T$, and with $\lambda = \sigma^2 d\ell_0$ and $d = K+1$ this is $O(\sigma^2 K^2 \ell_0/T)$. ∎

E.12. Proof of Proposition 13

The proof of Theorem 3 goes through with η carried along; we indicate the changes. Work on the same three events. (i) By Lemma 14(a), $R_T(\mathsf{E}) = \varepsilon\eta K n + \eta(T - Kn)$. (ii) Lemma 12 was proved for general η, and the induction is unchanged; each forced bad pull now costs η, so $R_{[1,Kn]}(\mathsf{C}) \geq \eta K m_0$. (iii) Lemma 13 gives at most $8\beta^2/\eta^2 + 1$ bad pulls in each direction at cost η, and probe pulls cost $\varepsilon\eta$, so $R_T(\mathsf{C}) \leq \eta K(8\beta^2/\eta^2+1) + \varepsilon\eta K n$. (iv) At handoff, Lemma 15 with $\alpha = (1-\varepsilon)\eta > 0$ gives $\|e_j\|^2_{V_{\tau_0+1}^{-1}} \leq 1/(n\gamma^2) + 1/(K\lambda) \leq 2\eta^2/(9\beta^2 L^2)$ under $n\gamma^2 \geq 9\beta^2 L^2/\eta^2$ and $K\lambda \geq 9\beta^2 L^2/\eta^2$, i.e. $n \geq 36\lambda L^2/(\eta^2\gamma^2)$ and $K \geq 36L^2/\eta^2$. Then $U_t(b_t) \leq 2\beta L\|e_j\|_{V_t^{-1}} \leq (2\sqrt{2}/3)\eta < \eta \leq U_t(g_\eta)$ for all $t > \tau_0$ by Lemmas 3 and 9, so the hybrid plays g_η and $R_T(\mathsf{H}_{\tau_0}) = \varepsilon\eta K n$. The ratio bounds follow: the lower bound is $\eta K m_0/(\varepsilon\eta K n) = m_0/(\varepsilon n)$, and at $n = \lceil 36\lambda L^2/(\eta^2\gamma^2) \rceil \leq 37\lambda L^2/(\eta^2\gamma^2)$ this is at least $(\lambda/60)\eta^2\gamma^2/(37\varepsilon\lambda L^2) = \eta^2\gamma^2/(2220\,\varepsilon L^2)$; the upper bound is $1 + (8\beta^2/\eta^2+1)/(\varepsilon n) \leq 1 + (32\lambda+\eta^2)\gamma^2/(36\varepsilon\lambda L^2) \leq 1 + \gamma^2/(\varepsilon L^2)$. ∎

E.13. Proof of Corollary 3

On the event of Proposition 3, the hybrid plays b_t at every $t > \tau$, so $R_{[\tau+1,T]}(\mathsf{H}_\tau) = T - \tau$, while the routed rounds cost nothing because the expert plays the optimal arm; hence $R_T(\mathsf{H}_\tau) = T - \tau$ and $R_T(\mathsf{E}) = 0$. Cold LinUCB also plays b_t at every round by Proposition 3 with $\tau = 0$, so $R_{[1,\tau]}(\mathsf{C}) = \tau$ and $R_{[\tau+1,T]}(\mathsf{C}) = T - \tau$. Therefore $A_\tau = \tau - 0$, $I_\tau = (T-\tau) - (T-\tau) = 0$, and $B_\tau = 0 - (T-\tau)$. ∎

E.14. Remarks on the constants

The constants 36, 30, 65, 961, 2220 come from three places: the factor $\frac{3}{4}$ by which noise can reduce a bad arm's index below its bonus, the factor $\frac{1}{2}+2\sqrt{2}$ by which the good arm's or the probe's index can exceed its mean at a cold start, and the choice $\beta = 2\sqrt{\lambda}$. None of them is tight. What matters in Theorem 3 is the structure of (7): the lower bound is (forced cold pulls per direction) divided by (probe cost per direction), the upper bound is (maximal cold pulls per direction) divided by the same, and both numerators are independent of ε, γ, and K. The gap between $m_0 \asymp \lambda$ and $8\beta^2 \asymp \lambda$ is a constant factor, and that is why the multiplier is pinned to the price ratio up to constants. The same structure with η gives the bounds in Proposition 13, where the looseness is a factor η^2 rather than a constant because the forced pulls in Lemma 12 are counted before the good arm's index has come down to its mean; a sharper count of cold LinUCB's pulls after the good arm is well estimated, as in the proof of Proposition 11, would remove it, but with probes available to cold LinUCB the required control of the estimate in a coupled coordinate is more delicate and we have not carried it out.

Appendix F. Proof of Theorem 2 and Numerical Checks

Proof of Theorem 2 Every offered arm is a multiple of a coordinate vector, so for every policy V_t is diagonal and each coordinate of $\widehat{\theta}_t$ is computed separately. If before round t the action D has been played n_D times with total reward s_D, and R has been played n_R times with total reward s_R, then

$$\mathrm{UCB}_t(D) = \frac{s_D}{\lambda + n_D} + \frac{\beta}{\sqrt{\lambda + n_D}}, \qquad \mathrm{UCB}_t(R) = \frac{s_R}{\lambda + n_R} + \frac{\beta}{\sqrt{\lambda + n_R}}, \qquad \mathrm{UCB}_t(b_t) = \omega,$$

the last because coordinate $t+2$ has not been touched before round t. Since $L > 1$, $\omega = \beta L/\sqrt{\lambda} > \beta/\sqrt{\lambda + n}$ for every $n \geq 0$. Rewards observed on a fresh arm change only its own coordinate, which never recurs, so they do not affect the indices of D, R or later fresh arms. The window is a fixed sequence of types; the linear model describes the rounds after it, and pseudo-regret is measured against $\theta^\star$ throughout.

(i) On the window, E_A plays D on the k rounds of type D (right) and the k of type R (wrong), and R on the $2k$ of type $\varnothing$ (wrong); E_B plays R on the k rounds of type D (wrong) and the k of type R (right), and D on the $2k$ of type $\varnothing$ (wrong). Each plays D and R exactly $2k$ times, so $M_\tau^A = M_\tau^B = 2k(e_1e_1^\top + e_2e_2^\top)$, each pays $2k(p-q)$ for its plays of R, and each is right k times. The response sums are ke_1 and ke_2, which gives the two estimates.

(ii) Cold LinUCB has $n_D = n_R = 0$, so $\mathrm{UCB}_t(D) = \mathrm{UCB}_t(R) = \beta/\sqrt{\lambda} < \omega$ and it plays b_t; by induction it does so at every round, paying $p - f$ each time. If the routed rounds contribute no rewards, then after τ we have $s_D = s_R = 0$, both indices equal $\beta/\sqrt{\lambda + 2k} < \omega$, and the same induction applies.

(iii) After the window H_τ^B has $n_D = 2k$ and $s_D = 0$, because E_B played D only on rounds of type $\varnothing$. Hence $\mathrm{UCB}_t(D) = \beta/\sqrt{\lambda + 2k} < \omega$, and as long as D is not played these statistics do not change; so D is never played. Every round after τ then costs at least $m := \min\{p-q, p-f\}$, and the window cost $2k(p-q) \geq 2km$, so $R_T(\mathsf{H}_\tau^B) \geq 2km + (T-4k)m = (T-2k)m$.

(iv) Let $X_i := \mathbf{1}\{y_{\tau+i} = D\}$, which are i.i.d. Bernoulli(p) (rounds of types R, F and $\varnothing$ all give D reward 0), $S_n := \sum_{i \leq n} X_i$, and $G := \{k + S_n > \omega(\lambda + 2k + n) \text{ for all } n \geq 0\}$. On G, H_τ^A plays D at every round after τ: if it has done so at rounds $\tau+1, \ldots, \tau+n$, then $n_D = 2k+n$ and $s_D = k + S_n$,

so $\mathrm{UCB}(D) > s_D/(\lambda + n_D) > \omega$; and $n_R = 2k$, $s_R = 0$, so $\mathrm{UCB}(R) = \beta/\sqrt{\lambda + 2k} < \omega$. It therefore plays D at round $\tau + n + 1$. Every round after τ then has gap 0, and $R_T(\mathsf{H}^A_\tau) = 2k(p-q)$.

For the probability, $n = 0$ is in G because $h > 0$. For $n \geq 1$, put $Y_i := \omega - X_i \in [\omega - 1, \omega]$, so that $\mathbb{E}Y_i = \omega - p < 0$ and G fails exactly when $\sum_{i \leq n} Y_i \geq k - \omega(\lambda + 2k) = h$ for some $n \geq 1$. By Hoeffding's lemma, $\mathbb{E}e^{\theta Y_i} \leq \exp(\theta(\omega - p) + \theta^2/8)$, which equals 1 at $\theta = 8(p - \omega) > 0$. So $\exp(\theta \sum_{i \leq n} Y_i)$ is a nonnegative supermartingale starting at 1, and Ville's maximal inequality gives $\mathbb{P}(G^c) \leq e^{-\theta h} = e^{-8(p-\omega)h}$. If also $\omega \leq p/2$ then $8(p - \omega) \geq 4p$, and if $\omega \leq 1/4$ then $h \geq (k - 2\omega\lambda)/2$, so $8(p - \omega)h \geq 2p(k - 2\omega\lambda)$.

(v) Off G, H^A_τ plays D (gap 0) until it first plays a fresh arm; from then on the statistics of D and R no longer change, both indices stay below ω, and it plays fresh arms at cost $p - f$ per round. So $R_T(\mathsf{H}^A_\tau) \leq 2k(p-q) + (p-f)T$ off G, and $\mathbb{E}R_T(\mathsf{H}^A_\tau) \leq 2k(p-q) + (p-f)T\,\mathbb{P}(G^c)$. The stated k gives $h \geq \ln T/(8(p-\omega))$, hence $\mathbb{P}(G^c) \leq 1/T$ and $\mathbb{E}R_T(\mathsf{H}^A_\tau) \leq 2k(p-q) + p - f$, while $R_T(\mathsf{C}) = (p-f)T$ is deterministic. The ratios follow from (ii)–(iv). ■

Remark 8 (Settling, and radius schedules) (a) With a constant radius the theorem also shows the all-or-nothing behaviour seen on AmazonCat (Appendix I): once H^A_τ plays a fresh arm, the statistics of D stop changing and every later fresh arm has index ω, so it never plays D again. It either settles on D for good or never does. (b) Parts (i)–(v) use the radius only through the comparisons above, so they hold for any non-increasing schedule β_t with $\omega := \sup_{t > \tau} \beta_t L/\sqrt{\lambda}$. The all-or-nothing behaviour of (a) needs the radius to be constant once the learner has left D; the schedule of our experiments is constant after $t \approx 1{,}500$.

An expert that sees only the features. Theorem 2 compares two experts that both see the type. A third expert plays D and R equally often, $2k$ times each, but chooses which window rounds get D without seeing the type, so that its right answers come at the natural rate. It has the same coverage and price as E_A and E_B and, on average, the same number of right answers. We have no theorem for it. Numerically, with $f = 0$, the parameters of the first row of Table 3 and 500 seeds, its hybrid spends the whole second half of the horizon on D in 62%, 72% and 84% of runs at $k = 2, 4, 8$, and its mean regret is 576, 412 and 238, against 28, 2 and 4 for E_A, 1,498, 1,495 and 1,491 for E_B, and 1,500 for cold LinUCB. The two informed experts sit on either side of it: selection toward the frequent action helps far more than unselected data, and selection away from it removes the gain.

Numerical checks. The instance is simple enough to simulate exactly. We ran LinUCB with the full $d \times d$ design matrix and, for long horizons, the scalar recursion used in the proof; the two chose the same action at every round of 36 runs with $T \leq 300$. With $f = 0$, $\lambda = 0.1$, $L = 2$ and $T = 5{,}000$, across $k \in \{1, \ldots, 32\}$ and the three settings of (p, q, ω) in Table 3: cold LinUCB's regret equalled pT in each of the 600 runs checked; H^B_τ never played D after round τ in 12,000 runs; a hybrid that left D never returned to it; when either hybrid received the routed covariance without the routed rewards, it never played D after τ (300 runs each); and with a radius large enough that $\beta(L-1)/\sqrt{\lambda} \geq 1$, so $\omega > p$, neither hybrid played D after τ (200 runs), as Remark 2 states. With $f = 0.25 > p - q$ ($p = 0.3$, $q = 0.1$, $\omega = 0.135$, $\lambda = 1$, $L = 2$), so that E_B is right on 35% of later rounds against E_A's 30%, cold LinUCB's regret equalled $(p-f)T$ in every run, H^B_τ never played D after τ, its regret was never below $(T - 2k)\min\{p - q, p - f\}$, and H^A_τ settled on D in 98% of runs at $k = 2$ and in every run at $k \in \{4, 8\}$ ($T \in \{2{,}000, 20{,}000\}$, 100–300 runs per cell). With k chosen as in (v), the mean regret of H^A_τ stayed below $2k(p-q) + p - f$ at $T = 2{,}000$, 20,000 and

200,000, and the ratio of cold LinUCB's regret to it grew from 28 to 227 to 1,923. Table 3 compares the frequency with which H^A_τ settles on D with the bound of Theorem 2(iv).

Table 3: Frequency with which H^A_τ settles on D after the routed window (2,000 seeds per cell, $\lambda = 0.1$, $L = 2$, $T = 5{,}000$) against the lower bound $1 - e^{-8(p-\omega)h}$ of Theorem 2(iv), with $f = 0$ and $h = k(1-2\omega) - \omega\lambda$. The bound holds in every cell.

(p, q, ω)		$k=1$	$k=2$	$k=3$	$k=4$	$k=6$
(0.3, 0.05, 0.135)	observed	0.862	0.979	0.996	1.000	1.000
	bound	0.612	0.852	0.943	0.978	0.997
(0.5, 0.1, 0.225)	observed	0.865	0.983	0.996	0.999	1.000
	bound	0.687	0.907	0.972	0.992	0.999
(0.2, 0, 0.09)	observed	0.870	0.979	0.997	1.000	1.000
	bound	0.510	0.762	0.884	0.944	0.987

Appendix G. Numerical Validation of the Catalog

Every claim of the catalog (Section 6 and Appendix C) is a statement about an explicitly constructed instance, so each can be checked by simulating that instance. We do so for the arrowhead identity and for each catalog row whose proof asserts a numerical bound. The simulations use the instances of the corresponding results, 10 seeds, and no tuning; they are reported in Table 4. The admissible sizes are out of reach (under (6) $L \geq 8$, so Theorem 3 needs $K \geq 2{,}304$ and $n \geq 138{,}240$), so several blocks use smaller parameters than the proofs require: $L = 3$, $K = 32$ and $n = 19{,}440$ for rows 4, 5 and 7, and $K = 4$, $n = 300$ for row $5'$. Those blocks check the mechanism and the direction of each inequality at a feasible size, not the constants. Their purpose is to catch an error in an inequality direction, not to estimate anything.

Two entries carry more weight than the rest. The certificate row checks the geometric condition of Theorem 3(iv) directly: with $K = 36L^2$ and $n = 36\lambda L^2/\gamma^2$ as in (iv), but at $L = 3$ rather than $L \geq 8$, the handoff certificate is $0.941 < 1$. At the $K = 32$ of the other row-5 entries it is 2.19, so those runs are not certified at handoff, although the hybrid still plays g on every tail round there. The ε-sweep row tests the mechanism rather than the conclusion. Theorem 3 predicts that the multiplier exceeds one by an amount of order $1/\varepsilon$; over a 64-fold range of ε the product $(\text{ratio} - 1) \cdot \varepsilon$ moves only between 0.0044 and 0.0039, so the predicted scaling is what the instance exhibits, not just the predicted sign.

The gaps between claim and simulation are all in the direction the proofs allow, and are the constants discussed in Appendix E, "Remarks on the constants": cold regret on the row-5 template comes in at 3,820 against a bound of 62,716, and the row-4 and row-7 multipliers at 0.343 and 0.99 against caps of 961 and 65. In every case the instance does what the row says it does—rows 4, 7 and 8 kill the multiplier, row $5'$ collects its saving only on the tail with an empty window—while the numerical bound is loose by the stated constant factors.

Table 4: Simulated instances against their proved claims. "Claim" is what the cited result asserts; "simulated" is what the instance produces. Row numbers refer to Table 2. Each block uses the instance of the corresponding result at the sizes stated in the text, which for rows 4–7 are below the admissible sizes, so the absolute magnitudes are not comparable across blocks; the first four row-5 entries are the instance at $\varepsilon = 0.002$ and the last two the ε-sweep instance.

Result	Quantity	Claim	Simulated
Lemma 10	max entrywise error, 200 trials	0	$< 10^{-10}$
Prop. 2	cold regret	$T = 2{,}000$	2,000
	warm regret at $m = 20$	0	0 (all seeds)
Prop. 3	tail regret, good expert	$T - \tau$	$T - \tau$ exactly
	tail regret, bad expert	$T - \tau$	$T - \tau$ exactly
Prop. 8 (row 4)	hybrid vs. cold, $\tau \le Km_0$	identical	identical
	multiplier	$\le 4\beta^2 + 1 = 961$	0.343
Thm. 3 (row 5)	expert regret, part (i)	$\varepsilon Kn + (T - Kn) = 201{,}244$	201,244
	cold regret, part (iii)	$\le K(8\beta^2{+}1) + \varepsilon Kn = 62{,}716$	3,820
	hybrid regret, part (iv)	$\varepsilon Kn = 1{,}244$	1,244
	non-good plays after handoff	0	0
	handoff certificate $2\beta L\|e_j\|_{V^{-1}}$	< 1	0.941
	$(\text{ratio} - 1) \cdot \varepsilon$, $64\times$ sweep in ε	constant	0.0044–0.0039
Prop. 9 (row 5′)	regret inside window	0	0.000
	probes played	0	0
	tail bad pulls per direction	$[m_0,\ 4\beta^2{+}1] = [2, 961]$	93–94
	tail regret	$\ge Km_0 = 8$	374
Prop. 11 (row 7)	multiplier	≤ 65	0.99
	bad pulls per direction	$\Theta(\beta^2)$	214
Cor. 4 (row 8)	multiplier at $\varepsilon \ge \gamma^2/L^2$	≤ 2	$0.63, 0.39, 0.06$

Appendix H. Additional Experimental Results

This appendix gives the full results summarized in Section 8: the decomposition and learning dynamics on AmazonCat/BERT, the complete coverage-versus-transfer analysis, the BGE confidence sweep, the WikiLinks results, a summary across configurations, and the comparison with other routers.

H.1. AmazonCat-13K with BERT Features

Choice of c_{LLM}. On AmazonCat the tuning seeds 11–20 select $c_{\text{LLM}} = 100$ for BERT features (mean Router accuracy 17.2, 19.5, 22.2, 21.4 and 5.4% at $c_{\text{LLM}} = 1, 10, 10^2, 10^3, 10^4$) and $c_{\text{LLM}} = 0.1$ for BGE features (Table 7). On WikiLinks the Flan-T5 grid is $c_{\text{LLM}} \in \{1, 10, 10^2, 10^3, 10^4\}$ and selects $c_{\text{LLM}} = 10^4$ (Table 9); the Qwen grid is $c_{\text{LLM}} \in \{0.01, 0.1, 1, 10, 100\}$ (Table 7). **[TODO:** WikiLinks Qwen sweep]

Table 5: The decomposition of Proposition 1 measured on the BERT-768d runs, in correct answers, mean over ten seeds. A_τ is summed over the routed rounds and I_τ, B_τ over the rest, using the routed set the Router actually selected (the identities hold for any routed set, by the remark after Proposition 1). Every configuration collects almost all of its advantage through I_τ: the expert's own answers are worth a few dozen correct rounds out of 1,578 routed, while the data they leave behind is worth thousands. I_τ is the only column that separates the three experts, and the multiplying configuration is also the only one whose I_τ is positive on every seed.

LLM	A_τ	I_τ	B_τ	seeds with $I_\tau > 0$
Flan-T5-small	47	$2{,}936 \pm 3{,}562$	4,582	9/10
Flan-T5-base	27	$7{,}178 \pm 4{,}772$	9,599	9/10
Flan-T5-large	53	$\mathbf{9{,}571 \pm 3{,}411}$	11,189	**10/10**

The multiplying configuration. Flan-T5-large reaches $27.64 \pm 5.56\%$ against cold LinUCB's $8.39 \pm 2.46\%$ with only 3.2% of rounds routed to the LLM overall; routing is concentrated in the first $\sim$ 5,000 rounds. Table 5 gives the shape of a positive case directly. $A_\tau = 53$ is negligible: it is positive, not negative, because the routed rounds fall early, while cold LinUCB is still near 2%, so even a 5% expert is marginally the better player there; but at 1,578 routed rounds the whole term is worth 53 correct answers against a total advantage of 9,624. $I_\tau = 9{,}571$ carries 99.4% of it, and is positive on all ten seeds. $B_\tau = 11{,}189$ is large because the expert stays at 5% while the warm-started learner passes 27%. This is the row-5′ signature of Proposition 9: the acquisition is nearly free, and the saving is collected on the tail, after routing has stopped. Figure 2 shows the same accounting in time: every routed round falls inside the first 5,000, and 96% of the 9,624 correct answers the Router gains over cold LinUCB are gained after its last routed round, while the expert itself stays frozen at 5.3%. The cumulative-accuracy curves part earlier than that only because an early lead weighs heavily in a short prefix.

H.2. Coverage and the Transfer

Section 7 says where that difference can live. The three experts cover the same directions to within 5%: summed over the routed rounds, the coverage of the directions cold LinUCB is trapped on is 1.40×10^5, 1.46×10^5 and 1.47×10^5, at prices 9.5×10^{-3}, 1.03×10^{-2} and 1.04×10^{-2}, so the price ratio of Theorem 1, measured after the run, is 1.73, 1.59 and 1.58 for the three—close to one another, and ordered opposite to the observed multipliers. We compute it from per-round logs, taking for U the leading directions of the features cold LinUCB played on rounds it lost after routing stopped, in place of certified arms. The streams are outside the theorem's setting, so this is an explanatory statistic rather than a bound the theorem certifies, and by construction it is blind to the transfer. It bounds the regret ratio, and the observed regret ratios lie below it for every expert; since the real stream is outside the theorem's setting we read this as consistency, not confirmation. Since coverage does not separate the experts, we test directly whether the transfer does, and Figure 1 shows that it does (full numbers and tests in Table 6, Appendix H.3).

The comparison is a replay in which the routed rounds are held fixed. For each expert and seed, three versions of the learner run in lockstep on the same stream: the full Router, which decides

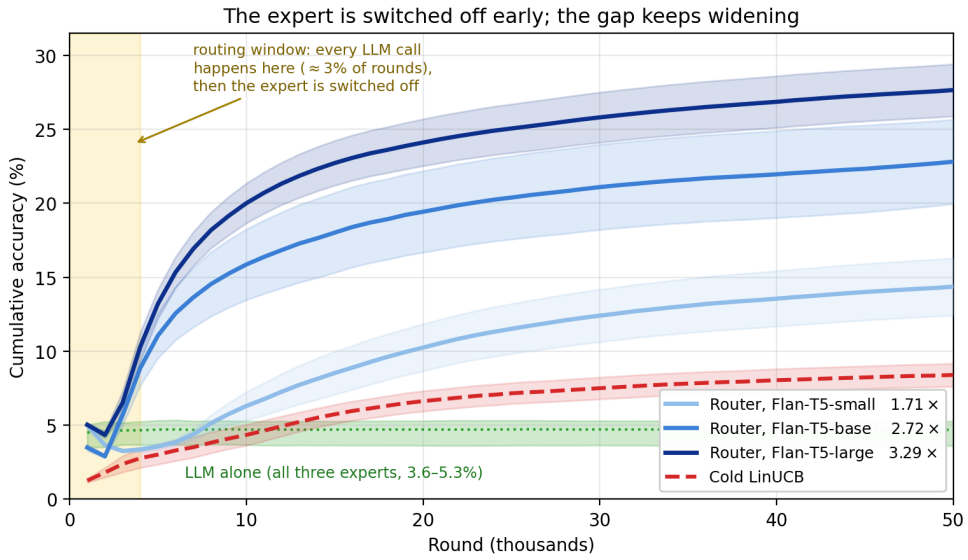


Figure 2: Learning dynamics on AmazonCat-13K (BERT-768d, $c_{\mathrm{LLM}} = 100$), mean over ten seeds with shaded standard error. The shaded window contains every routed round: all calls to the LLM fall inside the first $5{,}000$ rounds, after which the expert is switched off. The three Flan-T5 experts (held-out accuracies 5.20%, 3.61% and 5.27%, the band at the bottom) are consulted on equally few rounds, yet their Routers separate by nearly a factor of two; and 96–100% of each Router's advantage over cold LinUCB, counted in correct answers, accrues after its last routed round. That is the row-$5'$ signature: the saving is collected on the tail.

which rounds are routed, and two ablated learners that route exactly the same rounds and play the expert's answer there, whether or not their own confidence would have routed them. On every other round each learner plays its own LinUCB choice and sees the true reward. The three versions therefore have the same routed rounds, the same routed actions and the same routed covariance, and differ only in what the routed rounds add to the response sum; the full version reproduces the Router of Table 1 exactly. Removing m collapses the spread of the experts' mean per-seed multipliers from 1.74 to 0.50 and turns every significant pairwise difference into a null result: large over small falls from $t = 5.18$ to $t = 1.33$. Suppressing the response sum on the routed rounds gives the same collapse, to a spread of 0.01; and the two transfer-free conditions do not differ from each other on any expert (paired $t = -0.66, +0.64, +0.46$). Removing the transfer costs each expert in proportion to what it gained: the per-seed multiplier falls by 2.04 for large ($t = 4.57$), 1.27 for base ($t = 3.94$) and 0.70 for small ($t = 2.51$). Whether a run settles on the dominant category (Appendix I) follows the same pattern: 2, 8 and 9 of ten runs with the transfer, and at most 2 of ten for any expert without it. A third route to the same conclusion replaces the reward by $\langle \theta_{\mathrm{or}}, \phi \rangle + \mathcal{N}(0, \varsigma^2)$, making the model well specified so that Proposition 4 applies up to the Router's adaptive choice of routed rounds: at $\varsigma \in \{0.001, 0.01, 0.499\}$ the three experts' regret ratios, measured against the oracle's best action, span 0.019, 0.010 and 0.030, against 0.197 on the real reward; the ratios themselves lie between 0.97 and 1.09. Under linear rewards the experts are interchangeable, exactly as predicted.

Where the regret goes. On the regret scale coverage and the transfer have very different sizes. With the response sum suppressed on the routed rounds, coverage alone gives regret ratios 1.035, 1.029 and 1.019: about 2 to 3.5% less regret than cold LinUCB, indistinguishable across experts, and below the price ratio for each, consistent with Theorem 1 although the stream is outside its setting. The full Router gives 1.070, 1.187 and 1.266. For Flan-T5-base and Flan-T5-large, therefore, almost all of the regret saved is saved by the transfer, roughly 13 of 16 and 19 of 21 percentage points, and for Flan-T5-small about half, 3 of 6.5. The separation between experts survives the change of scale intact: on per-seed regret ratios, large beats small with paired $t = +5.65$ and base beats small with $t = +3.32$ when the transfer is present, and neither difference is significant with the transfer removed ($t = +1.25, +1.24$) or the response sum suppressed ($t = -0.47, -0.13$); Figure 1 shows the same collapse on the accuracy scale. The price ratio is about the same for the three experts, as is the coverage gain; it does not predict the size of that gain, which it bounds loosely.

Finally, the transfer is not a small correction to the estimate at handoff; on this data it is essentially the whole of it. The ridge oracle assigns a correct action a score of about 2.5×10^{-3} where the reward is 1, so the residuals dominate: $\|m\|/\|\widehat{\theta}_{\tau+1}\| = 0.9998$, and the identity of Definition 1 holds on the logged trajectories to 1.2×10^{-14}. Stripping m makes the three estimates agree to four decimals in held-out ranking quality (0.2058, 0.2058, 0.2057), while the transfers themselves separate the experts with the same resolution the multipliers have: base and large each beat small ($t = 3.12$, $t = 2.56$) and are indistinguishable from one another ($t = 0.36$), which is the pair the multipliers also cannot separate.

H.3. Coverage and the Transfer: Full Numbers

Table 6 gives the numbers behind Figure 1, together with the paired tests quoted in Section 8.2.

H.4. BGE Features

BGE-1024d. At $c_{\mathrm{LLM}} = 100$ the expert width $u^{(1)} = \|\phi\|_2/\sqrt{c_{\mathrm{LLM}}}$ is permanently below LinUCB's $u^{(0)}$, so the Router defers on every round and equals the LLM exactly: $I_\tau = 0$ with zero variance across seeds. This is a failure of the gate's hyperparameter, not evidence about I_τ, and it affected all six BGE configurations. Table 7 sweeps c_{LLM} down until the gate engages. The sweep is run on seeds 11–20 and selects $c_{\mathrm{LLM}} = 0.1$; the same value is then evaluated on seeds 1–10. There cold LinUCB reaches 32.59%, while the Router reaches 40.20% with Qwen-3B and 38.71% with Qwen-7B, accuracy multipliers of 1.23× and 1.19× (regret ratios 1.13 and 1.10), while routing 0.05% of rounds. Both gains are significant (paired $t = 3.92$ and 2.59 against cold, $t^\star = 2.26$). With Qwen-0.5B the gate engages just as often and the Router does not move at all (1.00×, $t = 0.01$), which is the expert-dependence the BERT rows also show. The accounting has the BERT signature where there is a gain: A_τ is 5 and 6 correct answers while I_τ is 3,801 and 3,056, positive on 9 and 8 of the ten seeds, against $I_\tau = 13$ for Qwen-0.5B. The multipliers are smaller than on BERT, which is what a stronger learner implies: when cold LinUCB already reaches a third of the correct answers, there is less left to buy: cold LinUCB on BGE features settles on the dominant category by itself in 7 of 20 seeds, so the expert has less to add (Appendix I). Pushing c_{LLM} higher drags the Router toward a weaker expert and below cold LinUCB. On WikiLinks the Qwen experts are better than the learner (29.47% and 38.74% against 18.88%), as in row 3 of Table 2: with the expert above the learner, $B_\tau \leq 0$ and the Router has little to add over the expert. [**TODO:** WikiLinks sweep]

Table 6: Separating coverage from the transfer on AmazonCat/BERT-768d, ten seeds, with the routed rounds held fixed: all three versions route exactly the rounds the full Router chose and play the expert's answer there, so their routed covariance is identical. In *full* a routed round also adds its observed reward to the response sum, as the Router does. In *transfer removed* it adds the expert-independent part: the learner is fed $\langle \theta_{\mathrm{or}}, \phi \rangle$ instead of the observed reward on the routed rounds, which deletes m of Definition 1. In *no response sum* it adds nothing to the response sum. On every other round each version plays its own LinUCB choice. Multipliers are accuracy multipliers, Router over cold LinUCB (8.40%), with the regret ratio in parentheses. The spread is the largest minus the smallest of the three experts' mean per-seed multipliers; t-statistics are paired over seeds, on each scale, $t^\star = 2.26$. With the transfer present the experts separate; without it, by either mechanism, no pairwise difference survives.

	Router accuracy			Multiplier over cold		
LLM	full	transfer removed	no response sum	full	transfer removed	no response sum
Flan-T5-small	$14.37 \pm 6.12\%$	$9.71 \pm 4.18\%$	$11.53 \pm 6.61\%$	$1.71\times$ (1.07)	$1.16\times$ (1.01)	$1.37\times$ (1.04)
Flan-T5-base	$22.80 \pm 9.00\%$	$12.56 \pm 5.66\%$	$11.02 \pm 6.86\%$	$2.72\times$ (1.19)	$1.50\times$ (1.05)	$1.31\times$ (1.03)
Flan-T5-large	$27.64 \pm 5.56\%$	$12.49 \pm 6.35\%$	$10.10 \pm 5.08\%$	$3.29\times$ (1.27)	$1.49\times$ (1.05)	$1.20\times$ (1.02)
runs settling on 1471 (small / base / large)				2/8/9	1/2/2	1/2/1
spread of mean per-seed multipliers				1.74	0.50	0.01
large − small, paired t				$+\mathbf{5.18}$	$+1.33$	$+0.01$
base − small, paired t				$+\mathbf{3.03}$	$+1.36$	$+0.03$
large − small, paired t, regret ratio				$+\mathbf{5.65}$	$+1.25$	-0.47
base − small, paired t, regret ratio				$+\mathbf{3.32}$	$+1.24$	-0.13

H.5. OneShotWikiLinks-311

Setup. 622,000 context–entity pairs over 311 entities. *Flan-T5 (BERT-768d features)*: embeddings from `bert-base-nli-mean-tokens`, joint feature $\phi(x, a) = x \odot a \in \mathbb{R}^{768}$, expert Flan-T5 (Chung et al., 2024) at 60M/250M/780M. *Qwen2.5 (BGE-1024d features)*: embeddings from `BAAI/bge-large-en-v1.5` (Xiao et al., 2024), expert Qwen2.5-Instruct (Hui et al., 2024) at 0.5B/3B/7B.

Results. Table 8 reports final cumulative accuracy at $c_{\mathrm{LLM}} = 100$. The Router routes to the LLM on $< 0.3\%$ of rounds for Flan-T5. Routing is driven by feature geometry, not by reward: all three Flan-T5 sizes produce nearly identical routing curves despite an order-of-magnitude spread in accuracy.

Confidence sweep. To rule out a suboptimal c_{LLM}, the Flan-T5 configurations are swept over $c_{\mathrm{LLM}} \in \{1, 10, 10^2, 10^3, 10^4\}$ on the tuning seeds 11–20 (Table 9), and the Qwen configurations over $c_{\mathrm{LLM}} \in \{0.01, 0.1, 1, 10, 100\}$ (Table 7).

[**TODO:** WikiLinks sweep] The decomposition explains the shape. For Flan-T5 the routed rounds do leave usable data—I_τ is 748, 832 and 886 correct answers and is positive on all ten seeds—so the expert's data is not aimed at the wrong directions, as in row 7 of Table 2. The gain stays small because there is little for the expert to steer the learner toward: the most frequent of the 311 entities covers 0.36% of contexts, whereas on AmazonCat one category covers 29.8%

Table 7: Confidence sweep on the BGE-1024d configurations, tuning seeds 11–20, which select c_{LLM}. [**TODO:** WikiLinks sweep] At the $c_{\mathrm{LLM}} = 100$ used for BERT the expert width $u^{(1)} = \|\phi\|/\sqrt{c_{\mathrm{LLM}}}$ is permanently below LinUCB's, the gate never engages, and the Router equals the LLM exactly on every seed ($I_\tau = 0$, zero variance). Lowering c_{LLM} raises the expert's width and restores the gate. On AmazonCat, where cold LinUCB is strong, the Router at $c_{\mathrm{LLM}} = 0.1$ beats it while routing 0.05% of rounds, and the same c_{LLM} reproduces the gain on the evaluation seeds (Table 1); at $c_{\mathrm{LLM}} \geq 1$ it is dragged down toward a weaker expert. On WikiLinks the expert is already better than the learner, as in row 3 of Table 2, and the Router has little to add.

		routed %			Router accuracy			
Dataset	c_{LLM}	0.5B	3B	7B	0.5B	3B	7B	best mult.
AmazonCat	0.01	0.0	0.0	0.0	34.77	34.77	34.77	1.00×
	0.1	0.04	0.05	0.05	33.84	39.68	**40.63**	**1.17**×
	1	75.2	71.9	73.5	28.09	29.26	36.50	1.05×
	10	100	100	100	21.38	22.39	27.17	0.78×
WikiLinks	0.01	—	—	—	—	—	—	—
	0.1	—	—	—	—	—	—	—
	1	—	—	—	—	—	—	—
	10	—	—	—	—	—	—	—
	100	—	—	—	—	—	—	—

Table 8: Final cumulative accuracy on OneShotWikiLinks-311, mean±std over seeds 1–10. Flan-T5 rows are at $c_{\mathrm{LLM}} = 100$ [**TODO:** replace with the $c_{\mathrm{LLM}} = 10^4$ runs selected by Table 9]; the Qwen rows are filled at the c_{LLM} chosen on seeds 11–20. LLM accuracy and cold LinUCB do not depend on c_{LLM}. Mult. is the accuracy multiplier Router/max(LLM, Cold); the regret ratio is within 0.02 of one on every Flan-T5 row. The lift column is the Router's gain over cold LinUCB in percentage points. [**TODO:** WikiLinks sweep]

Family	Size	LLM	Cold LinUCB	Router	Lift (pp)	Mult.
Flan-T5	small (60M)	7.16%	$10.71 \pm 0.04\%$	$10.84 \pm 0.04\%$	+0.13	1.01×
	base (250M)	10.56%	$10.71 \pm 0.04\%$	$10.87 \pm 0.05\%$	+0.16	1.01×
	large (780M)	12.43%	$10.71 \pm 0.04\%$	$10.88 \pm 0.04\%$	+0.17	0.88×
Qwen2.5	0.5B	18.72%	$18.88 \pm 0.04\%$	—	—	—
	3B	29.47%	$18.88 \pm 0.04\%$	—	—	—
	7B	38.74%	$18.88 \pm 0.04\%$	—	—	—

(Section 8.2). For Qwen the expert is far better than the learner (29–39% against 19%), so $B_\tau < 0$ and the Router has little to add over the expert. [**TODO:** WikiLinks sweep]

H.6. Summary Across Configurations

Twelve configurations span two datasets, two model families (60M to 7B), two embedding spaces and four orders of magnitude in c_{LLM}, with ten evaluation seeds per row and every c_{LLM} chosen on a disjoint block of ten tuning seeds. One configuration family multiplies clearly: AmazonCat with

Table 9: Confidence sweep on the WikiLinks Flan-T5 (BERT-768d) configurations, tuning seeds 11–20: share of rounds routed to the LLM and final Router accuracy at each c_{LLM}, mean over seeds. Cold LinUCB does not depend on c_{LLM} (10.70% on these seeds); the LLMs score 7.16%, 10.56% and 12.43%. Best mult. is the largest Router accuracy over $\max\{\text{LLM, cold}\}$ among the three experts. Mean Router accuracy over the experts is highest at $c_{\mathrm{LLM}} = 10^4$ (11.18%, against 11.05% at 10^3 and 10.86% at 10^2), which is therefore the value evaluated on seeds 1–10. At that value the Router consults the LLM on about 14% of rounds and moves toward the LLM's own accuracy rather than multiplying: it gains on the two experts that beat cold LinUCB and stays below Flan-T5-large itself.

	routed %			Router accuracy			
c_{LLM}	small	base	large	small	base	large	best mult.
1	0.10	0.09	0.10	10.77	10.79	10.82	1.01×
10	0.14	0.14	0.14	10.78	10.81	10.82	1.01×
10^2	0.29	0.27	0.27	10.83	10.85	10.90	1.01×
10^3	1.61	1.47	1.52	10.97	11.06	11.13	1.03×
$\mathbf{10^4}$	14.7	13.4	13.8	10.71	11.26	**11.55**	**1.05×**

BERT-768d and Flan-T5 (accuracy multipliers 1.71×, 2.72×, 3.29×; regret ratios 1.07, 1.19, 1.27). AmazonCat with BGE-1024d and Qwen, evaluated on seeds held out from the choice of c_{LLM}, gives a smaller but significant gain (up to 1.23× and 1.13, paired $t = 3.92$). The six WikiLinks configurations do not [**TODO:** WikiLinks sweep]; neither does BGE at the c_{LLM} that routes everything. In every configuration that multiplies, the accounting has the same shape: A_τ is at most a few dozen correct answers, I_τ is in the thousands, and B_τ is large because the expert stays poor while the warm-started learner does not. Where multiplication fails, the failing term of Proposition 1 can be named: on WikiLinks with Flan-T5 I_τ is positive but small, since no entity is frequent enough for the expert to steer the learner toward; with Qwen the expert is above the learner's ceiling and $B_\tau < 0$; at large c_{LLM} on BGE the gate never hands back and $I_\tau = 0$ identically.

H.7. Other Routers

Across the three Flan-T5 experts coverage is expert-independent, as Corollary 2 requires, and the transfer accounts for the whole of the observed separation and for most of the regret saved (Figure 1; Table 6 in Appendix H.3). The deterministic Router differs from the randomized CORRAL master of Agarwal et al. (2017) and the CORRAL-style routing of Chen et al. (2024) only in how it chooses the routed rounds; both feed every observation to the bandit learner, so the decomposition applies to both, and Proposition 4 adds that two routers with the same routed Gram matrix leave warm starts of the same law—any difference between routers must therefore appear in A_τ or in which residuals their routed sets happen to collect. Table 10 runs both alternatives.

Three things follow. First, no baseline beats the gate significantly on any expert, and the gate beats CORRAL on Flan-T5-small ($t = 2.64$) while issuing 16× fewer expert queries; on the other two experts CORRAL matches the gate at four to five times the query budget. Second, the offline arm at a matched budget of 1,600 pulls is statistically indistinguishable from the gate on all three experts even though its acquisition is free, so the online routing decision costs essentially nothing—which is the content of $A_\tau \approx 0$, now visible from the other side. Tripling the free budget to 5,000

Table 10: Alternative routing mechanisms on AmazonCat-13K (BERT-768d, $c_{\text{LLM}} = 100$), ten seeds, paired on seed with the gate. CORRAL is the randomized log-barrier master of Agarwal et al. (2017) with importance-weighted updates; the offline arms spend τ expert pulls on contexts drawn from *outside* the online stream and then run plain LinUCB, so their acquisition is not charged as regret and $A_\tau = 0$ identically. $\tau = 1{,}600$ matches the gate's routed count. Cold LinUCB is bit-identical across all four arms on every seed, so each comparison isolates the router. The last column is the paired t statistic for gate minus that arm over the ten shared seeds ($t^\star = 2.26$). Mult. is the accuracy multiplier, with the regret ratio in parentheses.

LLM	Router	queries	Router acc.	Mult.	A_τ	I_τ	t vs. gate
Flan-T5-small	Gate (Alg. 1)	1,406	$14.37 \pm 6.12\%$	$1.71\times$ (1.07)	47	2,936	—
	CORRAL	22,860	$8.87 \pm 1.37\%$	$1.06\times$ (1.01)	-508	744	**2.64**
	Offline τ=1,600	1,600	$11.17 \pm 5.71\%$	$1.33\times$ (1.03)	0	1,386	1.16
	Offline τ=5,000	5,000	$15.95 \pm 7.11\%$	$1.90\times$ (1.09)	0	3,778	-0.59
Flan-T5-base	Gate (Alg. 1)	1,570	$22.80 \pm 9.00\%$	$2.72\times$ (1.19)	27	7,178	—
	CORRAL	6,939	$24.44 \pm 2.39\%$	$2.91\times$ (1.21)	-23	8,045	-0.63
	Offline τ=1,600	1,600	$20.97 \pm 8.89\%$	$2.50\times$ (1.16)	0	6,286	0.42
	Offline τ=5,000	5,000	$27.29 \pm 5.17\%$	$3.25\times$ (1.26)	0	9,449	-2.00
Flan-T5-large	Gate (Alg. 1)	1,597	$27.64 \pm 5.56\%$	$3.29\times$ (1.27)	53	9,571	—
	CORRAL	7,488	$25.42 \pm 2.31\%$	$3.03\times$ (1.23)	50	8,460	1.49
	Offline τ=1,600	1,600	$26.27 \pm 7.61\%$	$3.13\times$ (1.24)	0	8,937	0.42
	Offline τ=5,000	5,000	$25.76 \pm 6.53\%$	$3.07\times$ (1.23)	0	8,681	0.67

still does not separate the two. Where an alternative does come out ahead of the gate in the mean, the margin is small, never significant, and paid for with a much larger budget: CORRAL on Flan-T5-base gains 1.6 points ($t = -0.63$) with $4.4\times$ the gate's expert queries, and the offline arm at $\tau = 5{,}000$ gains 1.6 and 4.5 points on Flan-T5-small and Flan-T5-base ($t = -0.59, -2.00$) only by drawing 3–$4\times$ as many expert answers, all collected before the online run begins. None of these trades buys a reliable improvement, so the gate is the cheapest router in the comparison and no alternative is significantly better on any expert.

Third, and most directly, the decomposition survives the change of master. In all twelve rows I_τ carries the value and A_τ is negligible beside it, and A_τ is exactly where the choice of router shows up: it is 0 by construction offline, a few dozen for the gate, and *negative* for CORRAL on the two weaker experts (-508 and -23), which is the price of routing 22,860 rounds to a 5.2% expert while cold LinUCB is improving. The comparison also shows that coverage is not bought by volume: CORRAL routes $16\times$ as many rounds as the gate on Flan-T5-small and ends with $I_\tau = 744$ against the gate's 2,936. What the gate supplies is not more expert pulls but pulls on the directions cold LinUCB is trapped on, which is what the price $\kappa(u)$ of Section 5 measures and what Proposition 4 says is the whole of what coverage contributes.

Appendix I. The Dominant Category on AmazonCat-13K

This appendix gives the evidence behind the paragraph "What the transfer carries" of Section 8.2. All numbers are on BERT-768d features unless stated.

The label distribution. On the 50,000 contexts of the online stream the four most frequent of the 13,330 categories are present on 29.8%, 17.0%, 11.3% and 7.8% of contexts (category identifiers 1471, 7961, 7892, 7891). A policy that always predicts the first scores 29.77%; it uses the label distribution, which an online learner is not given. Context-free bandits that must discover it do not: over the same stream, 50,000 rounds and seeds 1–10, UCB1 (one pull of every category first) scores 0.04% and an optimistic empirical-mean greedy rule 2.31%.

Where a run settles. We say a run *settles* on a category if that category accounts for more than 20% of the learner's plays in the last fifth of the horizon. For the dominant category the classification does not depend on the threshold: on the runs of Table 11, moving it anywhere between 10% and 30% changes the number of runs that settle on 1471 by at most one on BERT features and not at all on BGE features. The counts for the less frequent categories are more sensitive on BERT, where runs that settle on 7892 predict it on about a fifth of late rounds. Table 11 reports where runs settle and how accurate the runs that settle on each category are. On BERT features cold LinUCB settles on the dominant category in 1 of 20 seeds; the two multiplying experts move it there in 17 of 20. On BGE features cold LinUCB already settles there in 7 of 20 seeds, which is why the expert has less to add; there the runs that settle on the dominant category reach 41.2%, above the 29.8% of always predicting it, so the learner also uses the context. Knowing only where each run settles predicts mean Router accuracy well: predicting a configuration's accuracy as the frequency of each settled category times the mean accuracy of runs settling there in the *other* configurations gives $R^2 = 0.93$ over 25 BERT configurations and $R^2 = 0.93$ over 7 BGE configurations. These are every configuration that routes fewer than half the rounds: on BERT, the gate at $c_{\mathrm{LLM}} \in \{1, 10, 10^2, 10^3\}$, CORRAL, the two offline arms of Table 10 and the transfer-removed ablation of Table 6, for each of the three experts, plus cold LinUCB; on BGE, the gate at $c_{\mathrm{LLM}} \in \{0.01, 0.1\}$ for each expert, plus cold LinUCB.

A handful of routed answers decides it. Table 12 groups every Router run that routes fewer than half the rounds by the number k of routed rounds on which the expert proposed the dominant category and was correct, and reports the fraction that settle on it. On BERT features the fraction rises from 7% at $k = 0$ to 80% at $k = 4$, and every one of the 54 runs with $k \geq 5$ settles there. On BGE features the fraction at $k = 0$ is already 43%, the rate at which the learner finds the category by itself, and again every run with $k \geq 5$ settles there. The pooled runs span experts, values of c_{LLM} and, on BERT, the CORRAL master, so the curve is a property of the learner and the data rather than of one expert. The three BERT experts differ in exactly this count: in the $c_{\mathrm{LLM}} = 100$ runs the mean k is 0.20 for Flan-T5-small, 3.70 for Flan-T5-base and 4.00 for Flan-T5-large.

Controlled test. Table 13 changes the expert's answers on a fixed set of 15 contexts and nothing else. The runs use an exact re-implementation of the Router and of cold LinUCB, checked to make the same choice as the experimental runner on every round of a 3,000-round comparison, on 10,000 held-out contexts that do not appear in the online stream, with a horizon of 10,000 rounds and 20 seeds; the seed fixes the order of the contexts and is shared across conditions, so every comparison is paired. Multipliers are relative to cold LinUCB on the same runs (4.63%). +15 *on* 1471 replaces 15

Table 11: Where runs settle on AmazonCat-13K (seeds 1–20; BERT at $c_{\mathrm{LLM}} = 100$, BGE at $c_{\mathrm{LLM}} = 0.1$). Each cell counts runs that settle on that category; the percentage under each category is its frequency in the stream. The bottom block gives the mean final accuracy of the runs, pooled over the four rows, that settle on each category.

Features	Policy	1471 (29.8%)	7961 (17.0%)	7892 (11.3%)	7891 (7.8%)	none	Accuracy
BERT-768d	Cold LinUCB	1	0	9	1	9	8.34%
	Flan-T5-small	3	0	14	0	3	13.71%
	Flan-T5-base	17	0	3	0	0	25.10%
	Flan-T5-large	17	0	3	0	0	26.90%
BGE-1024d	Cold LinUCB	7	12	0	0	1	33.68%
	Qwen-0.5B	8	9	0	2	1	33.23%
	Qwen-3B	17	2	0	1	0	39.94%
	Qwen-7B	16	3	0	1	0	39.67%
Accuracy of settled runs, BERT		28.08%	—	11.61%	6.24%	5.93%	
Accuracy of settled runs, BGE		41.21%	30.84%	—	24.66%	25.96%	

Table 12: Runs that settle on the dominant category, by the number k of correct routed answers on it. BERT: 177 runs ($c_{\mathrm{LLM}} \in \{1, 10, 10^2, 10^3\}$ and CORRAL). BGE: 99 runs ($c_{\mathrm{LLM}} \in \{0.01, 0.1\}$).

k	0	1	2	3	4	≥ 5
BERT-768d	3/45 (7%)	11/29 (38%)	9/17 (53%)	13/17 (76%)	12/15 (80%)	54/54 (100%)
BGE-1024d	25/58 (43%)	3/3	5/9	6/8	8/9	12/12 (100%)

wrong answers of Flan-T5-small by the dominant category on contexts that contain it. $+15$ *random* gives the same number of correct answers, on the same contexts where possible, on a uniformly chosen true category other than 1471. $+15$ *on* 7961 gives 15 correct answers on the second most frequent category. -15 *on* 1471 replaces the 15 correct dominant-category answers of Flan-T5-large by wrong answers. The first three conditions raise held-out accuracy by the same 0.15 points; only the category differs. $+15$ *on* 1471*, reward hidden* uses the same answers as $+15$ *on* 1471, but when the Router routes one of the 15 changed contexts the learner adds the answer's feature vector to its design matrix and nothing to its response sum: the change adds coverage along the dominant category and no transfer. Only 2.15 of the 15 changed answers are routed per seed on average (none in one seed), so the whole effect of the intervention passes through two or three routed rounds.

Hiding the reward of the added answers removes the gain: the Router falls back to $1.48\times$, not significantly different from the unmodified expert, and significantly below the same answers with their rewards (paired $t = 3.46$, $p = 0.003$; settling $8/20$ against $2/20$, McNemar $p = 0.031$). Since only about two changed answers are routed per run, the comparison is between two or three rewarded answers on the dominant category, which are enough to move the learner there and double the multiplier, and the same two or three answers with their rewards hidden, which leave the learner

Table 13: Changing 15 of the expert's 10,000 held-out answers. Tests are paired over 20 seeds against the unmodified expert: a paired t-test on Router accuracy and an exact McNemar test on whether the run settles on 1471.

Expert	Change	Expert acc.	Router acc.	Mult.	Settle on 1471	p (acc. / settle)
Flan-T5-small	none	5.34%	6.14%	1.33×	1/20	—
	+15 random	5.49%	7.63%	1.65×	2/20	0.15 / 1.00
	+15 on 7961	5.49%	9.39%	2.03×	3/20	0.010 / 0.50
	+15 on 1471	5.49%	**12.10%**	**2.62×**	**8/20**	**0.002 / 0.016**
	+15 on 1471, reward hidden	5.49%	6.83%	1.48×	2/20	0.47 / 1.00
Flan-T5-large	none	5.32%	13.79%	2.98×	10/20	—
	−15 on 1471	5.17%	10.09%	2.18×	6/20	0.067 / 0.39

where it was. What the learner uses is the reward, not the coverage along the category. At equal accuracy, the gain grows with the frequency of the category the added answers point to: none for random categories, a significant gain for the second most frequent category (where 5 of 20 runs settle on that category instead), and the largest gain for the dominant one. Removing the dominant-category answers from Flan-T5-large lowers its multiplier but not significantly; its other answers still lead 6 of 20 runs to the dominant category. Individual seeds are sensitive to any change in the routed data, which can tip a run toward or away from a category, so we compare rates over seeds rather than reading individual pairs.

The same test on the evaluation stream. Table 14 repeats the test on the 50,000-round stream and seeds 1–10 of Table 1, with the same simulator. To keep the same share of the stream, 75 of Flan-T5-small's wrong answers are changed, again +0.15 points of expert accuracy (5.20% to 5.35%); the unmodified row is the Router of Table 1, and cold LinUCB is the same 8.40%. The conditions are those of Table 13: +75 *on* 1471, the same answers with their rewards hidden, and +75 correct answers on random categories other than 1471. The Router routes 2.6 of the 75 changed answers per seed on average (none in one seed), about the same number as on the held-out stream, since it consults the expert almost only in the first few thousand rounds. The result is the same: rewarded answers on the dominant category raise the multiplier from 1.71× to 2.56× and settling from 2 to 7 of ten runs, while the same answers with their rewards hidden give 1.84×, not significantly different from the unmodified expert and significantly below the same answers with their rewards (paired $t = 2.36$, $p = 0.042$; settling $7/10$ against $3/10$, McNemar $p = 0.13$). Answers on random categories give no significant gain either. With ten seeds the settling counts alone are not significant; the accuracy tests are.

Appendix J. Confidence-Gated Meta-Algorithm: Full Description

This appendix expands the brief description of Section 3 into the full round-by-round walkthrough, states the deterministic confidence-gated routing rule, and provides the pseudocode used in our experiments. The Router is the adaptive-τ version of the handoff hybrid H_τ of Section 3: it chooses, round by round, whether the expert or LinUCB acts, and feeds every observation to LinUCB. Proposition 5 applies to it with the acquisition term summed over expert-routed rounds.

Table 14: Changing 75 of the expert's answers on the 50,000-round evaluation stream, seeds 1–10. Tests are paired over the ten seeds against the unmodified expert: a paired t-test on Router accuracy and an exact McNemar test on whether the run settles on 1471. Multipliers and regret ratios are relative to cold LinUCB on the same seeds (8.40%).

Change	Expert acc.	Router acc.	Mult. (regret ratio)	Settle on 1471	Routed changed	p (acc. / settle)
none	5.20%	$14.37 \pm 6.12\%$	1.71× (1.07)	2/10	—	—
+75 random	5.35%	$15.42 \pm 9.41\%$	1.84× (1.08)	3/10	2.4	0.63 / 1.00
+75 on 1471	5.35%	$\mathbf{21.50 \pm 6.80\%}$	**2.56× (1.17)**	**7/10**	2.6	**0.007** / 0.063
+75 on 1471, reward hidden	5.35%	$15.45 \pm 5.71\%$	1.84× (1.08)	3/10	2.6	0.61 / 1.00

Rather than encoding prior knowledge through a single improved initialization, we assume access to *multiple frozen policy estimators* $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$. Each pair should be thought of as a constant policy obtained offline: $\widehat{\theta}_i$ is a fixed linear predictor and $V^{(i)}$ is the corresponding design matrix that captures the geometry of the data on which that predictor was trained. Our goal is not to compete with the single best expert in hindsight, an objective closer in spirit to online model-selection methods such as CORRAL (Agarwal et al., 2017; Chen et al., 2024), which provide robustness guarantees of the form "perform nearly as well as the best base algorithm." Here we want to collect the data-value term I_τ of Proposition 1 when it is available while paying as little acquisition regret as possible, and to fall back to LinUCB otherwise.

Setting. At each round $t = 1, \dots, T$, the learner observes a finite action set $\mathcal{A}_t$ with feature vectors $\{x_{t,a}\}_{a \in \mathcal{A}_t} \subseteq \mathbb{R}^d$, selects an action $A_t \in \mathcal{A}_t$, and observes reward $r_t = \langle \theta^\star, x_{t,A_t} \rangle + \eta_t$, with η_t conditionally mean-zero sub-Gaussian. We are given m frozen estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$ and an online LinUCB learner with state $(V_t, c_t, \widehat{\theta}_t)$, where $\widehat{\theta}_t = V_t^{-1} c_t$. The frozen estimators are never updated; only LinUCB is updated online.

Expert proposals and geometry-aware confidence. Each frozen expert $i \in [m]$ proposes the greedy action $a_t^{(i)} \in \arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_i, x_{t,a} \rangle$. To measure how trustworthy expert i is on the current round, we evaluate the uncertainty of its proposed action under its own design geometry,

$$u_t^{(i)} \;:=\; \big\| x_{t,a_t^{(i)}} \big\|_{(V^{(i)})^{-1}}, \qquad \|x\|_{M^{-1}} \;:=\; \sqrt{x^\top M^{-1} x}.$$

The interpretation is simple. If the proposed action lies in directions that are well covered by the offline data underlying expert i, then $u_t^{(i)}$ is small and the expert is more trustworthy; if the current context points in directions poorly represented in $V^{(i)}$, then $u_t^{(i)}$ is large and the expert should be trusted less. We treat LinUCB itself as an additional candidate, indexed by 0, with proposal $a_t^{(0)} \in \arg\max_a \langle \widehat{\theta}_t, x_{t,a} \rangle + \alpha_t \|x_{t,a}\|_{V_t^{-1}}$ and uncertainty score $u_t^{(0)} := \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$.

Deterministic confidence-gated selection. The natural rule is to prefer the frozen expert whose proposed action has the smallest uncertainty score, unless the online learner is already at least as confident. Concretely, with $i_t^\star \in \arg\min_{i \in [m]} u_t^{(i)}$,

$$I_t \;=\; \begin{cases} i_t^\star, & \text{if } u_t^{(i_t^\star)} < u_t^{(0)}, \\ 0, & \text{otherwise,} \end{cases} \qquad A_t \;:=\; a_t^{(I_t)}.$$

The master is entirely deterministic. Early in learning, LinUCB typically has a large confidence radius, so a frozen expert may be preferred; as data accumulate, V_t grows and the LinUCB uncertainty shrinks, and eventually the adaptive learner becomes the most confident option and takes over automatically.

Optional permanent handover to LinUCB. To formalize the idea that LinUCB should eventually dominate, we include a one-way switching criterion. With upper-confidence scores $U_t^{(i)} = \langle \widehat{\theta}_i, x_{t,a_t^{(i)}} \rangle + \alpha_i \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$ for each frozen expert and lower-confidence score $L_t^{(0)} = \langle \widehat{\theta}_t, x_{t,a_t^{(0)}} \rangle - \alpha_t \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$ for LinUCB, if $L_t^{(0)} \geq \max_{i \in [m]} U_t^{(i)}$ for K consecutive rounds we permanently hand control to LinUCB.

The instance used in our experiments. All experiments of Section 8 use a single frozen expert ($m = 1$), the LLM, with $V^{(1)} = c_{\mathrm{LLM}} I_d$, and do not use the permanent handover: the gate is evaluated on every round.

Distinction from randomized model selection. The deterministic, sample-by-sample geometric routing distinguishes Algorithm 1 from CORRAL-style randomized masters in two ways. First, it can exploit context-level information about which estimator is reliable on *this particular* action set; second, because it is deterministic, the choice of which rounds to route is a function of the current widths rather than of a sampled index. Both rules feed the played action and the observed reward into (V_t, c_t), so both produce the online warm start (M_τ, c_τ) whose value is I_τ in Proposition 1; the decomposition itself does not distinguish them, and any difference between them shows up in which rounds are routed, i.e., in A_τ and in the composition of M_τ.

Appendix K. Pre-Runtime Estimates

We tried two ways of anticipating the outcome of a handoff from a labeled held-out sample of n contexts and the expert's proposals on it, before any online round is played. The first is built only from the expert's held-out Gram matrix $\widehat{M}_{\mathsf{E}} = \sum_{s \leq n} x_{s,a_s^{\mathsf{E}}} x_{s,a_s^{\mathsf{E}}}^\top$, that is, from the coverage its answers would provide; the second also uses whether those answers are right.

Width-limited regret. The held-out average $\widehat{\Phi}(V) = \frac{1}{n} \sum_s \max_a \min\{\widehat{\Delta}_{s,a}, 2\beta \|x_{s,a}\|_{V^{-1}}\}$ of the quantity in Lemma 1, compared at $V = \lambda I$ and $V = \lambda I + (\tau/n)\widehat{M}_{\mathsf{E}}$, estimates how much the expert's data would lower the regret LinUCB can still incur. On AmazonCat it cannot move: with 13,330 actions in 768 dimensions the width term exceeds the unit gap for every held-out context, so $\widehat{\Phi}$ equals 1.0000 under both designs, for every expert and every $\tau \in \{1000, 2000, 5000, 10000\}$.

Dry run. Replaying the held-out contexts (resampled to $T' = 50{,}000$ rounds, ten seeds) and simulating cold LinUCB and LinUCB warm-started from $\tau = 1{,}600$ expert rounds ranks the three Flan-T5 experts correctly (regret ratios 1.021, 1.057 and 1.084; large over small, paired $t = 2.91$) but underestimates the size of the effect by a factor of about 3.2 on the regret scale (excesses over one of 2.2%, 5.7% and 8.4% against observed 7.0%, 18.7% and 26.6%).

What each estimate can see. The width-limited estimate sees only coverage, and on this data the experts cover the same directions to within 5% (Section 8.2); what separates them is the transfer, which depends on the residuals at the actions the expert chooses and is not a function of $\widehat{M}_{\mathsf{E}}$. The dry run does see the transfer, because its warm start adds the expert's observed rewards to the response

Algorithm 1: Confidence-Gated Master over Frozen Policies and LinUCB

Input: Frozen estimators $\{(\widehat{\theta}_i, V^{(i)})\}_{i=1}^m$, regularization $\lambda > 0$, confidence parameters $\{\alpha_t\}_{t=1}^T$

Initialize LinUCB state: $V_1 = \lambda I_d$, $c_1 = 0$, $\widehat{\theta}_1 = 0$
Set HANDOVER $\leftarrow$ False
for $t = 1, \ldots, T$ **do**
  Observe action set $\mathcal{A}_t$
  Compute LinUCB proposal $a_t^{(0)} \in \arg\max_{a \in \mathcal{A}_t} \left\{ \langle \widehat{\theta}_t, x_{t,a} \rangle + \alpha_t \|x_{t,a}\|_{V_t^{-1}} \right\}$
  Set $u_t^{(0)} \leftarrow \|x_{t,a_t^{(0)}}\|_{V_t^{-1}}$
  for $i = 1, \ldots, m$ **do**
    Compute frozen proposal $a_t^{(i)} \in \arg\max_{a \in \mathcal{A}_t} \langle \widehat{\theta}_i, x_{t,a} \rangle$
    Set $u_t^{(i)} \leftarrow \|x_{t,a_t^{(i)}}\|_{(V^{(i)})^{-1}}$
  end
  if HANDOVER $=$ True **then**
    $A_t \leftarrow a_t^{(0)}$
  else
    Choose $i_t^\star \in \arg\min_{i \in [m]} u_t^{(i)}$
    if $u_t^{(i_t^\star)} < u_t^{(0)}$ **then**
      $A_t \leftarrow a_t^{(i_t^\star)}$
    else
      $A_t \leftarrow a_t^{(0)}$
    end
  end
  Observe reward r_t
  Update LinUCB: $V_{t+1} \leftarrow V_t + x_{t,A_t} x_{t,A_t}^\top$, $c_{t+1} \leftarrow c_t + r_t x_{t,A_t}$, $\widehat{\theta}_{t+1} \leftarrow V_{t+1}^{-1} c_{t+1}$
  Compute $U_t^{(i)}$ for all $i \in [m]$ and $L_t^{(0)}$
  if $L_t^{(0)} \geq \max_{i \in [m]} U_t^{(i)}$ for K consecutive rounds **then**
    HANDOVER $\leftarrow$ True
  end
end

sum, and that is why it ranks the experts correctly. It still underestimates the size of the effect, and it needs labeled data and the expert's answers on them, which is most of the cost of running the experiment. Neither estimate therefore predicts the size of the multiplier cheaply, before the expert is run.